Hartshorne, Chapter V: Surfaces

Solutions to the exercises

Conventions. As in the book, k is an algebraically closed field and a surface is a nonsingular projective surface over k unless an exercise says otherwise. Results of the text are cited by their numbers in Hartshorne ("Thm. 1.9", "Prop. 3.4", ...), and results of other chapters as "II, ...", "III, ..." or "IV, ...". The numbering and statements of the exercises follow the edition scanned in this repository; each exercise is restated before its solution, and starred exercises are marked "Starred in the book" at the end of the statement.

Honest notes. Several printed statements are false as printed or need a hypothesis, and the solutions say what is proved instead: Ex. 2.15(c) claims a nonsingular curve Y which cannot exist (its arithmetic genus is 4 while it is birational to a curve of genus 3); Ex. 3.8 as printed has two curves with different multiplicity sequences, so the intended phenomenon is proved for a corrected pair; characteristic assumptions are made in Ex. 1.4, 1.8, 3.5, 4.3, 4.4, 4.16. **Ex. 4.15(c) is only partly proved:** it is proved for very general points (outside a countable union of proper closed sets) and, for every general-position tuple, under one extra condition; existence of general-position tuples of every length, which is all that (d) and (e) use, is proved. Ex. 4.6 is open-ended and is answered under stated hypotheses. Ex. 1.8 depends on a normalization of the trace map (the one used in III Ex. 7.4), stated as an assumption. Some solutions use standard facts beyond the text (Weierstrass preparation, Tits' theorem, Nagata's criterion), which are named where used.

1. Geometry on a surface

Throughout, k is algebraically closed, X is a (nonsingular projective) surface, K is a canonical divisor, a "curve" on X is an effective divisor, and $\equiv$ denotes numerical equivalence. We use freely the following consequences of the text.

- **(F1)** If $C \subseteq X$ is an irreducible nonsingular curve and D is any divisor, then $C.D = \deg_C(\mathcal{L}(D) \otimes \mathcal{O}_C)$. Indeed, both sides are additive in D and depend only on the linear equivalence class of D; writing $D \sim D_1 - D_2$ with D_i very ample (proof of (1.1)) and choosing, by (1.2), representatives of $|D_i|$ which are nonsingular and transversal to C, the formula is (1.3). (This is how the text uses it in (1.4.1) and (1.5).)
- **(F2)** If $C \neq C'$ are irreducible curves, then $C.C' = \sum_P (C.C')_P \geq 0$ by (1.4), and $C.C' \geq 1$ if $C \cap C' \neq \emptyset$ (each local intersection multiplicity is the length of a nonzero module).
- **(F3)** If H is ample, then $H.C > 0$ for every curve C and $H^2 > 0$: for $m \gg 0$, mH is

very ample, and $mH.C$ is the degree of C and m^2H^2 the degree of X in the embedding (Ex. 1.2).

- **(F4)** (1.9) says: H ample, $D \not\equiv 0$, $D.H = 0$ implies $D^2 < 0$.

Exercise 1.1

Problem. Let C, D be any two divisors on a surface X, and let the corresponding invertible sheaves be $\mathcal{L}, \mathcal{M}$. Show that

$$C.D = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}).$$

Solution. The proof of (1.6) shows (using Serre duality) that for every divisor E

$$\chi(\mathcal{L}(E)) = \tfrac{1}{2}E.(E - K) + \chi(\mathcal{O}_X). \qquad \text{(RR)}$$

Apply (RR) to $E = -C$, $-D$, $-C - D$, noting $\mathcal{L}^{-1} = \mathcal{L}(-C)$, $\mathcal{M}^{-1} = \mathcal{L}(-D)$, $\mathcal{L}^{-1} \otimes \mathcal{M}^{-1} = \mathcal{L}(-C - D)$:

$$\begin{aligned}
\chi(\mathcal{L}^{-1}) &= \tfrac{1}{2}(C^2 + C.K) + \chi(\mathcal{O}_X), \\
\chi(\mathcal{M}^{-1}) &= \tfrac{1}{2}(D^2 + D.K) + \chi(\mathcal{O}_X), \\
\chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}) &= \tfrac{1}{2}(C + D).(C + D + K) + \chi(\mathcal{O}_X) \\
&= \tfrac{1}{2}(C^2 + C.K) + \tfrac{1}{2}(D^2 + D.K) + C.D + \chi(\mathcal{O}_X).
\end{aligned}$$

Hence $\chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}) = \chi(\mathcal{O}_X)(1 - 1 - 1 + 1) + C.D = C.D$. □

Exercise 1.2

Problem. Let H be a very ample divisor on the surface X, corresponding to a projective embedding $X \subseteq \mathbf{P}^N$. If we write the Hilbert polynomial of X (III, Ex. 5.2) as

$$P(z) = \tfrac{1}{2}az^2 + bz + c,$$

show that $a = H^2$, $b = \frac{1}{2}H^2 + 1 - \pi$, where π is the genus of a nonsingular curve representing H, and $c = 1 + p_a$. Thus the *degree* of X in $\mathbf{P}^N$, as defined in (I, Sect. 7), is just H^2. Show also that if C is any curve in X, then the degree of C in $\mathbf{P}^N$ is just $C.H$.

Solution. Here $\mathcal{O}_X(1) = \mathcal{L}(H)$, and $P(z) = \chi(\mathcal{O}_X(z)) = \chi(\mathcal{L}(zH))$ (III, Ex. 5.2). By (RR) of Exercise 1.1,

$$P(z) = \tfrac{1}{2}zH.(zH - K) + \chi(\mathcal{O}_X) = \tfrac{1}{2}H^2z^2 - \tfrac{1}{2}(H.K)\,z + 1 + p_a.$$

So $a = H^2$, $c = 1 + p_a$, $b = -\frac{1}{2}H.K$. By Bertini (II, 8.18) H is linearly equivalent to a nonsingular hyperplane section C_0, which is connected (III, 7.9) and hence an irreducible nonsingular curve, of genus π; the adjunction formula (1.5) gives $2\pi - 2 = C_0.(C_0 + K) = H^2 + H.K$. Hence $H.K = 2\pi - 2 - H^2$ and $b = -\frac{1}{2}(2\pi - 2 - H^2) = \frac{1}{2}H^2 + 1 - \pi$. The degree of the surface X is 2! times the leading coefficient of P (I, Sect. 7), i.e. $a = H^2$.

Now let C be a curve (effective divisor) on X. Tensoring $0 \to \mathcal{L}(-C) \to \mathcal{O}_X \to \mathcal{O}_C \to 0$ with $\mathcal{L}(zH)$ gives

$$\chi(\mathcal{O}_C(z)) = \chi(\mathcal{L}(zH)) - \chi(\mathcal{L}(zH - C)).$$

By (RR),

$$\begin{aligned}\chi(\mathcal{L}(zH)) - \chi(\mathcal{L}(zH - C)) &= \tfrac{1}{2}\big[zH.(zH - K) - (zH - C).(zH - C - K)\big] \\ &= \tfrac{1}{2}\big[2z\, H.C - C^2 - C.K\big] = (H.C)\, z - \tfrac{1}{2}C.(C + K).\end{aligned}$$

Thus the Hilbert polynomial of $C \subseteq \mathbf{P}^N$ has leading coefficient $H.C$, i.e. $\deg C = H.C$ (I, Sect. 7; for a one-dimensional scheme the degree is 1! times the leading coefficient). □

Exercise 1.3

Problem. Recall that the *arithmetic genus* of a projective scheme D of dimension 1 is defined as $p_a = 1 - \chi(\mathcal{O}_D)$ (III, Ex. 5.3).

(a) If D is an effective divisor on the surface X, use (1.6) to show that $2p_a - 2 = D.(D + K)$.

(b) $p_a(D)$ depends only on the linear equivalence class of D on X.

(c) More generally, for *any* divisor D on X, we define the *virtual arithmetic genus* (which is equal to the arithmetic genus if D is effective) by the same formula: $2p_a - 2 = D.(D + K)$. Show that for any two divisors C, D we have

$$p_a(-D) = D^2 - p_a(D) + 2 \quad \text{and} \quad p_a(C + D) = p_a(C) + p_a(D) + C.D - 1.$$

Solution. (a) From $0 \to \mathcal{L}(-D) \to \mathcal{O}_X \to \mathcal{O}_D \to 0$ and additivity of χ, $\chi(\mathcal{O}_D) = \chi(\mathcal{O}_X) - \chi(\mathcal{L}(-D))$. By (RR) (Exercise 1.1; this is (1.6) rewritten via Serre duality), $\chi(\mathcal{L}(-D)) = \frac{1}{2}(-D).(-D - K) + \chi(\mathcal{O}_X) = \frac{1}{2}D.(D + K) + \chi(\mathcal{O}_X)$. Hence $\chi(\mathcal{O}_D) = -\frac{1}{2}D.(D + K)$ and $p_a(D) = 1 + \frac{1}{2}D.(D + K)$, i.e. $2p_a - 2 = D.(D + K)$.

(b) By (a), for effective D the number $p_a(D) = 1 + \frac{1}{2}(D^2 + D.K)$ is expressed through intersection numbers, which depend only on the linear equivalence class of D by (1.1)(4).

(c) With the definition $p_a(D) = 1 + \frac{1}{2}D.(D + K)$ for every divisor D:

$$p_a(-D) = 1 + \tfrac{1}{2}(-D).(-D + K) = 1 + \tfrac{1}{2}(D^2 - D.K),$$

while $D^2 - p_a(D) + 2 = D^2 + 2 - 1 - \frac{1}{2}D^2 - \frac{1}{2}D.K = 1 + \frac{1}{2}(D^2 - D.K)$. These agree. Next,

$$\begin{aligned}p_a(C + D) &= 1 + \tfrac{1}{2}(C + D).(C + D + K) = 1 + \tfrac{1}{2}\big[C.(C + K) + D.(D + K)\big] + C.D \\ &= \big(1 + \tfrac{1}{2}C.(C + K)\big) + \big(1 + \tfrac{1}{2}D.(D + K)\big) - 1 + C.D = p_a(C) + p_a(D) + C.D - 1.\end{aligned}$$ □

Exercise 1.4

Problem.

(a) If a surface X of degree d in $\mathbf{P}^3$ contains a straight line $C = \mathbf{P}^1$, show that $C^2 = 2 - d$.

(b) Assume $\operatorname{char} k = 0$. Show that for every $d \geq 1$, there exists a nonsingular surface X of degree d in $\mathbf{P}^3$ containing the line $x = y = 0$.

Solution. (a) X is a nonsingular hypersurface of degree d in $\mathbf{P}^3$, so $\omega_X \cong \mathcal{O}_X(d-4)$ (II, 8.20.3), i.e. $K \sim (d-4)H$ where H is the hyperplane class. By Exercise 1.2, $C.H = \deg C = 1$. Since $C \cong \mathbf{P}^1$ has genus 0, (1.5) gives $-2 = C.(C+K) = C^2 + (d-4)C.H = C^2 + d - 4$, so $C^2 = 2 - d$.

(b) For $d = 1$ take $X = \{x = 0\}$. For $d \geq 2$ let

$$F = xz^{d-1} + yw^{d-1} + x^d + y^d,$$

which lies in the ideal (x, y), so $X = V(F) \supseteq \{x = y = 0\}$. The partial derivatives are

$$F_x = z^{d-1} + dx^{d-1}, \quad F_y = w^{d-1} + dy^{d-1}, \quad F_z = (d-1)xz^{d-2}, \quad F_w = (d-1)yw^{d-2}.$$

Suppose they all vanish at a point. Since $\operatorname{char} k = 0$, $d, d-1 \neq 0$. If $d = 2$, $F_z = x$ and $F_w = y$ vanish, so $x = y = 0$ and then $F_x = z$, $F_y = w$ give $z = w = 0$. If $d \geq 3$: $F_z = 0$ gives $x = 0$ or $z = 0$; if $x = 0$ then $F_x = z^{d-1} = 0$ so $z = 0$; if $z = 0$ then $F_x = dx^{d-1} = 0$ so $x = 0$. Thus $x = z = 0$, and likewise $y = w = 0$. So there is no common zero in $\mathbf{P}^3$, and $V(F)$ is nonsingular (in characteristic 0 the singular points of $V(F)$ are the common zeros of the partials, F itself vanishing there by Euler's formula). A nonsingular hypersurface is irreducible (two components would meet along a curve at singular points), so X is a nonsingular surface of degree d containing the line $x = y = 0$. □

Exercise 1.5

Problem.

(a) If X is a surface of degree d in $\mathbf{P}^3$, then $K^2 = d(d-4)^2$.

(b) If X is a product of two nonsingular curves C, C', of genus g, g' respectively, then $K^2 = 8(g-1)(g'-1)$. Cf. (II, Ex. 8.3).

Solution. (a) As in Exercise 1.4, $K \sim (d-4)H$, and $H^2 = \deg X = d$ by Exercise 1.2. Hence $K^2 = (d-4)^2 H^2 = d(d-4)^2$.

(b) Let p_1, p_2 be the projections. By (II, Ex. 8.3), $\omega_X \cong p_1^*\omega_C \otimes p_2^*\omega_{C'}$. For points $P \in C$, $Q \in C'$ put $m_P = \{P\} \times C' = p_1^*(P)$ and $l_Q = C \times \{Q\} = p_2^*(Q)$. Write $K_C \sim \sum n_i P_i$, $K_{C'} \sim \sum n'_j Q_j$, with $\sum n_i = 2g - 2$, $\sum n'_j = 2g' - 2$. Then $K \sim \sum n_i m_{P_i} + \sum n'_j l_{Q_j}$.

Intersection numbers. m_P and l_Q are nonsingular and meet transversally in the single point (P, Q) (local equations $s - s(P)$ and $t - t(Q)$ in independent coordinates generate the maximal ideal), so $m_P.l_Q = 1$. Next, $\mathcal{L}(m_{P'}) = p_1^*\mathcal{L}_C(P')$, whose restriction to m_P is trivial, since p_1 is constant on m_P. By (F1), $m_{P'}.m_P = 0$ for all P, P' (also if $P = P'$); similarly $l_Q.l_{Q'} = 0$.

Hence

$$K^2 = 2\Big(\sum n_i\Big)\Big(\sum n'_j\Big)\, m.l = 2(2g-2)(2g'-2) = 8(g-1)(g'-1). \qquad \square$$

Exercise 1.6

Problem.

(a) If C is a curve of genus g, show that the diagonal $\Delta \subseteq C \times C$ has self-intersection $\Delta^2 = 2 - 2g$. (Use the definition of $\Omega_{C/k}$ in (II, Sect. 8).)

(b) Let $l = C \times \text{pt}$ and $m = \text{pt} \times C$. If $g \geq 1$, show that l, m, and Δ are linearly independent in $\text{Num}(C \times C)$. Thus $\text{Num}(C \times C)$ has rank ≥ 3, and in particular $\text{Pic}(C \times C) \neq p_1^* \operatorname{Pic} C \oplus p_2^* \operatorname{Pic} C$. Cf. (III, Ex. 12.6), (IV, Ex. 4.10).

Solution. (a) Let $X = C \times C$ and let $\mathcal{I}$ be the ideal sheaf of Δ in X. By (II, Sect. 8), $\Omega_{C/k}$ is $\Delta^*(\mathcal{I}/\mathcal{I}^2)$, i.e. $\mathcal{I}/\mathcal{I}^2$ regarded on $\Delta \cong C$. By (II, 6.18), $\mathcal{I} = \mathcal{L}(-\Delta)$, so $\mathcal{I}/\mathcal{I}^2 = \mathcal{L}(-\Delta) \otimes \mathcal{O}_\Delta$, and therefore $\mathcal{L}(\Delta) \otimes \mathcal{O}_\Delta \cong (\mathcal{I}/\mathcal{I}^2)^\vee \cong \Omega_{C/k}^\vee$. By (F1) and (IV, 1.3.3), $\Delta^2 = \deg \Omega_{C/k}^\vee = -(2g-2) = 2 - 2g$.

(b) Intersection numbers (notation of Exercise 1.5(b) with $C' = C$): $l^2 = m^2 = 0$ and $l.m = 1$. Moreover Δ meets $l = C \times \{Q\}$ only at (Q, Q), and transversally (local equations $s - t$ and $t - t(Q)$ generate the maximal ideal), so $\Delta.l = 1$, and likewise $\Delta.m = 1$. By (a), the matrix of the pairing on (l, m, Δ) is

$$\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 2-2g \end{pmatrix}, \qquad \det = -(2 - 2g - 1) + 1 = 2g \neq 0.$$

If $\alpha l + \beta m + \gamma \Delta \equiv 0$ in $\operatorname{Num} X$ ($\alpha, \beta, \gamma \in \mathbf{Z}$), intersecting with l, m, Δ shows that (α, β, γ) lies in the kernel of this matrix, so it is 0. Thus l, m, Δ are independent in $\operatorname{Num} X$, which has rank ≥ 3.

Now consider $p_1^* A + p_2^* B$ for divisors A, B on C. The divisors $m_P = p_1^*(P)$, $P \in C$, form a family parametrized by C: the divisor $\mathcal{D} = \{(x, y, t) : x = t\} = (p_{13})^* \Delta_C \subseteq X \times C$ is an effective Cartier divisor, irreducible and dominating C, hence flat over C (see Exercise 1.7), with fibre m_P over $t = P$. Hence the m_P are all prealgebraically equivalent, hence numerically equivalent by Exercise 1.7(c). Similarly the l_Q are numerically equivalent. So $p_1^* A \equiv (\deg A)\, m$ and $p_2^* B \equiv (\deg B)\, l$, and the image of $p_1^* \operatorname{Pic} C + p_2^* \operatorname{Pic} C$ in $\operatorname{Num} X$ lies in $\mathbf{Z}l + \mathbf{Z}m$, which does not contain the class of Δ. So $\Delta \notin p_1^* \operatorname{Pic} C + p_2^* \operatorname{Pic} C$, and $\operatorname{Pic}(C \times C) \neq p_1^* \operatorname{Pic} C \oplus p_2^* \operatorname{Pic} C$. □

Exercise 1.7

Problem. *Algebraic equivalence of divisors.* Let X be a surface. Recall that we have defined an algebraic family of effective divisors on X, parametrized by a nonsingular curve T, to be an effective Cartier divisor D on $X \times T$, flat over T (III, 9.8.5). In this case, for any two closed points $0, 1 \in T$, we say the corresponding divisors D_0, D_1 on X are *prealgebraically equivalent*. Two arbitrary divisors are prealgebraically equivalent if they are differences of prealgebraically equivalent effective divisors. Two divisors D, D' are *algebraically equivalent* if there is a finite sequence $D = D_0, D_1, \ldots, D_n = D'$ with D_i and D_{i+1} prealgebraically equivalent for each i.

(a) Show that the divisors algebraically equivalent to 0 form a subgroup of $\operatorname{Div} X$.

(b) Show that linearly equivalent divisors are algebraically equivalent. [*Hint:* If (f) is a principal divisor on X, consider the principal divisor $(tf - u)$ on $X \times \mathbf{P}^1$, where t, u are the homogeneous coordinates on $\mathbf{P}^1$.]

(c) Show that algebraically equivalent divisors are numerically equivalent. [*Hint:* Use (III, 9.9) to show that for any very ample H, if D and D' are algebraically equivalent, then $D.H = D'.H$.]

(The Note on the Neron–Severi group following the exercise is a remark, not an assignment.)

Solution. Reading of the definition. We read "differences of prealgebraically equivalent effective divisors" as: D, D' are prealgebraically equivalent (written $D \sim_p D'$) if $D = E_0 - F_0$, $D' = E_1 - F_1$ with E_0, E_1 effective and prealgebraically equivalent via a family, and likewise F_0, F_1 (the two families may have different parameter curves). Algebraic equivalence $\approx$ is the equivalence relation generated by $\sim_p$. ($\sim_p$ is reflexive, using the constant family $E \times T$, and symmetric, as $0, 1$ are arbitrary points of T.)

A flatness criterion. Let T be a nonsingular curve and $\mathcal{D}$ an effective Cartier divisor on the nonsingular variety $X \times T$. The local rings of $X \times T$ are regular, hence factorial, so locally $\mathcal{D}$ is defined by $f = \prod \pi_i^{n_i}$ with prime elements π_i, and the associated primes of $\mathcal{O}_{\mathcal{D}}$ are exactly the (π_i), i.e. the generic points of the irreducible components of $\mathcal{D}$. By (III, 9.7) $\mathcal{D}$ is flat over T iff no irreducible component of $\mathcal{D}$ maps to a closed point of T, i.e. iff no component of $\mathcal{D}$ is a whole fibre $X \times \{t\}$ (components have dimension 2 and fibres are irreducible). Consequences: (i) if $\mathcal{D}_1, \mathcal{D}_2$ are flat effective divisors over T, so is $\mathcal{D}_1 + \mathcal{D}_2$ (its components are those of $\mathcal{D}_1, \mathcal{D}_2$), and its fibre over t is the sum of the fibres (the restrictions of local equations to $X \times \{t\}$ are nonzero and multiply); (ii) for an effective divisor G on X, $p_1^* G = G \times T$ is flat with all fibres G.

(a) *Claim: if $D \sim_p D'$ and G is any divisor, then $D + G \sim_p D' + G$.* Write $D = E_0 - F_0$, $D' = E_1 - F_1$ as above and $G = G_1 - G_2$ with G_i effective. Then $D + G = (E_0 + G_1) - (F_0 + G_2)$ and $D' + G = (E_1 + G_1) - (F_1 + G_2)$. If $\mathcal{E} \subseteq X \times T$ is a family with fibres E_0, E_1, then $\mathcal{E} + p_1^* G_1$ is a family with fibres $E_0 + G_1, E_1 + G_1$ by (i),(ii); likewise for $F + G_2$. This proves the claim. Applying the claim term by term along a chain, $D \approx D'$ implies $D + G \approx D' + G$ for all G. Therefore, if $D \approx D'$ and $E \approx E'$, then $D + E \approx D' + E \approx D' + E'$; and if $D \approx D'$, adding $-D - D'$ gives $-D' \approx -D$. Consequently the set $\{D : D \approx 0\}$ contains 0 and is closed under sums (if $D \approx 0$, $E \approx 0$ then $D + E \approx 0$) and negatives ($D \approx 0 \Rightarrow -D \approx -0 = 0$): it is a subgroup of $\operatorname{Div} X$.

(b) *Step 1: effective $E \sim E'$ implies $E \sim_p E'$.* Write $E' = E + (f)$ with f a rational function; if f is constant, $E = E'$ and we are done, so let f be nonconstant. Then $s_0 = 1$ and $s_1 = f$ are global sections of $\mathcal{L}(E)$ (as subsheaf of the constant sheaf $\mathcal{K}$) with divisors E and $E + (f) = E'$ respectively. On $X \times \mathbf{P}^1$ consider the section

$$\sigma = t\, s_0 + u\, s_1 \in H^0\big(X \times \mathbf{P}^1,\ p_1^* \mathcal{L}(E) \otimes p_2^* \mathcal{O}(1)\big)$$

(this is the section $tf - u$ of the hint, up to renaming coordinates), and let $\mathcal{D}$ be its divisor of zeros, an effective Cartier divisor. Its restriction to the fibre $X \times \{(c_0 : c_1)\}$ is the section $c_0 + c_1 f$ of $\mathcal{L}(E)$, which is nonzero for every $(c_0 : c_1)$, because f is nonconstant. So no fibre is contained in $\mathcal{D}$, and $\mathcal{D}$ is flat over $\mathbf{P}^1$ by the criterion. The fibre of $\mathcal{D}$ over $(c_0 : c_1)$ is the divisor of $c_0 + c_1 f$, which is E at $(1 : 0)$ and E' at $(0 : 1)$. Hence $E \sim_p E'$.

Step 2. Let $D \sim D'$, $D = A - B$, $D' = A' - B'$ with A, B, A', B' effective. Then $A + B' \sim A' + B$ are effective and linearly equivalent, so $A + B' \sim_p A' + B$ by Step 1, hence $A + B' \approx A' + B$. Adding $-B - B'$ (by (a)) gives $A - B \approx A' - B'$, i.e. $D \approx D'$.

(c) Let H be very ample, $X \subseteq \mathbf{P}^N$. Let $\mathcal{E} \subseteq X \times T$ be a family of effective divisors; it is a closed subscheme of $\mathbf{P}^N \times T$ flat over T, so by (III, 9.9) the Hilbert polynomials of the fibres $\mathcal{E}_t \subseteq \mathbf{P}^N$ are independent of t. The fibre $\mathcal{E}_t$ is the divisor E_t, whose Hilbert polynomial has leading coefficient $E_t.H$ by Exercise 1.2. Hence $E_0.H = E_1.H$ whenever E_0, E_1 are prealgebraically equivalent effective divisors, and then $D.H = D'.H$ whenever $D \sim_p D'$ (by additivity, $D.H = E_0.H - F_0.H$), and so whenever $D \approx D'$.

Now let G be any divisor. As in the proof of (1.1), $G \sim H_1 - H_2$ with H_1, H_2 very ample. So $D.G = D.H_1 - D.H_2 = D'.H_1 - D'.H_2 = D'.G$. Thus $(D - D').G = 0$ for all G, i.e. $D \equiv D'$. $\square$

Exercise 1.8

Problem. *Cohomology class of a divisor.* For any divisor D on the surface X, we define its cohomology class $c(D) \in H^1(X, \Omega_X)$ by using the isomorphism $\operatorname{Pic} X \cong H^1(X, \mathcal{O}_X^*)$ of (III, Ex. 4.5) and the sheaf homomorphism $d\log : \mathcal{O}_X^* \to \Omega_X$ (III, Ex. 7.4c). Thus we obtain a group homomorphism $c : \operatorname{Pic} X \to H^1(X, \Omega_X)$. On the other hand, $H^1(X, \Omega)$ is dual to itself by Serre duality (III, 7.13), so we have a nondegenerate bilinear map

$$\langle\ ,\ \rangle : H^1(X, \Omega) \times H^1(X, \Omega) \to k.$$

(a) Prove that this is compatible with the intersection pairing, in the following sense: for any two divisors D, E on X, we have $\langle c(D), c(E)\rangle = (D.E) \cdot 1$ in k. [*Hint:* Reduce to the case where D and E are nonsingular curves meeting transversally. Then consider the analogous map $c : \operatorname{Pic} D \to H^1(D, \Omega_D)$, and the fact (III, Ex. 7.4) that c(point) goes to 1 under the natural isomorphism of $H^1(D, \Omega_D)$ with k.]

(b) If $\operatorname{char} k = 0$, use the fact that $H^1(X, \Omega_X)$ is a finite-dimensional vector space to show that $\operatorname{Num} X$ is a finitely generated free abelian group.

Solution. Set-up. The pairing is: cup product $H^1(X, \Omega_X) \times H^1(X, \Omega_X) \to H^2(X, \Omega_X^2) = H^2(X, \omega_X)$ followed by the trace $t_X : H^2(X, \omega_X) \to k$ (this is the pairing of (III, 7.13) for $n = 2$, $p = 1$). Serre duality determines the trace only up to a nonzero scalar, and the identity to be proved depends on this choice; so, as in (III, Ex. 7.4) (which the exercise invokes), we use trace maps normalized by: for a nonsingular projective variety Y and a point $P \in Y$, $t_Y(\eta(P)) = 1$. Ex. 7.4(a),(d) are taken as given for this normalization.

We use III, Ex. 7.4 in the following form. For a nonsingular irreducible projective variety Y of dimension n and a nonsingular subvariety $Z \subseteq Y$ of codimension p, the class $\eta(Z) \in H^p(Y, \Omega_Y^p)$ is the element corresponding under (III, 7.13) to the linear form $H^{n-p}(Y, \Omega_Y^{n-p}) \to H^{n-p}(Z, \Omega_Z^{n-p}) \xrightarrow{t_Z} k$. Ex. 7.4(a): $t_Y(\eta(P)) = 1$ for a point P; Ex. 7.4(d): for $p = 1$, $\eta(Z) = c(\mathcal{L}(Z))$. Concretely, for $p = 1$ and $Y = X$ a surface, $Z = D$ a nonsingular curve:

$$\langle c(D), y\rangle = t_D\big(y|_D\big) \qquad (y \in H^1(X, \Omega_X)), \tag{$*$}$$

where $y \mapsto y|_D$ is $H^1(X, \Omega_X) \to H^1(D, \Omega_X|_D) \to H^1(D, \Omega_D)$.

(a) *Reduction.* Both sides of the identity are bilinear in (D, E) (as c is a homomorphism) and depend only on the linear equivalence classes of D, E (the left side because c is defined on $\operatorname{Pic} X$, the right side by (1.1)(4)). As in the proof of (1.1), every class in $\operatorname{Pic} X$ is of the form $[C_1] - [C_2]$ with C_1, C_2 very ample, and by (1.2) we may take C_1, C_2 irreducible and nonsingular, and, writing $E \sim E_1 - E_2$ likewise, choose E_1, E_2 (applying (1.2) to C_1, C_2) nonsingular, irreducible,

and transversal to both C_1 and C_2. So it suffices to treat D, E irreducible nonsingular curves meeting transversally.

The case of transversal nonsingular curves. Let $D \cap E = \{P_1, \dots, P_r\}$, so that $D.E = r$ by (1.1)(1). By $(*)$,

$$\langle c(D), c(E)\rangle = t_D\big(c(E)|_D\big).$$

The class $c(E)$ is the image of the class of the cocycle (g_{ij}) of $\mathcal{L}(E)$ under $d\log$; since d commutes with restriction of forms, $d\log : \mathcal{O}_X^* \to \Omega_X$ is compatible with $\mathcal{O}_X^* \to \mathcal{O}_D^*$, $\Omega_X \to \Omega_D$, so $c(E)|_D = c_D(\mathcal{L}(E) \otimes \mathcal{O}_D)$, where $c_D : \operatorname{Pic} D \to H^1(D, \Omega_D)$ is the analogous map for the curve D. As in the proof of (1.3), since the intersection is transversal, $\mathcal{L}(E) \otimes \mathcal{O}_D \cong \mathcal{L}_D(P_1 + \cdots + P_r)$. Hence, c_D being a homomorphism and using (III, Ex. 7.4(d) and (a)) for the point P_i on the curve D,

$$t_D\big(c(E)|_D\big) = \sum_{i=1}^{r} t_D\big(c_D(\mathcal{L}_D(P_i))\big) = \sum_{i=1}^{r} t_D(\eta(P_i)) = r \cdot 1 = (D.E) \cdot 1.$$

This proves (a).

(b) Let $V = H^1(X, \Omega_X)$, $h = \dim_k V < \infty$ (III, 5.2), and $\operatorname{char} k = 0$, so $\mathbf{Z} \subseteq k$. By (a), $\langle c(D), c(E)\rangle = D.E \in \mathbf{Z} \subseteq k$.

Step 1: if $D_1, \dots, D_m \in \operatorname{Pic} X$ and $\det(D_i.D_j) \neq 0$, then $m \leq h$. Indeed, we claim $c(D_1), \dots, c(D_m)$ are k-linearly independent in V: if $\sum \lambda_i c(D_i) = 0$ with $\lambda_i \in k$, pairing with $c(D_j)$ gives $\sum_i \lambda_i (D_i.D_j) = 0$ for all j; the integer matrix $(D_i.D_j)$ has nonzero determinant, which stays nonzero in k (characteristic 0), so all $\lambda_i = 0$.

Hence there is a maximal $m \geq 0$ for which some $D_1, \dots, D_m$ has $G := (D_i.D_j)$ with $\det G \neq 0$ (for $m = 0$ the empty determinant is 1). Fix such $D_1, \dots, D_m$.

Step 2: if $N \in \operatorname{Pic} X$ and $N.D_j = 0$ for all j, then $N \equiv 0$. Suppose $N \not\equiv 0$ and choose E with $N.E \neq 0$. Let $\lambda = \operatorname{adj}(G) \cdot (E.D_j)_j \in \mathbf{Z}^m$, so that $\sum_i \lambda_i (D_i.D_j) = (\det G)(E.D_j)$, and put $E_0 = (\det G)E - \sum \lambda_i D_i$. Then $E_0.D_j = 0$ for all j and $N.E_0 = (\det G)(N.E) \neq 0$. For $s \in \mathbf{Z}$ the class $M_s = N + sE_0$ is orthogonal to all D_j and $M_s^2 = N^2 + 2s\, N.E_0 + s^2 E_0^2$ is a nonzero polynomial in s (its linear coefficient is nonzero), so $M_s^2 \neq 0$ for some s. Then the Gram matrix of $(D_1, \dots, D_m, M_s)$ is block diagonal with determinant $\det G \cdot M_s^2 \neq 0$, contradicting the maximality of m.

Step 3. The homomorphism $\operatorname{Num} X \to \mathbf{Z}^m$, $D \mapsto (D.D_1, \dots, D.D_m)$, is well defined and, by Step 2, injective. Thus $\operatorname{Num} X$ is isomorphic to a subgroup of $\mathbf{Z}^m$ and therefore is a finitely generated free abelian group (of rank $\leq h$). □

Exercise 1.9

Problem.

(a) If H is an ample divisor on the surface X, and if D is any divisor, show that $(D^2)(H^2) \leq (D.H)^2$.

(b) Now let X be a product of two curves $X = C \times C'$. Let $l = C \times \text{pt}$, and $m = \text{pt} \times C'$. For any divisor D on X, let $a = D.l$, $b = D.m$. Then we say D has *type* (a, b). If D has type (a, b), with $a, b \in \mathbf{Z}$, show that $D^2 \leq 2ab$, and equality holds if and

only if $D \equiv bl + am$. [*Hint:* Show that $H = l + m$ is ample, let $E = l - m$, let $D' = (H^2)(E^2)D - (E^2)(D.H)H - (H^2)(D.E)E$, and apply (1.9). This inequality is due to Castelnuovo and Severi. See Grothendieck [2].]

Solution. (a) By (F3), $H^2 > 0$. Put $D' = (H^2)D - (D.H)H$. Then $D'.H = (H^2)(D.H) - (D.H)(H^2) = 0$ and

$$D'^2 = (H^2)^2D^2 - 2(H^2)(D.H)^2 + (D.H)^2H^2 = H^2\big(H^2D^2 - (D.H)^2\big).$$

If $D' \not\equiv 0$, (1.9) gives $D'^2 < 0$, hence $H^2D^2 - (D.H)^2 < 0$ because $H^2 > 0$. If $D' \equiv 0$, then $D'^2 = 0$ and $H^2D^2 = (D.H)^2$. In either case $(D^2)(H^2) \leq (D.H)^2$.

(b) *Intersection numbers.* From Exercise 1.5(b): $l^2 = m^2 = 0$, $l.m = 1$.

$H = l + m$ is ample. Here $H^2 = 2l.m = 2 > 0$. By the Nakai–Moishezon criterion (1.10) it remains to see $H.Z > 0$ for every irreducible curve $Z \subseteq X$. Fix $l = C \times \{Q\}$ and $m = \{P\} \times C'$. If $p_2(Z)$ is a point Q_0, then $Z = C \times \{Q_0\}$ (an irreducible curve contained in the irreducible fibre) and $Z.m = 1$ (transversal at (P, Q_0)), $Z.l = 0$ (disjoint, or $l^2 = 0$ if $Q_0 = Q$), so $H.Z = 1$. Otherwise $p_2(Z) = C'$ ($p_2(Z)$ is closed and irreducible), so Z meets $p_2^{-1}(Q) = l$ and $Z \neq l$; by (F2), $Z.l \geq 1$. Also $Z.m \geq 0$: by (F2) if $Z \neq m$, and $m^2 = 0$ if $Z = m$. Hence $H.Z \geq 1 > 0$. So H is ample.

The inequality. Put $N = D - bl - am$. Then $N.l = a - 0 - a\,(m.l) = 0$ and $N.m = b - b = 0$, so $N.H = 0$. Also

$$N^2 = D^2 - 2D.(bl + am) + (bl + am)^2 = D^2 - 2(ab + ab) + 2ab = D^2 - 2ab.$$

If $N \not\equiv 0$, (1.9) gives $N^2 < 0$; if $N \equiv 0$, $N^2 = 0$. Hence $D^2 - 2ab \leq 0$, with equality if and only if $N \equiv 0$, i.e. $D \equiv bl + am$. (For $E = l - m$ one has $E^2 = -2$, $H^2 = 2$, $E.H = 0$, $D.H = a + b$, $D.E = a - b$, and the divisor D' of the hint is $-4D + 2(a + b)H - 2(a - b)E = -4N$.) □

Exercise 1.10

Problem. *Weil's Proof of the Analogue of the Riemann Hypothesis for Curves.* Let C be a curve of genus g defined over the finite field $\mathbf{F}_q$, and let N be the number of points of C rational over $\mathbf{F}_q$. Then $N = 1 - a + q$, with $|a| \leq 2g\sqrt{q}$. To prove this, we consider C as a curve over the algebraic closure k of $\mathbf{F}_q$. Let $f : C \to C$ be the k-linear Frobenius morphism obtained by taking qth powers, which makes sense since C is defined over $\mathbf{F}_q$ (IV, 2.4.1). Let $\Gamma \subseteq C \times C$ be the graph of f, and let $\Delta \subseteq C \times C$ be the diagonal. Show that $\Gamma^2 = q(2 - 2g)$, and $\Gamma.\Delta = N$. Then apply (1.9) to $D = r\Gamma + s\Delta$ for all r and s to obtain the result. See (App. C, Ex. 5.7) for another interpretation of this result.

Solution. *Setting.* $C = C_0 \times_{\mathbf{F}_q} k$ for a nonsingular projective geometrically irreducible curve $C_0/\mathbf{F}_q$, and $f = F_{C_0} \times \mathrm{id}_k$, where F_{C_0} is the absolute q-power Frobenius ($h \mapsto h^q$ on functions); f is a k-morphism. We write $X = C \times C$ and use the notation $m_P = \{P\} \times C$, $l_Q = C \times \{Q\}$ of Exercise 1.5(b). The reference "(1.9)" is used in the form of Exercise 1.9(b) (itself a consequence of Theorem 1.9).

Facts about f. (i) f is a finite morphism of degree q (the field extension $k(C) \supseteq f^*k(C) = k(C)^q$ has degree q, k being perfect). (ii) The fixed points of f on $C(k)$ are exactly the $\mathbf{F}_q$-rational points of C_0, so there are N of them. (iii) f^* kills differentials: $d(h^q) = 0$.

Intersection numbers with Γ. Γ is the image of the closed immersion $\gamma = (\mathrm{id}, f) : C \to X$, so $\Gamma \cong C$ is a nonsingular irreducible curve of genus g. Since $p_1\gamma = \mathrm{id}$ and $p_2\gamma = f$, (F1) and $\deg f^*M = \deg f \cdot \deg M$ (IV, Ex. 2.6) give

$$\Gamma.m_P = \deg \gamma^* p_1^* \mathcal{L}_C(P) = 1, \qquad \Gamma.l_Q = \deg f^* \mathcal{L}_C(Q) = q.$$

Likewise for $K = p_1^*K_C + p_2^*K_C$: $\Gamma.K = \deg K_C + \deg f^*K_C = (2g-2)(1+q)$. By adjunction (1.5), $2g-2 = \Gamma.(\Gamma + K)$, whence

$$\Gamma^2 = (2g-2) - (2g-2)(1+q) = q(2-2g).$$

For the diagonal the same argument ($\Delta \cong C$, f replaced by id) gives $\Delta.m_P = \Delta.l_Q = 1$ and $\Delta^2 = 2 - 2g$ (as in Exercise 1.6(a)).

$\Gamma.\Delta = N$. Set-theoretically $\Gamma \cap \Delta = \{(x,x) : f(x) = x\}$, which has N points by (ii). At such a point x (a rational point), choose a uniformizer $t \in \mathcal{O}_{C_0,x_0} \subseteq \mathcal{O}_{C,x}$. A local equation of Δ at (x,x) is $p_1^*t - p_2^*t$, and its pullback to $\Gamma \cong C$ via γ is $t - f^*t = t - t^q$, which is t times a unit; so the scheme $\Gamma \cap \Delta$ is reduced at (x,x), i.e. Γ and Δ meet transversally there. By (1.1)(1), $\Gamma.\Delta = \#(\Gamma \cap \Delta) = N$.

The inequality. Let $D = r\Gamma + s\Delta$ with $r, s \in \mathbf{Z}$. Its type is $a' = D.l = rq + s$, $b' = D.m = r + s$, and

$$D^2 = r^2 q(2-2g) + 2rsN + s^2(2-2g).$$

Exercise 1.9(b) gives $D^2 \le 2a'b' = 2(rq+s)(r+s)$. Therefore, with $a = q + 1 - N$ (so $N = 1 - a + q$),

$$0 \le 2a'b' - D^2 = 2gq\,r^2 + 2(q+1-N)\,rs + 2g\,s^2 = 2\big(gq\,r^2 + a\,rs + g\,s^2\big) \quad \text{for all } r, s \in \mathbf{Z}.$$

If $g = 0$ this reads $a\,rs \ge 0$ for all r, s; taking $(r,s) = (-a, 1)$ gives $-a^2 \ge 0$, so $a = 0$, as $|a| \le 0$ requires. If $g > 0$ take $(r,s) = (-a, 2gq)$: $gq\,a^2 - 2gq\,a^2 + 4g^3q^2 = gq(4g^2q - a^2) \ge 0$, so $a^2 \le 4g^2q$. In all cases $|a| \le 2g\sqrt{q}$ and $N = 1 - a + q$. □

Exercise 1.11

Problem. In this problem, we assume that X is a surface for which $\mathrm{Num}\,X$ is finitely generated (i.e., any surface, if you accept the Neron–Severi theorem (Ex. 1.7)).

(a) If H is an ample divisor on X, and $d \in \mathbf{Z}$, show that the set of effective divisors D with $D.H = d$, modulo numerical equivalence, is a finite set. [*Hint:* Use the adjunction formula, the fact that p_a of an irreducible curve is ≥ 0, and the fact that the intersection pairing is negative definite on $H^\perp$ in $\mathrm{Num}\,X$.]

(b) Now let C be a curve of genus $g \ge 2$, and use (a) to show that the group of automorphisms of C is finite, as follows. Given an automorphism σ of C, let $\Gamma \subseteq X = C \times C$ be its graph. First show that if $\Gamma \equiv \Delta$, then $\Gamma = \Delta$, using the fact that $\Delta^2 < 0$, since $g \ge 2$ (Ex. 1.6). Then use (a). Cf. (IV, Ex. 2.5).

Solution. Let $\Lambda = \mathrm{Num}\,X$, a finitely generated free abelian group by assumption (Λ is torsion free since $nD \equiv 0 \Rightarrow D.E = 0$ in $\mathbf{Z}$).

(a) Put $n = H^2 > 0$, $\kappa = K.H$.

Negative definiteness. Let $W = \{v \in \Lambda \otimes \mathbf{R} : v.H = 0\}$; this is the kernel of a linear form defined over $\mathbf{Q}$, so W has a basis in $\Lambda \otimes \mathbf{Q}$, and $W_{\mathbf{Q}} = W \cap (\Lambda \otimes \mathbf{Q})$ is dense. For $0 \neq v \in W_{\mathbf{Q}}$, a multiple Nv lies in Λ, is nonzero in $\operatorname{Num} X$ with $(Nv).H = 0$, so $(Nv)^2 < 0$ by (1.9); hence $q(v) < 0$ for the quadratic form $q(v) = v^2$ on $W_{\mathbf{Q}}$. Diagonalize q over $\mathbf{Q}$ in a $\mathbf{Q}$-basis $e_1, \dots, e_r$ of $W_{\mathbf{Q}}$: $q(\sum y_i e_i) = \sum d_i y_i^2$ with $d_i = q(e_i) < 0$. So q is negative definite on W over $\mathbf{R}$. In particular, Cauchy–Schwarz holds: $|u.v| \le \sqrt{-u^2}\sqrt{-v^2}$ for $u, v \in W$, and for each M, the set $\{v \in W : -v^2 \le M\}$ is compact, so it contains only finitely many points of the discrete subgroup $\Lambda \cap W$.

Irreducible curves of given degree. Fix an integer $e \ge 1$. We show that the classes of irreducible curves C with $C.H = e$ form a finite subset of Λ. Put $C' = nC - eH$ and $K' = nK - \kappa H$; both lie in $\Lambda \cap W$. Compute

$$C'^2 = n^2 C^2 - ne^2, \qquad C'.K' = n^2\, C.K - n\kappa e.$$

Since C is an irreducible curve, $H^0(\mathcal{O}_C) = k$, so $p_a(C) = h^1(\mathcal{O}_C) \ge 0$, and by Exercise 1.3(a), $C^2 + C.K = 2p_a - 2 \ge -2$. Thus

$$-C'^2 = ne^2 - n^2 C^2 \le ne^2 + 2n^2 + n^2\, C.K = ne^2 + 2n^2 + n\kappa e + C'.K' \le B + xy,$$

where $B = |ne^2 + 2n^2 + n\kappa e|$, $x = \sqrt{-C'^2}$, $y = \sqrt{-K'^2}$. So $x^2 \le B + xy$, which forces $x \le y + \sqrt{B}$ (if $x > y + \sqrt{B}$ then $x^2 - xy = x(x-y) > (y + \sqrt{B})\sqrt{B} \ge B$). Hence $-C'^2$ is bounded in terms of e, n, κ, K' only, so C' lies in a finite set; as $nC = C' + eH$ and Λ is torsion free, the class of C is determined by C'. This proves the claim.

Effective divisors. Let $D = \sum n_i C_i$ be effective with $D.H = d$, C_i distinct irreducible, $n_i \ge 1$. Then $e_i = C_i.H \ge 1$ by (F3) and $\sum n_i e_i = d$. So (if $d \le 0$ only $D = 0$ is possible and there is nothing to prove) there are finitely many possible (n_i, e_i) and, by the above, finitely many classes for each C_i with $C_i.H = e_i$; hence $[D] = \sum n_i [C_i]$ takes only finitely many values in Λ.

(b) Let $X = C \times C$, $g \ge 2$, l, m as before; $H = l + m$ is ample (Exercise 1.9(b)) and $\operatorname{Num} X$ is finitely generated by assumption. Let $\sigma \in \operatorname{Aut} C$ and $\Gamma_\sigma = \{(x, \sigma x)\}$ its graph, an irreducible nonsingular curve $\cong C$.

As in Exercise 1.10 (with f replaced by σ, of degree 1): $\Gamma_\sigma.m = \Gamma_\sigma.l = 1$, hence $\Gamma_\sigma.H = 2$, $\Gamma_\sigma.K = 2(2g-2)$, and by adjunction $\Gamma_\sigma^2 = (2g-2) - 2(2g-2) = 2 - 2g < 0$.

If $\Gamma_\sigma \equiv \Gamma_\tau$ then $\sigma = \tau$. Suppose $\sigma \neq \tau$. Then $\Gamma_\sigma \neq \Gamma_\tau$ are distinct irreducible curves, so $\Gamma_\sigma.\Gamma_\tau \ge 0$ by (F2). But $\Gamma_\sigma \equiv \Gamma_\tau$ gives $\Gamma_\sigma.\Gamma_\tau = \Gamma_\sigma.\Gamma_\sigma = 2 - 2g < 0$, a contradiction. (For $\tau = \mathrm{id}$, $\Gamma_\tau = \Delta$: this is the statement $\Gamma \equiv \Delta \Rightarrow \Gamma = \Delta$ of the exercise.)

Thus $\sigma \mapsto [\Gamma_\sigma] \in \operatorname{Num} X$ is injective. All Γ_σ are effective with $\Gamma_\sigma.H = 2$, so by (a) (with $d = 2$) their classes lie in a finite set. Hence $\operatorname{Aut} C$ is finite. □

Exercise 1.12

> **Problem.** If D is an ample divisor on the surface X, and $D' \equiv D$, then D' is also ample. Give an example to show, however, that if D is very ample, D' need not be very ample.

Solution. Ampleness. $D' \equiv D$ means $(D' - D).E = 0$ for all E. Taking $E = D'$ and $E = D$ gives $D'^2 = D.D' = D^2 > 0$ by (F3) (D ample), and $D'.C = D.C > 0$ for every irreducible curve C. By the Nakai–Moishezon criterion (1.10), D' is ample.

Example. Let C be a nonsingular projective curve of genus 3 which embeds in $\mathbf{P}^3$ with degree 6: e.g. a nonsingular irreducible member of $|\mathcal{O}_Q(2,4)|$ on the quadric $Q \cong \mathbf{P}^1 \times \mathbf{P}^1$ (a very ample class, so Bertini applies and the connectedness is by (III, 7.9)); by (1.5.2) its genus is $2 \cdot 4 - 2 - 4 + 1 = 3$ and its degree is $(2,4).(1,1) = 6$. Let $A = \mathcal{O}_C(1)$, a very ample line bundle of degree 6 on C. Choose distinct points $R, S \in C$ and put $A' = \omega_C(R+S)$, also of degree 6.

A' is not very ample. Since $6 > 2g - 2 = 4$, $h^0(A') = 6 - 3 + 1 = 4$. Yet $A'(-R-S) = \omega_C$ has $h^0 = 3$, whereas a very ample divisor satisfies $\dim |A' - R - S| = \dim |A'| - 2$, i.e. $h^0(A'(-R-S)) = h^0(A') - 2 = 2$ (IV, 3.1(b)). So A' is not very ample.

The surface. Let $X = C \times \mathbf{P}^1$ with projections p_1, p_2, and put $D = p_1^* A + p_2^* \mathcal{O}(1)$ (i.e. $\mathcal{L}(D) = p_1^* A \otimes p_2^* \mathcal{O}(1)$), $D' = p_1^* A' + p_2^* \mathcal{O}(1)$. Then D is very ample: it is the pullback of $\mathcal{O}(1,1)$ under the closed immersion $C \times \mathbf{P}^1 \hookrightarrow \mathbf{P}^3 \times \mathbf{P}^1$, followed by the Segre embedding. We have $D - D' = p_1^* B$ with $B = A - A'$ a divisor class of degree 0 on C. Writing $B = \sum n_i P_i$ with $\sum n_i = 0$, $p_1^* B = \sum n_i m_{P_i}$, and the divisors $m_P = \{P\} \times \mathbf{P}^1$ are all numerically equivalent (Exercise 1.6(b): they form a family over C, so Exercise 1.7(c) applies), so $p_1^* B \equiv (\sum n_i) m_{P_1} = 0$. Hence $D' \equiv D$.

D' is not very ample. If it were, then its restriction to the closed subscheme $C \times \{pt\} \cong C$ would be very ample (a closed immersion restricts to an embedding of the subscheme, and $\mathcal{O}(1)$ restricts accordingly). But that restriction is A', which is not very ample.

So D is very ample, $D' \equiv D$ (hence D' is ample by the first part), but D' is not very ample. □

2. Ruled surfaces

Throughout, k is algebraically closed, a "curve" means a nonsingular projective curve, a "surface" a nonsingular projective surface, and $\equiv$, $\sim$ denote numerical, resp. linear, equivalence. For a divisor D on a curve or surface, $\mathcal{L}(D)$ is the associated invertible sheaf, and for a sheaf $\mathcal{G}$ we write $\mathcal{G}(D) = \mathcal{G} \otimes \mathcal{L}(D)$. For a locally free sheaf $\mathcal{E}$ of rank 2 on C with $X = \mathbf{P}(\mathcal{E})$ we use the notation of (2.8.1): $\mathcal{E}$ normalized, C_0 the section with $\mathcal{L}(C_0) \cong \mathcal{O}_X(1)$, $\mathfrak{e}$ the divisor on C with $\mathcal{L}(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (so $\deg \mathfrak{e} = -e$), f a fibre. We use freely: sections $\sigma\colon C \to \mathbf{P}(\mathcal{E})$ correspond to surjections $\mathcal{E} \to \mathcal{M} \to 0$ with $\mathcal{M}$ invertible and $\mathcal{M} \cong \sigma^* \mathcal{O}_X(1)$ (II, 7.12); $\pi_* \mathcal{O}_X(n) = S^n \mathcal{E}$ (II, 7.11); a normalized $\mathcal{E}$ sits in an exact sequence $0 \to \mathcal{O}_C \to \mathcal{E} \to \mathcal{L}(\mathfrak{e}) \to 0$ (proof of (2.8)).

Exercise 2.1

> **Problem.** If X is a birationally ruled surface, show that the curve C, such that X is birationally equivalent to $C \times \mathbf{P}^1$, is unique (up to isomorphism).

Solution. Suppose X is birational to $C \times \mathbf{P}^1$ and to $C' \times \mathbf{P}^1$. Then there is a birational map $\theta\colon C \times \mathbf{P}^1 \dashrightarrow C' \times \mathbf{P}^1$. Let $p\colon C \times \mathbf{P}^1 \to C$ and $p'\colon C' \times \mathbf{P}^1 \to C'$ be the projections, and put $K = K(C \times \mathbf{P}^1) = K(C)(t)$. Via θ^* we regard $K(C' \times \mathbf{P}^1)$ as K itself, and consider the two subfields

$$A = p^* K(C) = K(C), \qquad B = (p' \circ \theta)^* K(C') \subseteq K.$$

Both have transcendence degree 1 over k, and $B \cong K(C')$.

Lemma. Let C be a curve and $B \subseteq K = K(C)(t)$ a subfield containing k, of transcendence degree 1, whose function field curve C'' has genus $g(C'') \geq 1$. Then $B \subseteq K(C)$.

Proof of the Lemma. The inclusion $B \subseteq K(C \times \mathbf{P}^1)$ is a dominant rational map $\varphi\colon C \times \mathbf{P}^1 \dashrightarrow C''$, defined on a dense open set U. Let $F_c = \{c\} \times \mathbf{P}^1$. The set $V = p(U) \subseteq C$ is open and nonempty (p is flat, hence open). For $c \in V$ the restriction of φ to the nonempty open subset $U \cap F_c$ of $F_c \cong \mathbf{P}^1$ is a rational map from a nonsingular curve to the projective curve C'', hence extends to a morphism $\mathbf{P}^1 \to C''$ (I, 6.8). If this morphism were nonconstant it would be finite and surjective, and (IV, 2.5.4) would give $0 = g(\mathbf{P}^1) \geq g(C'') \geq 1$, absurd. So φ is constant on $U \cap F_c$ for every $c \in V$.

Fix $c_0 \in V$. Since $U \cap F_{c_0}$ is a nonempty open subset of F_{c_0}, there is $a \in k$ with $(c_0, a) \in U$. Let $s_a\colon C \to C \times \mathbf{P}^1$, $c \mapsto (c, a)$. Then $s_a^{-1}(U)$ is a nonempty open subset of C, and $\varphi \circ s_a$ is a rational map $C \dashrightarrow C''$, which extends to a morphism $\rho\colon C \to C''$ (I, 6.8). Put $V' = V \cap s_a^{-1}(U)$ (nonempty, open). For $x \in U$ with $p(x) = c \in V'$, both x and $s_a(c)$ lie in $U \cap F_c$, on which φ is constant; so $\varphi(x) = \rho(p(x))$. Thus $\varphi = \rho \circ p$ on the dense open set $U \cap p^{-1}(V')$, whence $B = \varphi^* K(C'') = p^* \rho^* K(C'') \subseteq p^* K(C)$. (Lemma)

Now we prove $C \cong C'$. The situation is symmetric in C, C' (replace θ by θ^{-1}, and exchange A and B).

Case $g(C') \geq 1$. By the Lemma, $B \subseteq A$, i.e. $K(C')$ is isomorphic to a subfield of $K(C)$. If $g(C) = 0$, then $K(C) \cong k(u)$ and by Lüroth's theorem (IV, 2.5.5) every subfield of transcendence degree 1 is purely transcendental, so $g(C') = 0$, a contradiction. Hence $g(C) \geq 1$ and the Lemma applied to θ^{-1} gives $A \subseteq B$. Therefore $A = B$, i.e. $K(C) \cong K(C')$ as k-algebras, and $C \cong C'$ by (I, 6.12).

Case $g(C') = 0$. If $g(C) \geq 1$, the previous case with the roles of C and C' exchanged gives $g(C') \geq 1$, a contradiction. So $g(C) = 0$ and $C \cong \mathbf{P}^1 \cong C'$. □

Exercise 2.2

> **Problem.** Let X be the ruled surface $\mathbf{P}(\mathcal{E})$ over a curve C. Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Solution. Recall that the closed points of the fibre $\pi^{-1}(y) = \mathbf{P}(\mathcal{E}(y))$ are the one-dimensional quotients of the fibre $\mathcal{E}(y) = \mathcal{E} \otimes k(y)$, and that the point $\sigma(y)$ of a section σ corresponding to $\mathcal{E} \to \mathcal{M} \to 0$ is the quotient $\mathcal{E}(y) \to \mathcal{M}(y)$ (II, 7.12). Two quotients define the same point iff they have the same kernel.

($\Rightarrow$) Let $\mathcal{E} = \mathcal{L}_1 \oplus \mathcal{L}_2$. The two projections $\mathcal{E} \to \mathcal{L}_i$ are surjective and define sections σ_i with images C_i. For each y, the kernels of $\mathcal{E}(y) \to \mathcal{L}_i(y)$ are $\mathcal{L}_{3-i}(y)$, which are two different lines of $\mathcal{E}(y) = \mathcal{L}_1(y) \oplus \mathcal{L}_2(y)$. So $\sigma_1(y) \neq \sigma_2(y)$ for all y, i.e. $C_1 \cap C_2 = \emptyset$.

($\Leftarrow$) Let C', C'' be disjoint sections, corresponding to surjections $u'\colon \mathcal{E} \to \mathcal{L}'$, $u''\colon \mathcal{E} \to \mathcal{L}''$ (here $\mathcal{L}' = \sigma'^* \mathcal{O}_X(1)$ etc.). Disjointness means $\ker u'(y) \neq \ker u''(y)$ for every y, so $\mathcal{E}(y) \to \mathcal{L}'(y) \oplus \mathcal{L}''(y)$ is injective, hence bijective (both sides have dimension 2). Thus $u = (u', u'')\colon \mathcal{E} \to \mathcal{L}' \oplus \mathcal{L}''$ is a homomorphism of locally free sheaves of rank 2 which is an isomorphism on every fibre. By Nakayama it is surjective, and a surjection between locally free sheaves of the same finite rank is an isomorphism. Hence $\mathcal{E} \cong \mathcal{L}' \oplus \mathcal{L}''$ is decomposable. □

Exercise 2.3

Problem.

(a) If $\mathcal{E}$ is a locally free sheaf of rank r on a (nonsingular) curve C, then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \ldots, r$. We say that $\mathcal{E}$ is a *successive extension of invertible sheaves*. [*Hint:* Use (II, Ex. 8.2).]

(b) Show that this is false for varieties of dimension ≥ 2. In particular, the sheaf of differentials Ω on $\mathbf{P}^2$ is not an extension of invertible sheaves.

Solution. (a) Here C is a nonsingular projective curve (the argument works verbatim for affine C). We use induction on r; $r = 1$ is trivial. Let $r \geq 2$.

Claim: there is an injection $\mathcal{M} \hookrightarrow \mathcal{E}$ with $\mathcal{M}$ invertible and $\mathcal{E}/\mathcal{M}$ locally free of rank $r-1$. Choose $n \gg 0$ so that $\mathcal{E}(n)$ is generated by its global sections (II, 5.17), and let $V \subseteq H^0(C, \mathcal{E}(n))$ be a finite-dimensional subspace generating $\mathcal{E}(n)$. For each closed point y the evaluation $V \to \mathcal{E}(n)(y)$ is surjective onto a space of dimension r, so $Z = \{(s, y) \in V \times C : s(y) = 0\}$ is (the total space of) a vector bundle of rank $\dim V - r$ over C; hence $\dim Z = \dim V - r + 1 \leq \dim V - 1$ since $r \geq 2$. The image of Z under the projection to V is a constructible set of dimension $< \dim V$, hence not all of V. Choose $s \in V$ outside it (k is infinite): s vanishes at no point of C. Locally, $\mathcal{E}(n) \cong \mathcal{O}^r$ and $s = (s_1, \ldots, s_r)$ with $(s_1, \ldots, s_r) = (1)$ at each point, so that s is part of a local basis. Thus $s\colon \mathcal{O}_C \to \mathcal{E}(n)$ is injective with locally free cokernel; twisting by $\mathcal{O}(-n)$ gives an injection $\mathcal{M} = \mathcal{O}_C(-n) \hookrightarrow \mathcal{E}$ with locally free cokernel.

Now $\mathcal{E}' = \mathcal{E}/\mathcal{M}$ is locally free of rank $r - 1$; by induction it has a filtration $0 = \mathcal{E}'_0 \subseteq \cdots \subseteq \mathcal{E}'_{r-1} = \mathcal{E}'$ with invertible quotients. Put $\mathcal{E}_0 = 0$, $\mathcal{E}_1 = \mathcal{M}$ and $\mathcal{E}_i =$ the inverse image of $\mathcal{E}'_{i-1}$ in $\mathcal{E}$ for $i \geq 2$. Then $\mathcal{E}_i/\mathcal{E}_{i-1} \cong \mathcal{E}'_{i-1}/\mathcal{E}'_{i-2}$ for $i \geq 2$ are invertible, and so is $\mathcal{E}_1/\mathcal{E}_0 = \mathcal{M}$.

(b) We show that $\Omega = \Omega_{\mathbf{P}^2}$ is not an extension of invertible sheaves, even not in a single step: there is no exact sequence $0 \to \mathcal{L}_1 \to \Omega \to \mathcal{L}_2 \to 0$ with $\mathcal{L}_i$ invertible (a successive extension of rank 2 is exactly such a sequence, as $\mathcal{E}_1$ is invertible and $\Omega/\mathcal{E}_1$ invertible). Since $\operatorname{Pic} \mathbf{P}^2 = \mathbf{Z}$, we would have $\mathcal{L}_1 = \mathcal{O}(a)$, $\mathcal{L}_2 = \mathcal{O}(b)$ with $a, b \in \mathbf{Z}$. By the Euler sequence (II, 8.13) $0 \to \Omega \to \mathcal{O}(-1)^3 \to \mathcal{O} \to 0$, and $\chi(\mathcal{O}(m)) = \frac{1}{2}(m+1)(m+2)$, we get for all m

$$\chi(\Omega(m)) = 3\chi(\mathcal{O}(m-1)) - \chi(\mathcal{O}(m)) = \tfrac{1}{2}(m+1)\big(3m - (m+2)\big) = m^2 - 1.$$

On the other hand, additivity of χ gives

$$\chi(\Omega(m)) = \tfrac{1}{2}(m+a+1)(m+a+2) + \tfrac{1}{2}(m+b+1)(m+b+2) = m^2 + (a+b+3)m + \tfrac{1}{2}\big(a^2+b^2+3(a+b)+4\big).$$

Comparing the coefficients of m gives $a + b = -3$; then comparing constant terms gives $\frac{1}{2}(a^2 + b^2 - 5) = -1$, i.e. $a^2 + b^2 = 3$. This has no solution in integers (a sum of two squares is never $\equiv 3 \pmod 4$). So $\Omega_{\mathbf{P}^2}$ is not an extension of two invertible sheaves, and statement (a) fails for $\mathbf{P}^2$.

More generally, for $\mathbf{P}^n$ with $n \geq 2$ the sheaf $\Omega_{\mathbf{P}^n}$ is not a successive extension of invertible sheaves: if it were, with quotients $\mathcal{O}(a_i)$ ($\operatorname{Pic} \mathbf{P}^n = \mathbf{Z}$), the Chern polynomial (App. A, Sect. 3: the Chow ring of $\mathbf{P}^n$ is $\mathbf{Z}[H]/(H^{n+1})$, and c is multiplicative on exact sequences) would give

$$\prod_{i=1}^{n}(1 + a_i H) = c(\Omega) = (1 - H)^{n+1} \quad \text{in } \mathbf{Z}[H]/(H^{n+1}),$$

i.e. $\prod(1+a_it) = (1-t)^{n+1} - (-t)^{n+1}$ as polynomials in t (both sides have degree $\le n$). Its roots satisfy $\frac{1-t}{-t} = \zeta$ with $\zeta^{n+1} = 1$, $\zeta \ne 1$, i.e. $t = 1/(1-\zeta)$, and for $n \ge 2$ there are $\zeta \ne \pm 1$ giving non-real roots; but the left side has only the real roots $-1/a_i$. This contradiction shows that the statement of (a) fails in every dimension ≥ 2 (for $\mathbf{P}^n$). □

Exercise 2.4

> **Problem.** Let C be a curve of genus g, and let X be the ruled surface $C \times \mathbf{P}^1$. We consider the question, for what integers $s \in \mathbf{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.
>
> (a) Show that $r = 0$ and any $r \ge g+1$ are always possible. Cf. (IV, Ex. 6.8).
>
> (b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Solution. Here $X = \mathbf{P}(\mathcal{O} \oplus \mathcal{O})$, $e = 0$, and we may take $C_0 = C \times \{\text{pt}\}$, $C_0^2 = 0$, $C_0.f = 1$, $f^2 = 0$; by (2.3), $\operatorname{Num} X = \mathbf{Z}C_0 \oplus \mathbf{Z}f$. A *section* is the image $D = \sigma(C)$ of a morphism σ with $\pi\sigma = \mathrm{id}$, i.e. $\sigma = (\mathrm{id}, \varphi)$ with $\varphi\colon C \to \mathbf{P}^1$ a morphism, and D is the graph of φ. Let $n = \deg \varphi$ (with $n = 0$ if φ is constant). Then $D \equiv aC_0 + bf$ with $a = D.f = 1$, and, using (1.3) (as $D \cong C$ is nonsingular) and $\mathcal{L}(C_0) \cong p_2^*\mathcal{O}_{\mathbf{P}^1}(1)$:

$$b = D.C_0 = \deg_C\big(\sigma^* p_2^* \mathcal{O}(1)\big) = \deg \varphi^* \mathcal{O}(1) = n.$$

Hence $D \equiv C_0 + nf$ and $D^2 = 2n$. So $s = 2n$ is even, and

$$s = 2r \text{ is attained for a section } D \iff r = 0 \text{ or there is a morphism } C \to \mathbf{P}^1 \text{ of degree } r.$$

(Conversely every morphism φ of degree r gives a section with $D^2 = 2r$, $r = 0$ for constant φ.)

(a) $r = 0$: a constant φ, e.g. $D = C_0$. Now let $r \ge g+1$. We must produce a morphism $C \to \mathbf{P}^1$ of degree r, i.e. an invertible sheaf $\mathcal{L}$ of degree r together with two sections without common zero. This is the content of (IV, Ex. 6.8); for completeness here is a direct argument, using only Riemann–Roch. It suffices to find a divisor Δ of degree r with $h^0(\Delta) \ge 2$ such that $|\Delta|$ has no base points.

Step 1. Choose points $P_1, \dots, P_g$ one after the other such that P_{i+1} is not a base point of the linear system $|K - P_1 - \cdots - P_i|$ (nonempty as long as $i < g$, and with finitely many base points); then h^0 drops by exactly 1 at each step, so $\Delta' = P_1 + \cdots + P_g$ satisfies $h^0(K - \Delta') = g - g = 0$ ($\Delta' = 0$ if $g = 0$). By Riemann–Roch $h^0(\Delta') = 1$, i.e. $|\Delta'| = \{\Delta'\}$.

Step 2. For a point P the divisor $K - \Delta' + P$ has degree $g-1$, so Riemann–Roch gives $h^0(K - \Delta' + P) = h^0(\Delta' - P)$. This is 0 if $P \notin \{P_1, \dots, P_g\}$ (as $|\Delta'| = \{\Delta'\}$), and 1 if $P = P_i$ (since $\Delta' - P_i \ge 0$ and $h^0(\Delta' - P_i) \le h^0(\Delta') = 1$); in the latter case $|K - \Delta' + P_i| = \{E_i\}$ for a unique effective divisor E_i. Choose a point $Q_1 \notin E_1 \cup \cdots \cup E_g$ and arbitrary points $Q_2, \dots, Q_{r-g}$ (note $r - g \ge 1$), and put $\Delta = \Delta' + Q_1 + \cdots + Q_{r-g}$, of degree r.

Step 3: $|\Delta|$ has no base points and $h^0(\Delta) \ge 2$. Since $h^0(K - \Delta) \le h^0(K - \Delta') = 0$, Riemann–Roch gives $h^0(\Delta) = r + 1 - g \ge 2$. For any point P we claim $h^0(K - \Delta + P) = 0$: if $P \notin \{P_i\}$ this follows from $h^0(K - \Delta' + P) = 0$; if $P = P_i$, a nonzero section of $\mathcal{L}(K - \Delta + P_i)$ would give an effective divisor in $|K - \Delta' + P_i| = \{E_i\}$ containing Q_1, contradicting the choice of Q_1.

Hence by Riemann–Roch $h^0(\Delta - P) = (r-1) + 1 - g = h^0(\Delta) - 1$ for every P, so no point is a base point of $|\Delta|$.

Put $L = \mathcal{L}(\Delta)$ and $V = H^0(L)$, $\dim V \geq 2$, base-point-free. Pick $s_1 \neq 0$ with zeros $R_1, \ldots, R_q$ (finitely many). Each $V(-R_i) = \{s : s(R_i) = 0\}$ is a proper subspace, so, k being infinite, there is $s_2 \in V \setminus \bigcup_i V(-R_i)$. Then s_1, s_2 have no common zero and $\varphi = (s_1 : s_2) \colon C \to \mathbf{P}^1$ has degree $\deg L = r$. Hence r occurs.

(b) Let $g = 3$. A morphism of degree 1 to $\mathbf{P}^1$ is an isomorphism (a birational finite morphism of nonsingular curves), impossible as $g = 3 > 0$. So $r = 1$ does not occur. By definition (IV, Sect. 5), C is hyperelliptic iff there is a degree 2 morphism $C \to \mathbf{P}^1$, so $r = 2$ occurs iff C is hyperelliptic. It remains to show: $r = 3$ occurs iff C is not hyperelliptic.

C not hyperelliptic $\Rightarrow$ $r = 3$ occurs. Fix $P \in C$. Since $\deg K = 4$ and $h^0(K) = 3$, Riemann–Roch gives $h^0(K - P) = h^0(P) - 1 - 1 + 3 = 2$ (here $h^0(P) = 1$ as $g > 0$). If Q were a base point of $|K - P|$, then $h^0(K - P - Q) = 2$; but Riemann–Roch gives $h^0(K - P - Q) = h^0(P + Q) - 2 - 1 + 3 = h^0(P + Q)$, and $h^0(P + Q) = 1$: indeed if $h^0(P + Q) \geq 2$, then $|P + Q|$ has no base point (a base point R would give $h^0(P + Q - R) \geq 2$ for a divisor of degree 1, forcing $C \cong \mathbf{P}^1$), so it defines a morphism of degree 2 to $\mathbf{P}^1$ and C would be hyperelliptic. Hence $|K - P|$ has no base points, and so $|K - P|$ is a base-point-free pencil of degree 3, giving a morphism $C \to \mathbf{P}^1$ of degree 3.

C hyperelliptic $\Rightarrow$ $r = 3$ does not occur. Let $\pi \colon C \to \mathbf{P}^1$ have degree 2 and let $h = \pi^*\mathcal{O}(1)$ (degree 2). The pullbacks of the three independent sections of $\mathcal{O}_{\mathbf{P}^1}(2)$ are independent sections of $2h$, so $h^0(2h) \geq 3$; by Riemann–Roch $h^0(2h) = 4 + 1 - 3 + h^0(K - 2h)$ and $\deg(K - 2h) = 0$, so $K \sim 2h$ and $H^0(K) = \pi^* H^0(\mathcal{O}_{\mathbf{P}^1}(2))$ (both of dimension 3). Suppose $\psi \colon C \to \mathbf{P}^1$ has degree 3 and put $L = \psi^*\mathcal{O}(1)$: it has degree 3 and $|L|$ has no base points. Since $\deg(K - L) = 1$ we have $h^0(K - L) \leq 1$, so $h^0(L) = 3 + 1 - 3 + h^0(K - L) \leq 2$; hence $h^0(L) = 2$ and $K - L \sim Q$ for a point Q: $L \sim K - Q$. But every section of $K = \pi^*\mathcal{O}(2)$ vanishing at Q has the form $\pi^* s$ with $s(\pi(Q)) = 0$, so it vanishes on the whole divisor $\pi^*(\pi(Q)) = Q + Q'$ (Q' the conjugate point, $Q' = Q$ if Q is ramified). Therefore Q' is a base point of $|K - Q|$, contradicting that $|L| = |K - Q|$ has no base points. So no degree 3 morphism exists.

Thus for $g = 3$: $r = 0$, $r \geq 4$ always occur, $r = 1$ never, and exactly one of $r = 2$ (hyperelliptic) and $r = 3$ (non-hyperelliptic) occurs. □

Exercise 2.5

Problem. *Values of e.* Let C be a curve of genus $g \geq 1$.

(a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled surface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $e < 0$, let D be a divisor of degree $d = -e$, and let $\xi \in H^1(\mathcal{L}(-D))$ be a nonzero element defining an extension

$$0 \to \mathcal{O}_C \to \mathcal{E} \to \mathcal{L}(D) \to 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(\mathcal{L}(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$. Cf. proof of (2.15).

(c) Now show that if $-g \le e < 0$, there exists a ruled surface X over C with invariant e. [*Hint:* For any given D in (b), show that a suitable ξ exists, using an argument similar to the proof of (II, 8.18).]

(d) For $g = 2$, show that $e \ge -2$ is also necessary for the existence of X.

Note. It has been shown that $e \ge -g$ for any ruled surface (Nagata [8]).

Solution. In (b) the letter E denotes an effective divisor, as in the book; to avoid confusion with the sheaf $\mathcal{E}$, we write Z for it in the proof. Recall that $\mathcal{E}$ is normalized if $H^0(\mathcal{E}) \neq 0$ and $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$ for all invertible $\mathcal{M}$ of negative degree, and then $e = -\deg \mathcal{E}$.

(a) Let $0 \le e \le 2g-2$. Choose an effective divisor Z_0 of degree $2g-2-e \ge 0$ and let $\mathcal{N} = \mathcal{L}(Z_0 - K)$ be the invertible sheaf of the divisor $Z_0 - K$, so $\deg \mathcal{N} = -(2g-2) + (2g-2-e) = -e$ and $\mathcal{N}^{-1} = \mathcal{L}(K - Z_0)$. By Serre duality $H^1(\mathcal{N}^{-1}) = H^1(\mathcal{L}(K-Z_0)) \cong H^0(\mathcal{L}(Z_0))^\vee \neq 0$. Take $0 \neq \xi \in H^1(\mathcal{N}^{-1}) = \operatorname{Ext}^1(\mathcal{N}, \mathcal{O}_C)$ and the corresponding nonsplit extension

$$0 \to \mathcal{O}_C \xrightarrow{j} \mathcal{E} \to \mathcal{N} \to 0.$$

$\mathcal{E}$ is normalized: $H^0(\mathcal{E}) \supseteq H^0(\mathcal{O}_C) \neq 0$; if $\deg \mathcal{M} < 0$ then $H^0(\mathcal{M}) = 0$ and $H^0(\mathcal{N} \otimes \mathcal{M}) = 0$ (degree $-e + \deg \mathcal{M} < 0$), so $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$. Since $\deg \mathcal{E} = \deg \mathcal{N} = -e$, the invariant is e.

$\mathcal{E}$ is indecomposable: suppose $\mathcal{E} \cong \mathcal{A} \oplus \mathcal{B}$ with $\mathcal{A}, \mathcal{B}$ invertible. One of the components of j is nonzero, say $j_A \colon \mathcal{O} \to \mathcal{A}$; then $\deg \mathcal{A} \ge 0$. On the other hand $\deg \mathcal{A} \le 0$ since otherwise $\mathcal{E} \otimes \mathcal{A}^{-1} \supseteq \mathcal{A} \otimes \mathcal{A}^{-1} = \mathcal{O}$ would have a nonzero section with $\deg \mathcal{A}^{-1} < 0$, contradicting normalization. So $\deg \mathcal{A} = 0$ and the nonzero map j_A between invertible sheaves of degree 0 is an isomorphism. Then $j_A^{-1} \circ \mathrm{pr}_A$ is a retraction of j and the extension splits, contradicting $\xi \neq 0$.

(b) We have $\operatorname{Ext}^1(\mathcal{L}(D), \mathcal{O}) = H^1(\mathcal{L}(-D))$ and, by Serre duality, $H^1(\mathcal{L}(-D)) \cong H^0(\mathcal{L}(D+K))^\vee$; this identification is meant in the statement. Note $\deg(D+K) > 2g-2$, so $\dim V = g + d - 1$ for $V = H^0(\mathcal{L}(D+K))$.

First, $H^0(\mathcal{E}) \neq 0$ always. Moreover if $H^0(\mathcal{E} \otimes \mathcal{M}) \neq 0$ for some $\mathcal{M}$ with $\deg \mathcal{M} < 0$, choose an effective divisor Z'' with $\deg \mathcal{M}(Z'') = -1$; multiplication by the section $1_{Z''}$ of $\mathcal{L}(Z'')$ embeds $\mathcal{E} \otimes \mathcal{M} \hookrightarrow \mathcal{E} \otimes \mathcal{M}(Z'')$, so $H^0(\mathcal{E} \otimes \mathcal{M}(Z'')) \neq 0$. Hence *$\mathcal{E}$ is normalized iff $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$ for all invertible $\mathcal{M}$ of degree -1.*

Fix $\mathcal{M}$ of degree -1. From $0 \to \mathcal{M} \to \mathcal{E} \otimes \mathcal{M} \to \mathcal{L}(D) \otimes \mathcal{M} \to 0$ and $H^0(\mathcal{M}) = 0$ we get

$$H^0(\mathcal{E} \otimes \mathcal{M}) = \ker\bigl(\delta_{\mathcal{M}} \colon H^0(\mathcal{L}(D) \otimes \mathcal{M}) \to H^1(\mathcal{M})\bigr), \qquad \delta_{\mathcal{M}}(s) = s \cup \xi,$$

the cup product with $\xi \in H^1(\mathcal{L}(-D))$. Now $\mathcal{L}(D) \otimes \mathcal{M}$ has degree $d - 1 \ge 0$. A nonzero $s \in H^0(\mathcal{L}(D) \otimes \mathcal{M})$ has an effective divisor of zeros Z of degree $d-1$, and $\mathcal{M} \cong \mathcal{L}(Z - D)$, s corresponding to the canonical section of $\mathcal{L}(Z)$. Under this identification $s \cup \xi$ is the image of ξ under the map $\iota_Z \colon H^1(\mathcal{L}(-D)) \to H^1(\mathcal{L}(Z-D))$ induced by the inclusion $\mathcal{L}(-D) \subseteq \mathcal{L}(Z-D)$. Conversely every effective Z of degree $d-1$ arises (with $\mathcal{M} = \mathcal{L}(Z-D)$). Hence

$$\mathcal{E} \text{ is not normalized} \iff \exists\, Z \ge 0,\ \deg Z = d-1,\ \iota_Z(\xi) = 0.$$

By Serre duality (functorial), ι_Z is dual to the inclusion $H^0(\mathcal{L}(D+K-Z)) \hookrightarrow H^0(\mathcal{L}(D+K))$ (multiplication by the canonical section of $\mathcal{L}(Z)$), whose image is the subspace V_Z of sections

vanishing on Z, projectivizing to L_Z. So $\iota_Z(\xi) = 0$ iff ξ annihilates V_Z, i.e. iff $V_Z \subseteq \ker \xi$, i.e. iff $L_Z \subseteq H$. This proves (b). (Note H is a hyperplane because $\xi \neq 0$.)

(c) Let $-g \leq e < 0$, $d = -e \in \{1, \dots, g\}$, and fix any divisor D of degree d. By (b) we must find $0 \neq \xi \in V^\vee$, $V = H^0(\mathcal{L}(D+K))$ ($\dim V = g+d-1$), such that ξ does not annihilate V_Z for any effective Z of degree $d-1$. We have $\deg(D+K-Z) = 2g-1 > 2g-2$, so $H^1(\mathcal{L}(D+K-Z)) = 0$ and $\dim V_Z = g$ for every Z; the annihilator $\Lambda_Z = \mathrm{Ann}(V_Z) \subseteq V^\vee$ has dimension $d-1$. We must show that $\bigcup_Z \Lambda_Z \neq V^\vee$ (then a vector $\xi \notin \bigcup \Lambda_Z$ is automatically nonzero because $0 \in \Lambda_Z$).

Let $T = C^{d-1}$ (ordered $(d-1)$-tuples, $T = \mathrm{pt}$ if $d = 1$), $\Delta \subseteq T \times C$ the divisor $\sum_i \{y = P_i\}$, finite and flat of degree $d-1$ over T, and $\mathcal{N} = p_2^* \mathcal{L}(D+K)$. Consider $0 \to \mathcal{N}(-\Delta) \to \mathcal{N} \to \mathcal{N}|_\Delta \to 0$ and apply p_{1*}. Since $H^1(\mathcal{L}(D+K-Z)) = 0$ for all Z, cohomology and base change (III, 12.11) shows that $R^1 p_{1*}\mathcal{N}(-\Delta) = 0$ and $p_{1*}\mathcal{N}(-\Delta)$ is locally free with fibres V_Z. We get an exact sequence of vector bundles on T

$$0 \to p_{1*}\mathcal{N}(-\Delta) \to V \otimes \mathcal{O}_T \to \mathcal{G} \to 0, \qquad \mathcal{G} = p_{1*}(\mathcal{N}|_\Delta) \text{ of rank } d-1.$$

Dualizing, $\mathcal{G}^\vee \subseteq V^\vee \otimes \mathcal{O}_T$ is a subbundle of rank $d-1$ with fibre Λ_Z at $Z = \sum P_i$. Its total space $\mathbf{V}(\mathcal{G}^\vee)$ has dimension $(d-1)+(d-1) = 2d-2$ and maps to $V^\vee$ with image $\bigcup_Z \Lambda_Z$. This image is constructible of dimension $\leq 2d-2$, while $\dim V^\vee = g+d-1$. As $d \leq g$ we have $2d-2 \leq g+d-2 < g+d-1$, so the image is a proper subset of $V^\vee$, and a suitable $\xi \neq 0$ exists. By (b) the corresponding $\mathcal{E}$ is normalized, with $\deg \mathcal{E} = \deg \mathcal{L}(D) = d$, so $\mathbf{P}(\mathcal{E})$ is a ruled surface with invariant $e = -d$.

(d) Let $g = 2$. By (a) and (c) the values $e \geq 0$ and $-2 \leq e < 0$ do occur. We show $e \leq -3$ is impossible. Suppose a ruled surface $X = \mathbf{P}(\mathcal{E})$, $\mathcal{E}$ normalized, with $e = -d < 0$ exists.

Step 1: $\mathcal{E}$ is a nonsplit extension $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L}(D) \to 0$ with $\deg D = d$. By the proof of (2.8) there is such a sequence ($\mathcal{L}(D) = \bigwedge^2 \mathcal{E}$); it is nonsplit because $e < 0$ and a decomposable normalized $\mathcal{E}$ has $e \geq 0$ by (2.12)(a).

Step 2: if $d \geq 2$ then there is a normalized $\mathcal{E}'$ with $e' = e+1$. Choose a point Q and let $\mathcal{E}' \subseteq \mathcal{E}$ be the inverse image of $\mathcal{L}(D-Q) \subseteq \mathcal{L}(D)$. Then $0 \to \mathcal{O} \to \mathcal{E}' \to \mathcal{L}(D-Q) \to 0$ is exact, $H^0(\mathcal{E}') \supseteq H^0(\mathcal{O}) \neq 0$, and $H^0(\mathcal{E}' \otimes \mathcal{M}) \subseteq H^0(\mathcal{E} \otimes \mathcal{M}) = 0$ if $\deg \mathcal{M} < 0$; so $\mathcal{E}'$ is normalized with $\deg \mathcal{E}' = d-1$, i.e. $e' = -(d-1) = e+1$.

By Step 2, if $e \leq -3$ occurs, then $e = -3$ occurs. So it suffices to exclude $e = -3$, i.e. $d = 3$.

Step 3: $d = 3$ is impossible. Let $\mathcal{E}$ be as in Step 1 with $d = 3$; $\deg(D+K) = 5 = 2g+1$, so $|D+K|$ is very ample (IV, 3.2) and embeds C in $\mathbf{P}(V^\vee) = \mathbf{P}^3$ ($\dim V = g+d-1 = 4$) as a curve of degree 5; the hyperplane sections are the divisors of $|D+K|$. The functional $\xi \in V^\vee \setminus 0$ is a point $p \in \mathbf{P}^3$. For an effective divisor Z of degree $d-1 = 2$, $\Lambda_Z = \mathrm{Ann}(V_Z)$ is the line $\overline{Z}$ spanned by Z in $\mathbf{P}^3$ (a secant line if $Z = Q+Q'$, $Q \neq Q'$, the tangent line at Q if $Z = 2Q$): the hyperplanes containing Z are those containing the line. By the argument in (b)/(c), $\mathcal{E}$ is normalized iff $p \notin \overline{Z}$ for all Z of degree 2. We show every point $p \in \mathbf{P}^3$ lies on such a line.

If $p \in C$, take $Z = p + Q$ with $Q \in C$ arbitrary. Suppose $p \notin C$ lies on no line $\overline{Z}$. Let $W = \ker \xi \subseteq V$ (the sections whose hyperplane passes through p, $\dim W = 3$); it defines the projection $\pi_p \colon C \to \mathbf{P}^2$ from p. For Z of degree 2, $W \to V/V_Z$ ($\dim V/V_Z = 2$) is surjective iff $V_Z \not\subseteq W$ iff $\xi \notin \Lambda_Z$ iff $p \notin \overline{Z}$. By assumption this holds for all Z of degree 2; taking $Z = Q+Q'$ with $Q \neq Q'$ and $Z = 2Q$ we see (as $V \to H^0(\mathcal{L}(D+K)|_Z)$ is onto with kernel V_Z, so that $W \to H^0(\mathcal{L}(D+K)|_Z)$ is onto) that W is base-point-free and separates points and tangent

vectors. By the criterion (II, 7.8.2) π_p is a closed immersion, so C would be isomorphic to a nonsingular plane curve of degree 5 (degree $= \deg(D+K) = 5$). But a nonsingular plane curve of degree 5 has genus $\frac{1}{2}\cdot 4\cdot 3 = 6 \neq 2$ (I, Ex. 7.2). This contradiction shows that every $p \in \mathbf{P}^3 \setminus C$ also lies on some line $\overline{Z}$.

Hence for every $\xi \neq 0$ the sheaf $\mathcal{E}$ fails to be normalized, so no ruled surface over a genus 2 curve has $e = -3$, and consequently none has $e \leq -3$. Therefore $e \geq -2$ is necessary (and, by (c), sufficient). □

Exercise 2.6

> **Problem.** Show that every locally free sheaf of finite rank on $\mathbf{P}^1$ is isomorphic to a direct sum of invertible sheaves. [*Hint:* Choose a subinvertible sheaf of maximal degree, and use induction on the rank.]

Solution. Let $\mathcal{E}$ be locally free of rank r on $\mathbf{P}^1$; induct on r, the case $r = 1$ being trivial. Recall $\operatorname{Pic}\mathbf{P}^1 = \{\mathcal{O}(n)\}$ and $\deg\mathcal{O}(n) = n$.

There is a maximal degree. For $m \gg 0$, $\mathcal{E}^\vee(m)$ is generated by finitely many global sections, so there is a surjection $\mathcal{O}^N \to \mathcal{E}^\vee(m)$; dualizing, $\mathcal{E}(-m) \hookrightarrow \mathcal{O}^N$. Hence $H^0(\mathcal{E}(-m-1)) \subseteq H^0(\mathcal{O}(-1)^N) = 0$. Thus there is no nonzero map $\mathcal{O}(c) \to \mathcal{E}$ for $c > m$. On the other hand $\mathcal{E}(-c)$ is generated by global sections for $-c \gg 0$, so such maps exist for some c. Let a be the largest integer with $\operatorname{Hom}(\mathcal{O}(a), \mathcal{E}) \neq 0$, and fix a nonzero map $\mathcal{O}(a) \to \mathcal{E}$; it is injective since $\mathcal{O}(a)$ is torsion-free of rank 1 and $\mathbf{P}^1$ is integral. Put $\mathcal{L} = \mathcal{O}(a)$.

$\mathcal{Q} = \mathcal{E}/\mathcal{L}$ is locally free. If $\mathcal{Q}$ had torsion, the inverse image $\mathcal{L}' \subseteq \mathcal{E}$ of the torsion subsheaf would be torsion-free of rank 1, hence invertible, $\mathcal{L} \subsetneq \mathcal{L}'$, $\mathcal{L}' \cong \mathcal{O}(c)$ with $c > a$ (the quotient $\mathcal{L}'/\mathcal{L}$ has finite positive length), contradicting maximality of a. So $\mathcal{Q}$ is torsion-free, hence locally free (the local rings are DVRs), of rank $r-1$, and by induction $\mathcal{Q} \cong \bigoplus_{i=2}^{r} \mathcal{O}(a_i)$.

All $a_i \leq a$. Twist $0 \to \mathcal{L} \to \mathcal{E} \to \mathcal{Q} \to 0$ by $\mathcal{O}(-a-1)$: since $H^1(\mathcal{O}(-1)) = 0$, the map $H^0(\mathcal{E}(-a-1)) \to H^0(\mathcal{Q}(-a-1)) = \bigoplus_i H^0(\mathcal{O}(a_i - a - 1))$ is surjective. If some $a_i \geq a+1$, the right side is nonzero, so $H^0(\mathcal{E}(-a-1)) \neq 0$, i.e. there is a nonzero map $\mathcal{O}(a+1) \to \mathcal{E}$, contradicting the choice of a.

The sequence splits. It is classified by an element of $\operatorname{Ext}^1(\mathcal{Q}, \mathcal{L}) = \bigoplus_i H^1(\mathbf{P}^1, \mathcal{O}(a - a_i))$, and $a - a_i \geq 0$ gives $H^1 = 0$ (II or III, 5.1). So $\mathcal{E} \cong \mathcal{L} \oplus \mathcal{Q} \cong \mathcal{O}(a) \oplus \bigoplus_{i\geq 2} \mathcal{O}(a_i)$. □

Exercise 2.7

> **Problem.** On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C, and that no two of them are linearly equivalent.

Solution. Recall (2.11.6): C is an elliptic curve, $P \in C$, and $\mathcal{E}$ is the nonsplit extension $0 \to \mathcal{O} \to \mathcal{E} \xrightarrow{q} \mathcal{L}(P) \to 0$ given by $0 \neq \xi \in H^1(\mathcal{L}(-P))$ (dim = 1); $\mathcal{E}$ is normalized with $e = -1$, $\bigwedge^2 \mathcal{E} = \mathcal{L}(P)$, $\mathfrak{e} \sim P$, $C_0^2 = -e = 1$.

Step 1: description of the sections with $D^2 = 1$. A section D has $D \equiv C_0 + mf$ (since $D.f = 1$), so $D.C_0 = C_0^2 + m = 1 + m$ and $D^2 = 1 + 2m$. By (2.9) $D.C_0 = \deg \mathcal{M}$, where $\mathcal{E} \to \mathcal{M} \to 0$ is the quotient corresponding to D. Hence $D^2 = 1$ iff $m = 0$ iff $\deg \mathcal{M} = 1$ iff $\mathcal{M} \cong \mathcal{L}(Q)$ for a (unique) point $Q \in C$ (IV, 1.3.7 / $\operatorname{Pic}^1 C \cong C$ on an elliptic curve). So we must study surjections $\mathcal{E} \to \mathcal{L}(Q)$.

Step 2: $\operatorname{Hom}(\mathcal{E}, \mathcal{L}(Q))$ *is one-dimensional for each $Q \in C$, and every nonzero element is surjective.* Apply $\operatorname{Hom}(-, \mathcal{M})$, $\mathcal{M} = \mathcal{L}(Q)$, to the defining sequence:

$$0 \to \operatorname{Hom}(\mathcal{L}(P), \mathcal{M}) \to \operatorname{Hom}(\mathcal{E}, \mathcal{M}) \to H^0(\mathcal{M}) \xrightarrow{\partial} H^1(\mathcal{M}(-P)), \qquad \partial(t) = t \cup \xi.$$

Here $h^0(\mathcal{M}) = 1$ and the first term is $H^0(\mathcal{L}(Q - P))$.

- If $Q \neq P$: $\mathcal{L}(Q - P)$ is of degree 0 and nontrivial, so $H^0(\mathcal{L}(Q - P)) = 0 = H^1(\mathcal{L}(Q - P))$, and $\operatorname{Hom}(\mathcal{E}, \mathcal{M}) \cong H^0(\mathcal{M})$ is one-dimensional. A generator φ restricts on $\mathcal{O} \subseteq \mathcal{E}$ to the section t_Q of $\mathcal{L}(Q)$ with divisor Q. So $\varphi(y) \neq 0$ at every $y \neq Q$. If $\varphi(Q)$ were 0, φ would factor through $\mathcal{M}(-Q) = \mathcal{O}$, a nonzero element of $\operatorname{Hom}(\mathcal{E}, \mathcal{O})$. But $\operatorname{Hom}(\mathcal{E}, \mathcal{O}) = 0$: from the sequence, it injects into $\ker(H^0(\mathcal{O}) \to H^1(\mathcal{L}(-P)), 1 \mapsto \xi) = 0$, as $\operatorname{Hom}(\mathcal{L}(P), \mathcal{O}) = H^0(\mathcal{L}(-P)) = 0$. So φ is surjective.
- If $Q = P$: $\operatorname{Hom}(\mathcal{L}(P), \mathcal{L}(P)) = k$ gives the quotient q. The map ∂ sends the section 1_P of $\mathcal{L}(P)$ to the image of ξ under $H^1(\mathcal{L}(-P)) \to H^1(\mathcal{O})$ induced by $\mathcal{L}(-P) \subseteq \mathcal{O}$; this map is injective (its kernel is the image of $H^0(\mathcal{O}_P)$, which is zero as $H^0(\mathcal{O}) \to H^0(\mathcal{O}_P)$ is onto). So ∂ is injective and $\operatorname{Hom}(\mathcal{E}, \mathcal{L}(P)) = k \cdot q$.

Thus for each $Q \in C$ there is, up to a scalar, exactly one surjection $\varphi_Q \colon \mathcal{E} \to \mathcal{L}(Q)$, hence exactly one section σ_Q and one curve $D_Q = \sigma_Q(C) \cong C$ with $D_Q^2 = 1$ (and $D_P = C_0$). By Step 1 every section with self-intersection 1 is one of the D_Q, and $D_Q \neq D_{Q'}$ for $Q \neq Q'$ (the sheaf $\mathcal{L}(Q) = \sigma_Q^* \mathcal{O}_X(1)$ is determined by D_Q). So $Q \mapsto D_Q$ is a bijection from C onto the set of sections with $D^2 = 1$.

Step 3: they form an algebraic family. On $C \times C$ (coordinates (Q, y), projections p_1, p_2, diagonal Δ) consider $\mathcal{H} = p_{1*} \mathcal{H}om(p_2^* \mathcal{E}, \mathcal{L}(\Delta))$, whose fibre over Q should be $\operatorname{Hom}(\mathcal{E}, \mathcal{L}(Q))$. We have $H^1(\mathcal{E}^\vee(Q)) = 0$ for every Q: indeed $\mathcal{E}^\vee \cong \mathcal{E} \otimes (\bigwedge^2 \mathcal{E})^{-1} = \mathcal{E}(-P)$, and from $0 \to \mathcal{L}(Q - P) \to \mathcal{E}(Q - P) \to \mathcal{L}(Q) \to 0$ we get $H^1(\mathcal{E}(Q - P)) = 0$ when $Q \neq P$ (both ends have vanishing H^1), while for $Q = P$ the sequence $H^0(\mathcal{L}(P)) \to H^1(\mathcal{O}) \to H^1(\mathcal{E}) \to H^1(\mathcal{L}(P)) = 0$ and the isomorphism $H^0(\mathcal{L}(P)) \to H^1(\mathcal{O})$ ($1_P \mapsto$ image of $\xi \neq 0$, as above) give $H^1(\mathcal{E}) = 0$. Hence by cohomology and base change (III, 12.11) $\mathcal{H}$ is invertible with fibres $\operatorname{Hom}(\mathcal{E}, \mathcal{L}(Q))$. The tautological map $p_1^* \mathcal{H} \otimes p_2^* \mathcal{E} \to \mathcal{L}(\Delta)$ is surjective on each fibre $\{Q\} \times C$ by Step 2, hence surjective, and gives a surjection $p_2^* \mathcal{E} \to \mathcal{L}(\Delta) \otimes p_1^* \mathcal{H}^{-1}$, i.e. (II, 7.12) a morphism $\Sigma \colon C \times C \to X$ with $\pi \circ \Sigma = p_2$ and $\Sigma(Q, \cdot) = \sigma_Q$. The morphism $(p_1, \Sigma) \colon C \times C \to C \times X$ is a closed immersion with image a divisor $\mathcal{D} \subseteq C \times X$, flat over C (via the first factor) with fibre D_Q over Q. This is an algebraic family of curves on X, parametrized by C, whose members are exactly the sections with $D^2 = 1$.

Step 4: no two are linearly equivalent. By (2.9), the section D_Q corresponding to the quotient $\mathcal{L}(Q)$ satisfies $D_Q \sim C_0 + (Q - \mathfrak{e})f \sim C_0 + (Q - P)f$. If $D_Q \sim D_{Q'}$ then $\pi^*(Q - Q') \sim 0$ on X, so $Q - Q' \sim 0$ on C because $\pi^* \colon \operatorname{Pic} C \to \operatorname{Pic} X$ is injective (2.3). Since $g(C) = 1$, $Q = Q'$. $\square$

Exercise 2.8

Problem. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \to \mathcal{F} \to 0$, $\mathcal{F} \neq 0$, $\mathcal{F} \neq \mathcal{E}$, we have

$$(\deg \mathcal{F})/\operatorname{rank} \mathcal{F} > (\deg \mathcal{E})/\operatorname{rank} \mathcal{E}.$$

Replacing $>$ by $\geq$ defines *semistable*.

(a) A decomposable $\mathcal{E}$ is never stable.

(b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E} > 0$ (respectively, ≥ 0).

(c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g-2$, (2) an element $\mathcal{L} \in \operatorname{Pic} C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(\mathcal{L}^\vee)$, determined up to a nonzero scalar multiple.

Solution. Write $\mu(\mathcal{G}) = \deg \mathcal{G}/\operatorname{rank} \mathcal{G}$.

(a) Let $\mathcal{E} = \mathcal{E}_1 \oplus \mathcal{E}_2$ with $\mathcal{E}_i \neq 0$ locally free. Then $\mathcal{E}_1$, $\mathcal{E}_2$ are locally free quotients of $\mathcal{E}$, different from 0 and $\mathcal{E}$, and $\mu(\mathcal{E}) = \frac{r_1}{r_1+r_2}\mu(\mathcal{E}_1) + \frac{r_2}{r_1+r_2}\mu(\mathcal{E}_2)$ is a weighted average of $\mu(\mathcal{E}_1), \mu(\mathcal{E}_2)$. So $\mu(\mathcal{E}_1) \leq \mu(\mathcal{E})$ or $\mu(\mathcal{E}_2) \leq \mu(\mathcal{E})$, and $\mathcal{E}$ is not stable.

(b) Let $\mathcal{E}$ have rank 2 and be normalized. A locally free quotient $\mathcal{E} \to \mathcal{F}$ with $\mathcal{F} \neq 0, \mathcal{E}$ has rank 1 (if the rank were 2 the kernel would be a torsion-free sheaf of rank 0, i.e. zero, and $\mathcal{F} = \mathcal{E}$); it is invertible with invertible kernel $\mathcal{N}$, $\deg \mathcal{N} + \deg \mathcal{F} = \deg \mathcal{E}$. The condition for stability is $\deg \mathcal{F} > \frac{1}{2}\deg \mathcal{E}$ for all such quotients (semistability: $\geq$).

($\Leftarrow$) Since $\mathcal{E}$ is normalized, $\deg \mathcal{N} \leq 0$: otherwise $\mathcal{N} \otimes \mathcal{N}^{-1} = \mathcal{O} \subseteq \mathcal{E} \otimes \mathcal{N}^{-1}$ gives $H^0(\mathcal{E} \otimes \mathcal{N}^{-1}) \neq 0$ with $\deg \mathcal{N}^{-1} < 0$. Hence $\deg \mathcal{F} \geq \deg \mathcal{E}$. If $\deg \mathcal{E} > 0$, then $\deg \mathcal{F} \geq \deg \mathcal{E} > \frac{1}{2}\deg \mathcal{E}$, so $\mathcal{E}$ is stable; if $\deg \mathcal{E} \geq 0$ then $\deg \mathcal{F} \geq \deg \mathcal{E} \geq \frac{1}{2}\deg \mathcal{E}$, so $\mathcal{E}$ is semistable.

($\Rightarrow$) By the proof of (2.8), a nonzero section of $\mathcal{E}$ gives an exact sequence $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{F} \to 0$ with $\mathcal{F}$ invertible (of degree $\deg \mathcal{E}$). If $\mathcal{E}$ is stable, $\deg \mathcal{E} = \deg \mathcal{F} > \frac{1}{2}\deg \mathcal{E}$ so $\deg \mathcal{E} > 0$; if semistable, $\deg \mathcal{E} \geq \frac{1}{2}\deg \mathcal{E}$ so $\deg \mathcal{E} \geq 0$.

(c) *Convention.* Tensoring by an invertible sheaf does not change (in)decomposability or (semi)stability type of a rank-2 sheaf (if $\mathcal{F}$ is a quotient of $\mathcal{E}$ then $\mathcal{F} \otimes \mathcal{M}$ is a quotient of $\mathcal{E} \otimes \mathcal{M}$ and μ shifts by $\deg \mathcal{M}$), and $\mathbf{P}(\mathcal{E}) \cong \mathbf{P}(\mathcal{E} \otimes \mathcal{M})$ (2.2). So the classification is understood up to twisting; we fix the twist by requiring (see below) that $\mathcal{E}$ be *normalized*, as in (2.8).

Let $\mathcal{E}$ be locally free of rank 2, not semistable. Then there is an invertible quotient $\mathcal{F}$ with $\deg \mathcal{F} < \frac{1}{2}\deg \mathcal{E}$ and kernel $\mathcal{N}$, an invertible *subbundle* with $\deg \mathcal{N} > \frac{1}{2}\deg \mathcal{E}$. *Claim:* every invertible subsheaf $\mathcal{N}' \subseteq \mathcal{E}$ with $\deg \mathcal{N}' > \frac{1}{2}\deg \mathcal{E}$ is contained in $\mathcal{N}$. Indeed, if $\mathcal{N}' \not\subseteq \mathcal{N}$, the composite $\mathcal{N}' \to \mathcal{E} \to \mathcal{F}$ is a nonzero map of invertible sheaves, so $\deg \mathcal{N}' \leq \deg \mathcal{F} < \frac{1}{2}\deg \mathcal{E}$. If $\mathcal{N}'$ is itself a subbundle contained in $\mathcal{N}$, then $\mathcal{N}/\mathcal{N}' \subseteq \mathcal{E}/\mathcal{N}'$ is a torsion subsheaf of a locally free sheaf, hence zero, so $\mathcal{N}' = \mathcal{N}$. Thus $\mathcal{N} = \mathcal{N}(\mathcal{E})$ is *the unique invertible subsheaf of maximal degree*, and this subbundle is determined by $\mathcal{E}$ (it is preserved by isomorphisms). Replacing $\mathcal{E}$ by $\mathcal{E} \otimes \mathcal{N}^{-1}$ we may assume $\mathcal{N}(\mathcal{E}) = \mathcal{O}$. Such $\mathcal{E}$ are normalized: $H^0(\mathcal{E}) \supseteq H^0(\mathcal{O}) \neq 0$, and if $\deg \mathcal{M} < 0$ and $\mathcal{M}^{-1} \to \mathcal{E}$ were a nonzero map ($\deg \mathcal{M}^{-1} > 0$), it could not factor through $\mathcal{N} = \mathcal{O}$ (no nonzero maps from positive degree to $\mathcal{O}$), so it maps nontrivially to $\mathcal{F} = \mathcal{E}/\mathcal{O}$ of degree $\deg \mathcal{E} < 0$, again impossible. (Conversely, if $\mathcal{E}$ is normalized and not semistable, then $\deg \mathcal{E} < 0$ by (b), the saturation of $\mathcal{O} \hookrightarrow \mathcal{E}$ is an invertible subsheaf of degree ≥ 0, hence 0 by normalization, hence equal to $\mathcal{O}$, and it has degree $0 > \frac{1}{2}\deg \mathcal{E}$; so $\mathcal{N}(\mathcal{E}) = \mathcal{O}$.) Let $\mathfrak{N}$ be the

set of isomorphism classes of indecomposable locally free $\mathcal{E}$ of rank 2 which are not semistable and satisfy $\mathcal{N}(\mathcal{E}) \cong \mathcal{O}$.

The map $\mathfrak{N} \to \{\textit{triples}\}$. For $\mathcal{E} \in \mathfrak{N}$ we have $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L} \to 0$ with $\mathcal{L} = \mathcal{E}/\mathcal{O}$ invertible. Non-semistability gives $0 = \deg \mathcal{N} > \frac{1}{2}\deg \mathcal{E} = \frac{1}{2}\deg \mathcal{L}$, so $e := -\deg \mathcal{L} > 0$. The sequence does not split (otherwise $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$ is decomposable), so its class $\xi \in \operatorname{Ext}^1(\mathcal{L}, \mathcal{O}) = H^1(\mathcal{L}^\vee)$ is nonzero; then $H^1(\mathcal{L}^\vee) \cong H^0(\mathcal{L} \otimes \omega_C)^\vee \neq 0$ forces $\deg(\mathcal{L} \otimes \omega_C) = 2g - 2 - e \geq 0$, i.e. $e \leq 2g - 2$. The triple $(e, \mathcal{L}, [\xi])$ is well defined: an isomorphism $\mathcal{E} \to \mathcal{E}'$ in $\mathfrak{N}$ carries $\mathcal{N} = \mathcal{O}$ to $\mathcal{N}' = \mathcal{O}$ by a scalar $c \in k^*$ and induces an isomorphism $\mathcal{L} \to \mathcal{L}'$; the classes of the two extensions then correspond up to the scalar c (standard property of extension classes under morphisms of extensions).

Injectivity: the iso class of an extension is determined by $(\mathcal{L}, [\xi])$ (up to scalar): two extensions with the same $\mathcal{L}$ and proportional classes are isomorphic extensions.

Surjectivity: given $(e, \mathcal{L}, [\xi])$ with $0 < e \leq 2g - 2$, $\deg \mathcal{L} = -e$, $0 \neq \xi \in H^1(\mathcal{L}^\vee)$ (nonzero space, as above), let $0 \to \mathcal{O} \xrightarrow{j} \mathcal{E} \to \mathcal{L} \to 0$ be the corresponding nonsplit extension. Then $\mathcal{O} \subseteq \mathcal{E}$ is a subbundle of degree $0 > -\frac{e}{2} = \frac{1}{2}\deg \mathcal{E}$, so $\mathcal{E}$ is not semistable and, by the Claim (the invertible subsheaf $\mathcal{O}$ has degree $> \frac{1}{2}\deg \mathcal{E}$ so lies in $\mathcal{N}(\mathcal{E})$, and a subbundle inside $\mathcal{N}(\mathcal{E})$ equals it), $\mathcal{N}(\mathcal{E}) = \mathcal{O}$. If $\mathcal{E} = \mathcal{A} \oplus \mathcal{B}$ were decomposable (invertible summands), one component of j, say j_A, would be nonzero, so $\deg \mathcal{A} \geq 0$; but $\deg \mathcal{A} \leq \deg \mathcal{N}(\mathcal{E}) = 0$. So j_A is an isomorphism and j splits, contradicting $\xi \neq 0$. Hence $\mathcal{E} \in \mathfrak{N}$ and maps to the given triple.

So the indecomposable non-semistable locally free sheaves of rank 2, normalized so that their maximal destabilizing subsheaf is $\mathcal{O}$ (equivalently: normalized in the sense of (2.8), with $e > 0$), are classified by $(e, \mathcal{L}, [\xi])$ as stated; an arbitrary indecomposable non-semistable $\mathcal{E}$ is $\mathcal{E}_0 \otimes \mathcal{N}(\mathcal{E})$ with $\mathcal{E}_0 \in \mathfrak{N}$, so is determined by the triple together with $\mathcal{N}(\mathcal{E}) \in \operatorname{Pic} C$. (In the book's statement, (1)–(3) classify up to the twisting which does not change $\mathbf{P}(\mathcal{E})$.) □

Exercise 2.9

> **Problem.** Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbf{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or, $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$. Cf. (IV, 6.4.1). [*Hint:* Use (2.11.4).]

Solution. Let P_0 be the vertex of X_0, the cone over a conic. By (2.11.4) (with C = conic $\cong \mathbf{P}^1$, $d = 2$), the blow-up $\mu\colon X \to X_0$ of P_0 is the ruled surface $X = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(-2))$ over $\mathbf{P}^1$, which is normalized with $e = 2$; $C_0 = \mu^{-1}(P_0)$ is the section with $C_0^2 = -2$. Moreover μ is an isomorphism $X \setminus C_0 \to X_0 \setminus P_0$. We use $\operatorname{Num} X = \mathbf{Z}C_0 \oplus \mathbf{Z}f = \operatorname{Pic} X$ (as $\operatorname{Pic} \mathbf{P}^1 = \mathbf{Z}$), $C_0.f = 1$, $f^2 = 0$, and by (2.11) $K \equiv -2C_0 + (2 \cdot 0 - 2 - 2)f = -2C_0 - 4f$.

The hyperplane class. Let H be a plane section of X_0 not through P_0 (a conic) and $\tilde{H} = \mu^* H$. Then $\tilde{H} \equiv aC_0 + bf$ with $a = \tilde{H}.f = 1$ (a plane meets a ruling once) and $0 = \tilde{H}.C_0 = -2a + b$, so $b = 2$: $\tilde{H} \sim C_0 + 2f$.

The strict transform. Let $\tilde{Y} \subseteq X$ be the strict transform of Y; it is the blow-up of Y in the closed subscheme $Y \times_{X_0} P_0$ (II, 7.15), and if $P_0 \in Y$ this is the reduced point P_0, a Cartier divisor on the nonsingular curve Y, so $\tilde{Y} \cong Y$; if $P_0 \notin Y$, also $\tilde{Y} \cong Y$. Write $\tilde{Y} \equiv aC_0 + bf$.

If $\tilde{Y} = f$ (a line through the vertex) then Y is a line, $\deg Y = 1 = 2 \cdot 0 + 1$, $g = 0 = 0^2 - 0$, in agreement with the second alternative with $a = 0$. Otherwise $\tilde{Y} \neq f$, and $\tilde{Y} \neq C_0$ (it is not contracted), so $a = \tilde{Y}.f > 0$.

Degree and genus in terms of a, b. $\deg Y = \tilde{Y}.\tilde{H} = (aC_0 + bf).(C_0 + 2f) = -2a + 2a + b = b$. By adjunction (1.5), since $\tilde{Y} \cong Y$ is nonsingular,

$$2g(Y)-2 = \tilde{Y}.(\tilde{Y}+K) = (aC_0+bf).((a-2)C_0+(b-4)f) = -2a(a-2)+a(b-4)+b(a-2) = 2ab-2a^2-2b,$$

so $g(Y) = 1 + ab - a^2 - b = (a-1)(b-a-1)$. Finally $\tilde{Y}.C_0 = b - 2a$.

Claim: $\tilde{Y}.C_0 = 0$ if $P_0 \notin Y$, and $\tilde{Y}.C_0 = 1$ if $P_0 \in Y$. The first is clear since then $\tilde{Y} \cap C_0 = \emptyset$. Suppose $P_0 \in Y$. The inverse image ideal $\mu^{-1}\mathfrak{m}_{P_0} \cdot \mathcal{O}_X$ is the ideal sheaf $\mathcal{O}_X(-E')$ of the exceptional divisor $E' = \operatorname{Proj} \bigoplus \mathfrak{m}^n/\mathfrak{m}^{n+1}$; the associated graded ring of the vertex is the homogeneous coordinate ring of the conic, a domain, so E' is the reduced irreducible curve C_0: $\mathcal{O}_X(-C_0) = \mathfrak{m}_{P_0}\mathcal{O}_X$. Restricting to $\tilde{Y} \cong Y$: the map $\mathcal{O}_X(-C_0) \otimes \mathcal{O}_{\tilde{Y}} \to \mathcal{O}_{\tilde{Y}}$ is injective (nonzero map of an invertible sheaf on an integral curve) with image $\mathfrak{m}_{P_0}\mathcal{O}_Y = \mathfrak{m}_{Y,P_0} = \mathcal{O}_Y(-P_0)$, as $Y \subseteq X_0$ is a closed subscheme. So $\mathcal{O}_X(-C_0)|_{\tilde{Y}} \cong \mathcal{O}_Y(-P_0)$ has degree -1, and by (1.3) $\tilde{Y}.C_0 = 1$.

Case $P_0 \notin Y$: $b = 2a$, so $\deg Y = 2a$, $g(Y) = (a-1)(a-1) = (a-1)^2$, with $a \geq 1$. We show Y is a complete intersection of X_0 with a surface of degree a. Since $\tilde{Y} \sim aC_0 + 2af = a\tilde{H}$, $\tilde{Y} \in |a\tilde{H}|$. Now $H^0(X, \mathcal{O}_X(a\tilde{H})) = H^0(\mathbf{P}^1, S^a(\mathcal{O} \oplus \mathcal{O}(-2)) \otimes \mathcal{O}(2a)) = \bigoplus_{i=0}^{a} H^0(\mathcal{O}(2a-2i))$ has dimension $\sum_{i=0}^{a}(2a-2i+1) = (a+1)^2$. The restriction map $H^0(\mathbf{P}^3, \mathcal{O}(a)) \to H^0(X_0, \mathcal{O}_{X_0}(a))$ has kernel the degree a part of the ideal (q) of X_0, which has dimension $\binom{a+1}{3}$, so its image has dimension $\binom{a+3}{3} - \binom{a+1}{3} = (a+1)^2$. The pull-back $\mu^*\colon H^0(X_0, \mathcal{O}(a)) \to H^0(X, \mathcal{O}(a\tilde{H}))$ is injective (μ is birational and X_0 is integral), hence, comparing dimensions, every divisor in $|a\tilde{H}|$ is of the form $\operatorname{div}(\mu^* s)$ with $s = G|_{X_0}$ for a form G of degree a. For our $\tilde{Y}$ this reads $\tilde{Y} = \operatorname{div}(\mu^* s)$ with $s = G|_{X_0}$. As $\tilde{Y} \cap C_0 = \emptyset$, the section $\mu^* s$ has no zero on C_0, so $s(P_0) \neq 0$ and $\operatorname{div}(s) = X_0 \cap \{G = 0\}$ lies in $X_0 \setminus P_0 \cong X \setminus C_0$, where it corresponds to the reduced curve $\tilde{Y}$. Hence $X_0 \cap \{G = 0\} = Y$ scheme-theoretically: Y is the complete intersection of X_0 with the surface $G = 0$ of degree a (consistent with (IV, 6.4.1): $\deg Y = 2a$, $g = 1 + \frac{1}{2} \cdot 2a\,(a + 2 - 4) = (a-1)^2$).

Case $P_0 \in Y$: $b = 2a + 1$, so $\deg Y = 2a + 1$ is odd and $g(Y) = (a-1)(2a+1-a-1) = (a-1)a = a^2 - a$. □

Exercise 2.10

> **Problem.** For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbf{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve of genus g which has a g^1_3. Cf. (IV, Sect. 5).

Solution. Here $X = X_e = \mathbf{P}(\mathcal{O} \oplus \mathcal{O}(-e))$ over $\mathbf{P}^1$ with $\operatorname{Pic} X = \mathbf{Z}C_0 \oplus \mathbf{Z}f = \operatorname{Num} X$ (as $\operatorname{Pic} \mathbf{P}^1 = \mathbf{Z}$; (2.3)), $C_0^2 = -e$, and the embedding is given by $|H|$, $H = C_0 + nf$, which is very ample since $n > e$ (2.17(c)); $\dim H^0(\mathcal{L}(H)) = d + 2$ (2.19). By (2.11), $K \sim -2C_0 - (e+2)f$.

The curve. Let $\mathfrak{D} = H - K \sim 3C_0 + (n + e + 2)f$. Put $a = 3$, $b = n + e + 2$. By Cor. 2.18(b), $|aC_0 + bf|$ contains an irreducible nonsingular curve if $a > 0$ and either $b > ae$ or ($e > 0$ and

$b = ae$). Here $b - 3e = n - 2e + 2 \geq 0$ by hypothesis $n \geq 2e - 2$, with equality only if $n = 2e - 2$; and in that case $e > 2$ (since $n > e$), in particular $e > 0$. If $e = 0$ then $b = n + 2 > 0 = ae$. So in all cases Cor. 2.18(b) provides a nonsingular irreducible curve $Y \in |\mathfrak{D}|$.

Genus. By adjunction (1.5), $K_Y = (K + Y)|_Y$ and $K + Y \sim K + H - K = H$. So $K_Y \cong \mathcal{O}_X(H)|_Y = \mathcal{O}_Y(1)$. Moreover $2g(Y) - 2 = Y.(Y + K) = Y.H = (3C_0 + bf).(C_0 + nf) = -3e + 3n + b = 4n - 2e + 2 = 2d + 2$, so $g(Y) = d + 2$.

Canonical embedding. From $0 \to \mathcal{O}_X(H - Y) \to \mathcal{O}_X(H) \to \mathcal{O}_Y(H) \to 0$ and $H - Y \sim K$ we get

$$0 \to H^0(\mathcal{L}(K)) \to H^0(\mathcal{L}(H)) \to H^0(\mathcal{O}_Y(1)) \to H^1(\mathcal{L}(K)) \ .$$

By (2.5), $p_g(X) = \dim H^0(\mathcal{L}(K)) = 0$ and $q = \dim H^1(\mathcal{O}_X) = g(\mathbf{P}^1) = 0$, so by Serre duality $H^1(\mathcal{L}(K)) \cong H^1(\mathcal{O}_X)^\vee = 0$. Hence $H^0(X, \mathcal{L}(H)) \to H^0(Y, \mathcal{O}_Y(1)) = H^0(Y, K_Y)$ is an isomorphism (both have dimension $d + 2 = g$). Therefore the embedding $Y \subseteq X \subseteq \mathbf{P}^{d+1} = \mathbf{P}^{g-1}$ is given by the complete canonical linear system $|K_Y|$: Y is a canonical curve.

Conclusion. Since $|K_Y|$ is very ample, Y is nonhyperelliptic (IV, 5.2). The projection $\pi|_Y : Y \to \mathbf{P}^1$ has degree $Y.f = 3$, so Y has a g^1_3. Which genera occur? $g = d+2 = 2n-e+2$ with $n > e \geq 0$, $n \geq 2e - 2$. For $e = 0$: $n \geq 1$ gives all even $g = 2n + 2 \geq 4$. For $e = 1$: $n \geq 2$ (conditions $n > 1$, $n \geq 0$) gives all odd $g = 2n + 1 \geq 5$. So for every $g \geq 4$ there is a nonhyperelliptic curve of genus g with a g^1_3 (a trigonal canonical curve lying on a rational normal scroll). □

Exercise 2.11

Problem. Let X be a ruled surface over the curve C, defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{L}(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C.

(a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no base points.

(b) If $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \mathfrak{e} - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

Solution. Put $\mathcal{G} = \mathcal{E}(\mathfrak{b}) = \mathcal{E} \otimes \mathcal{L}(\mathfrak{b})$, $F = \mathcal{O}_X(1) \otimes \pi^*\mathcal{L}(\mathfrak{b}) = \mathcal{L}(C_0 + \mathfrak{b}f)$. By (II, 7.11) and the projection formula, $\pi_* F = \mathcal{G}$ and $H^0(X, F) = H^0(C, \mathcal{G})$. Tensoring $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L}(\mathfrak{e}) \to 0$ with $\mathcal{L}(\mathfrak{b})$:

$$0 \to \mathcal{L}(\mathfrak{b}) \to \mathcal{G} \to \mathcal{L}(\mathfrak{b} + \mathfrak{e}) \to 0. \tag{1}$$

We use the following three lemmas (valid for any normalized $\mathcal{E}$ and any divisor $\mathfrak{b}$ on C); they are also used in Exercises 2.12, 2.13. Here "no base points" for a complete linear system $|\mathfrak{c}|$ means that $\mathcal{L}(\mathfrak{c})$ is generated by global sections (in particular $H^0 \neq 0$), and "nonspecial" means $H^1(\mathcal{L}(\mathfrak{c})) = 0$.

Lemma A (sections). Assume $H^1(\mathcal{L}(\mathfrak{b})) = 0$, $H^0(\mathcal{L}(\mathfrak{b})) \neq 0$ and $|\mathfrak{b} + \mathfrak{e}|$ has no base points. Then there is a section $D \sim C_0 + \mathfrak{b}f$ of π.

Proof. By (2.9), a section $D \sim C_0 + \mathfrak{b}f$ corresponds to a surjection $\mathcal{E} \to \mathcal{M} = \mathcal{L}(\mathfrak{b} + \mathfrak{e})$ (then $\mathcal{M} \cong \mathcal{L}(\mathfrak{b}')$ with $D \sim C_0 + (\mathfrak{b}' - \mathfrak{e})f \sim C_0 + \mathfrak{b}f$). Let $j : \mathcal{O} \to \mathcal{E}$, $q : \mathcal{E} \to \mathcal{L}(\mathfrak{e})$ be as in the sequence above. Since $\mathrm{Ext}^1(\mathcal{L}(\mathfrak{e}), \mathcal{M}) = H^1(\mathcal{L}(\mathfrak{b})) = 0$, the map $\mathrm{Hom}(\mathcal{E}, \mathcal{M}) \to \mathrm{Hom}(\mathcal{O}, \mathcal{M}) = H^0(\mathcal{M})$, $\psi \mapsto \psi \circ j$, is surjective. Let B be the (finite) base locus of $|\mathfrak{b}|$. For each $Q \in B$ the set

$\{t \in H^0(\mathcal{M}) : t(Q) = 0\}$ is a proper subspace (as $|\mathfrak{b} + \mathfrak{e}|$ has no base points), so we can choose $t \in H^0(\mathcal{M})$ with $t(Q) \neq 0$ for all $Q \in B$ (k is infinite); $t \neq 0$ and its zeros $y_1, \ldots, y_m$ are finitely many points, none in B. Choose ψ_0 with $\psi_0 \circ j = t$ and for $u \in H^0(\mathcal{L}(\mathfrak{b}))$ put $\psi_u = \psi_0 + \mu_u \circ q$, where $\mu_u \colon \mathcal{L}(\mathfrak{e}) \to \mathcal{M}$ is multiplication by u. Then $\psi_u \circ j = t$. A map to an invertible sheaf is surjective iff it is nonzero on every fibre. At $y \notin \{y_i\}$, $\psi_u(j(1)) = t(y) \neq 0$. At y_i, $\psi_u(y_i)$ vanishes on the line $j(\mathcal{O}(y_i))$ and factors through $\mathcal{L}(\mathfrak{e})(y_i)$; it is an element $\alpha_i + u(y_i)$ of the one-dimensional space $\mathrm{Hom}(\mathcal{L}(\mathfrak{e})(y_i), \mathcal{M}(y_i)) = \mathcal{L}(\mathfrak{b})(y_i)$, with α_i determined by ψ_0. So ψ_u is surjective iff $u(y_i) \neq -\alpha_i$ for all i. Since $y_i \notin B$, evaluation at y_i is a nonzero linear form on $H^0(\mathcal{L}(\mathfrak{b}))$, so $\{u : u(y_i) = -\alpha_i\}$ is a proper affine hyperplane, and a finite union of proper affine hyperplanes cannot cover the vector space. Choose u outside them. (Lemma A)

Lemma B (very ampleness). Suppose that for every closed subscheme $Z' \subseteq C$ of length 2 (i.e. every effective divisor of degree 2) the restriction $H^0(C, \mathcal{G}) \to H^0(Z', \mathcal{G}|_{Z'})$ is surjective. Then $F = \mathcal{L}(C_0 + \mathfrak{b}f)$ is very ample; in particular it has no base points.

Proof. By (II, 7.8.2) it suffices to show that for every closed subscheme $Z \subseteq X$ of length 2 the map $H^0(X, F) \to H^0(Z, F|_Z)$ is surjective (taking Z reduced of two points this gives global generation and separation of points; taking Z supported at one point, it gives separation of tangent vectors, since a subspace of the 2-dimensional $F \otimes \mathfrak{m}/\mathfrak{m}^2$ which surjects onto all its one-dimensional quotients is everything). Let $Z \subseteq X$ have length 2, and Z' its scheme-theoretic image in C; then either $Z \subseteq$ a fibre $\pi^{-1}(y)$, or $\pi|_Z \colon Z \to Z'$ is an isomorphism with Z' of length 2.

Case A: $Z \subseteq \pi^{-1}(y) = \mathbf{P}(\mathcal{E}(y))$. The restriction of F to the fibre is $\mathcal{O}_{\mathbf{P}^1}(1)$, with $H^0 = \mathcal{G}(y)$, and $H^0(X, F) = H^0(C, \mathcal{G}) \to H^0(\pi^{-1}(y), F) = \mathcal{G}(y)$ is the evaluation. This is surjective: take $Z' = y + y'$, $y' \neq y$; surjectivity onto $\mathcal{G}|_{Z'} = \mathcal{G}(y) \oplus \mathcal{G}(y')$ gives surjectivity onto $\mathcal{G}(y)$. And $H^0(\mathcal{O}_{\mathbf{P}^1}(1)) \to H^0(\mathcal{O}_Z(1))$ is surjective ($\mathcal{O}(1)$ is very ample on $\mathbf{P}^1$). Composition gives the claim.

Case B: $Z \cong Z'$. Then Z is a section of $\mathbf{P}(\mathcal{E}) \times_C Z' \to Z'$, given by a quotient $\mathcal{E}|_{Z'} \to \mathcal{M}_Z \to 0$ with $\mathcal{M}_Z$ invertible on Z', and $F|_Z = \mathcal{M}_Z \otimes \mathcal{L}(\mathfrak{b})|_{Z'}$. The restriction $H^0(X, F) = H^0(\mathcal{G}) \to H^0(Z, F|_Z)$ is the composite $H^0(\mathcal{G}) \to H^0(\mathcal{G}|_{Z'}) \to H^0(\mathcal{M}_Z \otimes \mathcal{L}(\mathfrak{b})|_{Z'})$, the first map surjective by hypothesis, the second surjective because $\mathcal{G}|_{Z'} \to \mathcal{M}_Z \otimes \mathcal{L}(\mathfrak{b})|_{Z'}$ is surjective and Z' is affine. (Lemma B)

Lemma C (criteria for the hypothesis of B). The hypothesis of Lemma B holds if either

(i) $H^1(C, \mathcal{G}(-Z')) = 0$ for every effective divisor Z' of degree 2; or

(ii) $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C and $H^1(\mathcal{L}(\mathfrak{b})) = 0$.

Moreover if $H^1(\mathcal{G}(-y)) = 0$ for all points y, then $\mathcal{G}$ is generated by global sections, hence so is F.

Proof. (i) The kernel of $\mathcal{G} \to \mathcal{G}|_{Z'}$ is $\mathcal{G}(-Z')$, and $H^0(\mathcal{G}) \to H^0(\mathcal{G}|_{Z'}) \to H^1(\mathcal{G}(-Z')) = 0$. (ii) Tensor (1) with $\mathcal{O}_{Z'}$ (exact, all sheaves locally free): $0 \to \mathcal{L}(\mathfrak{b})|_{Z'} \to \mathcal{G}|_{Z'} \to \mathcal{L}(\mathfrak{b} + \mathfrak{e})|_{Z'} \to 0$. Given $w \in H^0(\mathcal{G}|_{Z'})$, its image in $H^0(\mathcal{L}(\mathfrak{b}+\mathfrak{e})|_{Z'})$ lifts to a global section of $\mathcal{L}(\mathfrak{b}+\mathfrak{e})$ (very ample), which lifts to $s \in H^0(\mathcal{G})$ since $H^1(\mathcal{L}(\mathfrak{b})) = 0$; then $w - s|_{Z'} \in H^0(\mathcal{L}(\mathfrak{b})|_{Z'})$ is the restriction of a global section of $\mathcal{L}(\mathfrak{b})$ (very ample). So w is in the image. For the last statement: $H^0(\mathcal{G}) \to \mathcal{G}(y)$ is onto when $H^1(\mathcal{G}(-y)) = 0$, so $\mathcal{G}$ is generated by global sections; then $F = \mathcal{O}_X(1) \otimes \pi^*\mathcal{L}(\mathfrak{b})$

is a quotient of $\pi^*\mathcal{G}$, which is generated by global sections (recall $\pi^*\pi_*F \to F$ is surjective, as $\mathcal{O}_X(1)$ is a quotient of $\pi^*\mathcal{E}$), so F is generated by global sections. (Lemma C)

Proof of (a). By Lemma A there is a section $D \sim C_0 + \mathfrak{b}f$ (the hypotheses $H^1(\mathfrak{b}) = 0$, $H^0(\mathfrak{b}) \neq 0$, $|\mathfrak{b}+\mathfrak{e}|$ bpf hold). Next, $\mathcal{L}(\mathfrak{b})$ and $\mathcal{L}(\mathfrak{b}+\mathfrak{e})$ are generated by global sections and $H^1(\mathcal{L}(\mathfrak{b})) = 0$, so from (1) $H^0(\mathcal{G}) \to H^0(\mathcal{L}(\mathfrak{b}+\mathfrak{e}))$ is onto; hence at each point y the fibre $\mathcal{G}(y)$ is spanned by the images of global sections of $\mathcal{L}(\mathfrak{b})$ (viewed in $\mathcal{G}$ via (1)) and of lifts of global sections of $\mathcal{L}(\mathfrak{b}+\mathfrak{e})$, and so $\mathcal{G}$ is generated by global sections. Then F has no base points as in the last statement of Lemma C, i.e. $|D| = |C_0 + \mathfrak{b}f|$ has no base points.

Proof of (b). The hypothesis "$\mathfrak{b} - P$ nonspecial for all P" gives $H^1(\mathcal{L}(\mathfrak{b})) = 0$ (the map $H^1(\mathcal{L}(\mathfrak{b}-P)) \to H^1(\mathcal{L}(\mathfrak{b}))$ is surjective). So (ii) of Lemma C holds, Lemma B applies, and $F = \mathcal{L}(C_0 + \mathfrak{b}f)$ is very ample. (We used only that $\mathfrak{b}, \mathfrak{b}+\mathfrak{e}$ are very ample and $\mathfrak{b}$ is nonspecial.) □

Exercise 2.12

Problem. Let X be a ruled surface with invariant e over an elliptic curve C, and let $\mathfrak{b}$ be a divisor on C.

(a) If $\deg \mathfrak{b} \geq e + 2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.

(b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geq e + 3$.

Note. The case $e = -1$ will require special attention.

Solution. We use Lemmas A, B, C of Exercise 2.11. Here $g = 1$, and by (2.12), (2.15), $e \geq -1$; $\deg \mathfrak{e} = -e$; on an elliptic curve a divisor class of degree > 0 is nonspecial and a divisor of degree ≥ 2 (resp. ≥ 3) has no base points (resp. is very ample) (IV, 3.2). Also $h^0(\mathcal{L}(\mathfrak{c})) = \deg \mathfrak{c}$ for $\deg \mathfrak{c} > 0$. We need the following fact for $e = -1$: then $\mathcal{E}$ is the nonsplit extension $0 \to \mathcal{O} \to \mathcal{E} \to \mathcal{L}(P) \to 0$ of (2.15) ($\mathfrak{e} \sim P$), and

$$H^1(\mathcal{E}) = 0. \tag{2}$$

Indeed, $H^0(\mathcal{L}(P)) \to H^1(\mathcal{O}) \to H^1(\mathcal{E}) \to H^1(\mathcal{L}(P)) = 0$, and the connecting map sends 1_P to the image of $\xi \neq 0$ under $H^1(\mathcal{L}(-P)) \to H^1(\mathcal{O})$, which is injective (see 2.7), hence an isomorphism of one-dimensional spaces.

(a) Let $\deg \mathfrak{b} \geq e + 2$. Then $\deg \mathfrak{b} \geq 1$ (as $e \geq -1$), so $H^1(\mathcal{L}(\mathfrak{b})) = 0$ and $H^0(\mathcal{L}(\mathfrak{b})) \neq 0$; and $\deg(\mathfrak{b}+\mathfrak{e}) = \deg \mathfrak{b} - e \geq 2$, so $|\mathfrak{b}+\mathfrak{e}|$ has no base points. By Lemma A there is a section $D \sim C_0 + \mathfrak{b}f$. For base points of $|D| = |C_0 + \mathfrak{b}f|$ we apply the last part of Lemma C: we show $H^1(\mathcal{G}(-y)) = 0$ for every y, where $\mathcal{G} = \mathcal{E}(\mathfrak{b})$. From $0 \to \mathcal{L}(\mathfrak{b}-y) \to \mathcal{G}(-y) \to \mathcal{L}(\mathfrak{b}+\mathfrak{e}-y) \to 0$: $\deg(\mathfrak{b}+\mathfrak{e}-y) = \deg \mathfrak{b} - e - 1 \geq 1$, so its H^1 vanishes. And $\deg(\mathfrak{b}-y) = \deg \mathfrak{b} - 1 \geq e + 1$. If $e \geq 0$ this is ≥ 1 and $H^1(\mathcal{L}(\mathfrak{b}-y)) = 0$. If $e = -1$, then $\deg(\mathfrak{b}-y) \geq 0$ and $H^1(\mathcal{L}(\mathfrak{b}-y)) \neq 0$ only if $\mathfrak{b} - y \sim 0$, in which case $\mathcal{G}(-y) \cong \mathcal{E}$ and $H^1 = 0$ by (2). So in all cases $H^1(\mathcal{G}(-y)) = 0$, $\mathcal{G}$ is generated by global sections, and $|D|$ has no base points.

(b) ($\Leftarrow$) Let $\deg \mathfrak{b} \geq e + 3$ and let Z' be an effective divisor of degree 2. Using the sequence $0 \to \mathcal{L}(\mathfrak{b}-Z') \to \mathcal{G}(-Z') \to \mathcal{L}(\mathfrak{b}+\mathfrak{e}-Z') \to 0$: $\deg(\mathfrak{b}+\mathfrak{e}-Z') = \deg \mathfrak{b} - e - 2 \geq 1$ gives $H^1 = 0$. And $\deg(\mathfrak{b}-Z') = \deg \mathfrak{b} - 2 \geq e + 1$: for $e \geq 0$ this is ≥ 1, so $H^1(\mathcal{L}(\mathfrak{b}-Z')) = 0$; for $e = -1$

it is ≥ 0, nonvanishing of H^1 requires $\mathfrak{b} - Z' \sim 0$, and then $\mathcal{G}(-Z') \cong \mathcal{E}$ has $H^1 = 0$ by (2). So $H^1(\mathcal{G}(-Z')) = 0$ for all Z', and by Lemma C(i) and Lemma B, $|C_0 + \mathfrak{b}f|$ is very ample. (Note that for $e \geq 0$ one could also use 2.11(b), but for $e = -1$, $\deg \mathfrak{b} = 2$ the divisor $\mathfrak{b}$ is not very ample, so that exercise does not apply; the cohomological criterion handles it.)

($\Rightarrow$) Suppose $D = C_0 + \mathfrak{b}f$ is very ample. Then $D|_{C_0}$ is a very ample divisor on $C_0 \cong C$ (restriction of a very ample sheaf to a closed subvariety), of degree $D.C_0 = C_0^2 + \deg \mathfrak{b} = \deg \mathfrak{b} - e$ (by (1.3)). A very ample sheaf $\mathcal{M}$ of degree $\delta > 0$ on an elliptic curve has $h^0(\mathcal{M}) = \delta$ and embeds C_0 in $\mathbf{P}^{\delta-1}$; for $\delta \leq 2$ this is $\mathbf{P}^0$ or $\mathbf{P}^1$, which contain no curve of genus 1. Hence $\deg \mathfrak{b} - e \geq 3$, i.e. $\deg \mathfrak{b} \geq e + 3$. □

Exercise 2.13

Problem. For every $e \geq -1$ and $n \geq e + 3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbf{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbf{P}^4$.

Solution. Let C be an elliptic curve. For each $e \geq -1$ there is a normalized $\mathcal{E}$ on C with invariant e (for $e = -1$ the indecomposable bundle of (2.15), for $e = 0$, $\mathcal{O} \oplus \mathcal{O}$, and for $e \geq 1$, $\mathcal{O} \oplus \mathcal{L}$ with $\deg \mathcal{L} = -e$ by (2.12)(a), (2.11.1), (2.11.3)); let $X = \mathbf{P}(\mathcal{E})$. Choose a divisor $\mathfrak{b}$ on C of degree n and put $D = C_0 + \mathfrak{b}f$. By Ex. 2.12(b), D is very ample as $n \geq e + 3$; so $|D|$ embeds X in $\mathbf{P}^N$, $N + 1 = h^0(X, \mathcal{L}(D)) = h^0(C, \mathcal{E}(\mathfrak{b}))$. We have $D.f = 1$, so each fibre is mapped to a line: X is a scroll. Its degree is $D^2 = C_0^2 + 2\deg \mathfrak{b} = -e + 2n = d$. Finally $\mathcal{G} = \mathcal{E}(\mathfrak{b})$ has rank 2 and degree $-e + 2n = d$ and (as in the proof of (2.12)) $H^1(\mathcal{G}) = 0$ (from (1): $\deg \mathfrak{b} > 0$ and $\deg(\mathfrak{b} + \mathfrak{e}) = n - e > 0$), so by Riemann–Roch $h^0(\mathcal{G}) = \deg \mathcal{G} + 2(1 - g) = d$. Thus $N = d - 1$ and $X \subseteq \mathbf{P}^{d-1}$ is an elliptic scroll of degree $d = 2n - e$.

For $e = -1$, $n = 2$ (allowed: $n \geq e + 3 = 2$) we get $d = 5$: an elliptic scroll of degree 5 in $\mathbf{P}^4$. □

Exercise 2.14

Problem. Let X be a ruled surface over a curve C of genus g, with invariant $e < 0$, and assume that $\operatorname{char} k = p > 0$ and $g \geq 2$.

(a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0, f$, then either $a = 1$, $b \geq 0$, or $2 \leq a \leq p-1$, $b \geq \frac{1}{2}ae$, or $a \geq p$, $b \geq \frac{1}{2}ae + 1 - g$.

(b) If $a > 0$ and $b > a\big(\frac{1}{2}e + (1/p)(g-1)\big)$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

Solution. We use $C_0^2 = -e$, $C_0.f = 1$, $f^2 = 0$, $K \equiv -2C_0 + (2g - 2 - e)f$ (2.11), and $\operatorname{Num} X = \mathbf{Z}C_0 \oplus \mathbf{Z}f$.

(a) Since $Y \neq f$ and $\pi|_Y$ is surjective, $Y.f = a > 0$.

Case $a = 1$. Then $\pi|_Y : Y \to C$ has degree $Y.f = 1$, so it is a finite birational morphism onto the nonsingular curve C, hence an isomorphism: Y is a section, corresponding to a surjection $\mathcal{E} \to \mathcal{M} \to 0$ with kernel $\mathcal{N}$ invertible, $\deg \mathcal{N} + \deg \mathcal{M} = \deg \mathcal{E} = -e$. By normalization $\deg \mathcal{N} \leq 0$ (otherwise $\mathcal{N} \otimes \mathcal{N}^{-1} = \mathcal{O} \subseteq \mathcal{E} \otimes \mathcal{N}^{-1}$ contradicts $H^0(\mathcal{E} \otimes \mathcal{N}^{-1}) = 0$). So $\deg \mathcal{M} \geq -e$. By (2.9) $\deg \mathcal{M} = C_0.Y = -e + b$. Hence $b \geq 0$.

Case $a \geq 2$. By adjunction (1.5), $2p_a(Y) - 2 = Y.(Y + K)$, and

$$Y^2 = -a^2e+2ab, \qquad Y.K = ae+a(2g-2)-2b, \qquad \text{so} \qquad 2p_a(Y)-2 = (a-1)(2b-ae)+a(2g-2). \tag{3}$$

Let $\tilde{Y} \to Y$ be the normalization and $\varphi\colon \tilde{Y} \to C$ the finite morphism of degree a induced by π. Then $p_a(Y) \geq g(\tilde{Y})$ (IV, Ex. 1.8).

If $a \leq p-1$, the extension $K(\tilde{Y})/K(C)$ has degree $a < p$, so its inseparable degree (a power of p dividing a) is 1: φ is separable and by Hurwitz (IV, 2.4) $2g(\tilde{Y})-2 = a(2g-2)+\deg R \geq a(2g-2)$. Hence $2p_a(Y) - 2 \geq a(2g-2)$ and by (3) $(a-1)(2b-ae) \geq 0$, so $b \geq \frac{1}{2}ae$ (since $a \geq 2$).

In any case (in particular for $a \geq p$) $g(\tilde{Y}) \geq g$ by (IV, 2.5.4), so $2p_a(Y) - 2 \geq 2g - 2$, and (3) gives $(a-1)(2b-ae) \geq -(a-1)(2g-2)$, i.e. $2b - ae \geq -(2g-2)$, i.e. $b \geq \frac{1}{2}ae + 1 - g$. This proves (a).

(b) Suppose $a > 0$ and $b > a\left(\frac{e}{2} + \frac{g-1}{p}\right)$. Then $b - \frac{1}{2}ae > a(g-1)/p > 0$ (as $g \geq 2$). We apply Nakai's criterion (1.10) to $D \equiv aC_0 + bf$. First $D^2 = a(2b-ae) > 0$. Next, $D.f = a > 0$ and $D.C_0 = b - ae = (b - \frac{1}{2}ae) + \frac{1}{2}a(-e) > 0$ since $e < 0$. Let $Y \equiv a'C_0 + b'f$ be an irreducible curve $\neq C_0, f$; then $D.Y = ab' + a'b - aa'e$ and, by (a):

- if $a' = 1$, $b' \geq 0$: $D.Y \geq b - ae > 0$;
- if $2 \leq a' \leq p-1$, $b' \geq \frac{1}{2}a'e$: $D.Y \geq \frac{1}{2}aa'e + a'b - aa'e = a'(b - \frac{1}{2}ae) > 0$;
- if $a' \geq p$, $b' \geq \frac{1}{2}a'e + 1 - g$: $D.Y \geq a'(b - \frac{1}{2}ae) - a(g-1) > a'a\frac{g-1}{p} - a(g-1) = a(g-1)(\frac{a'}{p} - 1) \geq 0$.

So $D.Y > 0$ for all irreducible curves Y, and D is ample by (1.10).

Conversely, if $D \equiv aC_0 + bf$ is ample then $D.f = a > 0$ and $D^2 = a(2b-ae) > 0$, so $2b > ae$, i.e. $b > \frac{1}{2}ae$. □

Exercise 2.15

Problem. *Funny behavior in characteristic p.* Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3 (IV, Ex. 2.4).

(a) Show that the action of the k-linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0 (Cf. (IV, 4.21)).

(b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(\mathcal{L}(-P))$ such that $f^*\xi = 0$ in $H^1(\mathcal{L}(-3P))$.

(c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \to \mathcal{O}_C \to \mathcal{E} \to \mathcal{L}(P) \to 0,$$

and let X be the corresponding ruled surface over C. Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi\colon Y \to C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypotheses of (2.21b), but is not ample.

Solution. The curve C is nonsingular: the partial derivatives of $F = x^3y + y^3z + z^3x$ in characteristic 3 are z^3, x^3, y^3, which have no common zero in $\mathbf{P}^2$. So C is a plane quartic and $g = 3$. The coefficients of F lie in $\mathbf{F}_3$.

Frobenius. Let $F_{\text{abs}}\colon C \to C$ be the absolute Frobenius morphism (identity on points, $u \mapsto u^3$ on functions; IV, Sect. 2). For any invertible sheaf $\mathcal{M}$ it induces $F_{\text{abs}}^*\colon H^1(C, \mathcal{M}) \to H^1(C, F_{\text{abs}}^*\mathcal{M}) = H^1(C, \mathcal{M}^{\otimes 3})$, computed on Cech cocycles $(m_{ij}) \mapsto (m_{ij}^{\otimes 3})$; it is additive and 3-semilinear ($\lambda \mapsto \lambda^3$). The k-linear Frobenius f of the exercise (IV, 2.4.1) is $(x : y : z) \mapsto (x^3 : y^3 : z^3)$ (as C is defined over $\mathbf{F}_3$, $C_p \cong C$); it differs from F_{abs} by the semilinear automorphism of $C = C_0 \times_{\mathbf{F}_3} k$ induced by $\lambda \mapsto \lambda^3$ on k, so that f^* and F_{abs}^* agree on classes defined over $\mathbf{F}_3$ and differ by a bijective semilinear map in general; in particular $f^* = 0$ *iff* $F_{\text{abs}}^* = 0$, and similarly for the maps in (b). We write f^* for either and compute with F_{abs}.

(a) From $0 \to \mathcal{O}_{\mathbf{P}^2}(-4) \xrightarrow{F} \mathcal{O}_{\mathbf{P}^2} \to \mathcal{O}_C \to 0$ and $H^1(\mathcal{O}_{\mathbf{P}^2}) = H^2(\mathcal{O}_{\mathbf{P}^2}) = 0$ we get an isomorphism $\delta\colon H^1(\mathcal{O}_C) \to H^2(\mathbf{P}^2, \mathcal{O}(-4))$, and by (III, 5.1) the latter has basis the Cech classes of the Laurent monomials $x^{-i}y^{-j}z^{-k}$ with $i, j, k \geq 1$, $i + j + k = 4$, i.e. $c_1 = x^{-2}y^{-1}z^{-1}$, $c_2 = x^{-1}y^{-2}z^{-1}$, $c_3 = x^{-1}y^{-1}z^{-2}$ (a monomial with some nonnegative exponent is zero in H^2).

Frobenius in terms of δ. Let $u = (u_{ij})$ be a Cech 1-cocycle of $\mathcal{O}_C$ and $\tilde{u}_{ij}$ lifts to $\mathcal{O}_{\mathbf{P}^2}$. Then $\delta[u]$ is represented by the 2-cocycle $\partial\tilde{u}/F$, where ∂ is the Cech coboundary (which takes values in $F \cdot \mathcal{O}(-4)$). The cocycle u^3 has the lifts $\tilde{u}_{ij}^3$ and $\partial(\tilde{u}^3) = (\partial\tilde{u})^3$ (characteristic 3), so $\delta[u^3]$ is represented by $(\partial\tilde{u})^3/F = F^2 \cdot (\partial\tilde{u}/F)^3$. Hence, under δ, F_{abs}^* corresponds to the map $c \mapsto F^2c^3$ on $H^2(\mathbf{P}^2, \mathcal{O}(-4))$ (degrees: $8 - 12 = -4$).

Now $F^2 = x^6y^2 + y^6z^2 + z^6x^2 + 2x^3y^4z + 2xy^3z^4 + 2x^4yz^3$. For $c = c_1 = x^{-2}y^{-1}z^{-1}$, $c^3 = x^{-6}y^{-3}z^{-3}$, and multiplying out, each term of F^2c^3 has a nonnegative exponent: $y^{-1}z^{-3}$, $x^{-6}y^3z^{-1}$, $x^{-4}y^{-3}z^3$, $2x^{-3}y^1z^{-2}$, $2x^{-5}y^0z^1$, $2x^{-2}y^{-2}z^0$. All vanish in H^2. Since F is invariant under the cyclic permutation $x \to y \to z \to x$, which permutes c_1, c_2, c_3, also $F^2c_i^3 = 0$ for $i = 2, 3$. As F_{abs}^* is semilinear and kills a basis, it is zero. This proves (a).

(b) Let $\mathcal{I} = \mathcal{L}(-P) \subseteq \mathcal{O}_C$ be the ideal of P. The cube map $\mathcal{I} \to \mathcal{I}^{\otimes 3} \cong \mathcal{L}(-3P) \subseteq \mathcal{O}_C$, $u \mapsto u^3$, induces the map $f^*\colon H^1(\mathcal{L}(-P)) \to H^1(\mathcal{L}(-3P))$ on cohomology. Consider the diagram with the maps $\iota_1\colon H^1(\mathcal{L}(-P)) \to H^1(\mathcal{O}_C)$, $\iota_3\colon H^1(\mathcal{L}(-3P)) \to H^1(\mathcal{O}_C)$ induced by the inclusions; since cubing commutes with inclusions, $\iota_3 \circ f^* = f^* \circ \iota_1 = 0$ by (a). So f^* maps $H^1(\mathcal{L}(-P))$ into $\ker \iota_3$. Now compute dimensions. From $0 \to \mathcal{L}(-P) \to \mathcal{O} \to \mathcal{O}_P \to 0$: $H^0(\mathcal{O}) \to H^0(\mathcal{O}_P)$ is onto, so $H^1(\mathcal{L}(-P)) \cong H^1(\mathcal{O}_C)$ has dimension 3. From $0 \to \mathcal{L}(-3P) \to \mathcal{O} \to \mathcal{O}_{3P} \to 0$, with $\dim H^0(\mathcal{O}_{3P}) = 3$ and $H^0(\mathcal{L}(-3P)) = 0$: the cokernel of $H^0(\mathcal{O}) = k \to H^0(\mathcal{O}_{3P})$ has dimension 2 and equals $\ker \iota_3$. Thus $f^*\colon H^1(\mathcal{L}(-P))$ ($\dim 3$) $\to \ker \iota_3$ ($\dim 2$) is an additive 3-semilinear map. Choose a basis ξ_1, ξ_2, ξ_3 of $H^1(\mathcal{L}(-P))$ and put $w_i = f^*\xi_i \in \ker \iota_3$. For $\xi = \sum \lambda_i\xi_i$ we have $f^*\xi = \sum \lambda_i^3 w_i$. Since $\dim \ker \iota_3 = 2 < 3$, the w_i are linearly dependent: there are $\mu_i \in k$, not all 0, with $\sum \mu_i w_i = 0$. As $\lambda \mapsto \lambda^3$ is bijective on k, we may write $\mu_i = \lambda_i^3$; then $\xi = \sum \lambda_i\xi_i \neq 0$ and $f^*\xi = 0$.

(c) Let $\mathcal{E}$ be the extension defined by this $\xi \in \text{Ext}^1(\mathcal{L}(P), \mathcal{O}) = H^1(\mathcal{L}(-P))$, $X = \mathbf{P}(\mathcal{E})$, $\pi\colon X \to C$.

$\mathcal{E}$ is normalized with $e = -1$: $H^0(\mathcal{E}) \neq 0$; for $\mathcal{M}$ of negative degree, $H^0(\mathcal{M}) = 0$ and $H^0(\mathcal{L}(P) \otimes \mathcal{M}) = 0$ unless $\mathcal{M} \cong \mathcal{L}(-P)$; and $H^0(\mathcal{E}(-P)) = 0$ because $H^0(\mathcal{E}(-P)) \hookrightarrow \ker(H^0(\mathcal{O}) \to H^1(\mathcal{L}(-P)), 1 \mapsto \xi) = 0$. So $\mathcal{E}$ is normalized, $\deg \mathcal{E} = 1$, $e = -1$, $C_0^2 = 1$.

The curve Y. Apply F_{abs}^* (flat, as C is a nonsingular curve) to the extension: $0 \to \mathcal{O} \to$

$F^*_{\mathrm{abs}}\mathcal{E} \to \mathcal{L}(3P) \to 0$, whose class is $f^*\xi = 0 \in H^1(\mathcal{L}(-3P))$ (a class given by transition data (u_{ij}) becomes (u_{ij}^3)). Thus $F^*_{\mathrm{abs}}\mathcal{E} \cong \mathcal{O} \oplus \mathcal{L}(3P)$ and in particular there is a surjection $F^*_{\mathrm{abs}}\mathcal{E} \to \mathcal{O}$ (a retraction of $\mathcal{O} \to F^*\mathcal{E}$). By (II, 7.12) this defines a morphism $\nu\colon C \to X$ with $\pi \circ \nu = F_{\mathrm{abs}}$ and $\nu^*\mathcal{O}_X(1) \cong \mathcal{O}_C$. Let $Y = \nu(C)$ (irreducible, closed) and $\Gamma = \nu_*[C] = m[Y]$ with $m = \deg(C \to Y)$. Numerically $\Gamma \equiv a'C_0 + b'f$ where, by (1.3) and the projection formula, $a' = \Gamma.f = \deg \nu^*\pi^*\mathcal{O}(y) = \deg F^*_{\mathrm{abs}}\mathcal{O}(y) = 3$ and $\Gamma.C_0 = \deg \nu^*\mathcal{O}_X(C_0) = \deg \nu^*\mathcal{O}_X(1) = 0$ (here $\mathcal{L}(C_0) \cong \mathcal{O}_X(1)$, (2.8)). With $C_0^2 = 1$: $0 = \Gamma.C_0 = 3 + b'$, so $b' = -3$: $\Gamma \equiv 3C_0 - 3f$.

Now $3 = \deg F_{\mathrm{abs}} = m \cdot \deg(Y \to C)$. If $m = 3$, then $Y \to C$ has degree 1, Y is a section with $3Y \equiv 3C_0 - 3f$, so $Y \equiv C_0 - f$ and by (2.9) $\deg \mathcal{M} = Y.C_0 = 1 - 1 = 0$ for the corresponding quotient $\mathcal{E} \to \mathcal{M}$, whose kernel then has degree $1 > 0$, contradicting normalization (as above). So $m = 1$: $C \to Y$ is birational, $Y \equiv 3C_0 - 3f$, $\pi|_Y\colon Y \to C$ has degree 3 and $K(Y) = K(C)$ (the source) contains $K(C)^3$ (the target), so it is purely inseparable of degree 3.

Remark on "nonsingular". The book says Y is nonsingular; this cannot be right. By (3) of Exercise 2.14 (valid in any characteristic) with $a = 3$, $b = -3$, $e = -1$, $g = 3$: $2p_a(Y) - 2 = 2(-6 + 3) + 3 \cdot 4 = 6$, so $p_a(Y) = 4$, while Y is birational to C, of genus 3. (Equivalently, $d\nu\colon T_C \to \nu^*T_{X/C} = (\nu^*\Omega_{X/C})^\vee$ is a nonzero map $\mathcal{O}(-K) \to \mathcal{O}(-3P)$ between sheaves of degrees -4 and -3 and so vanishes at one point: ν is not an immersion there.) So Y has one (unibranch) singular point, with δ-invariant 1. In fact no nonsingular curve Y' with $\pi|_{Y'}$ purely inseparable of degree 3 can exist at all: then $K(Y')^3 = K(C)$ (as $[K(Y') : K(Y')^3] = 3$), so Y' is isomorphic to C up to the Frobenius twist of k and has genus 3, whereas any curve $\equiv 3C_0 - 3f$ has $p_a = 4$. So the word "nonsingular" in the printed statement has to be dropped. This does not affect what is needed below: only the irreducible curve $Y \equiv 3C_0 - 3f$ with $\pi|_Y$ purely inseparable is used.

$D = 2C_0$. The hypotheses of (2.21b) are that $D \equiv aC_0 + bf$ has $a > 0$ and $b > \frac{1}{2}ae$ (the other hypothesis there, $\operatorname{char} k = 0$ or $g \le 1$, fails here, which is the point). For $D = 2C_0$: $a = 2$, $b = 0 > \frac{1}{2} \cdot 2 \cdot (-1) = -1$. So the numerical condition of (2.21b) holds. But $D.Y = 2C_0.(3C_0 - 3f) = 6C_0^2 - 6C_0.f = 6 - 6 = 0$, so D is not ample ($D.Y > 0$ is necessary for ampleness; (1.10) or the definition via the degree of Y in a projective embedding). Thus Prop. 2.21(b) fails in characteristic p (the case $\operatorname{char} k = p > 0$, $g \ge 2$, $e < 0$, cf. Ex. 2.14). □

Exercise 2.16

> **Problem.** Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(C)$ (II, Ex. 6.10) and (II, Ex. 6.11) are the same. This is false for a projective curve.

Solution. Here $K(C)$ is the quotient of the free abelian group on isomorphism classes of locally free sheaves by the relations $[\mathcal{E}] = [\mathcal{E}'] + [\mathcal{E}'']$ for exact sequences $0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{E}'' \to 0$ (for regular C this agrees with the group generated by coherent sheaves, II, Ex. 6.9–6.11; we only use that rank and det are defined on locally free sheaves).

($\Rightarrow$) is trivial. For ($\Leftarrow$): for an exact sequence of locally free sheaves, $\det \mathcal{E} \cong \det \mathcal{E}' \otimes \det \mathcal{E}''$, and $\det\colon \{\text{loc. free}\} \to \operatorname{Pic} C$ therefore induces a group homomorphism $\det\colon K(C) \to \operatorname{Pic} C$. If $[\mathcal{E}] = [\mathcal{E}']$ then $\det \mathcal{E} \cong \det \mathcal{E}'$. So it suffices to prove:

Claim. If $C = \operatorname{Spec} A$ is a nonsingular affine curve and $\mathcal{E}$ is locally free of rank $r \ge 1$, then $\mathcal{E} \cong \mathcal{O}_C^{r-1} \oplus \det \mathcal{E}$.

Then, if the ranks agree and the determinants agree, $\mathcal{E} \cong \mathcal{O}^{r-1} \oplus \det \mathcal{E} \cong \mathcal{O}^{r-1} \oplus \det \mathcal{E}' \cong \mathcal{E}'$.

Proof of the Claim. Induct on r; $r = 1$ is trivial. Let $r \geq 2$. $\mathcal{E}$ is generated by finitely many sections; let $V \subseteq H^0(\mathcal{E})$ be a finite-dimensional subspace generating $\mathcal{E}$. As in 2.3(a) (with C affine): $Z = \{(s, y) \in V \times C : s(y) = 0\}$ is a vector bundle of rank $\dim V - r$ over C of dimension $\dim V - r + 1 \leq \dim V - 1$, so there is $s \in V$ not vanishing at any point; $s\colon \mathcal{O} \to \mathcal{E}$ is then injective with locally free cokernel $\mathcal{E}_1$ of rank $r - 1$. Since C is affine, $\operatorname{Ext}^1(\mathcal{E}_1, \mathcal{O}) = H^1(C, \mathcal{E}_1^\vee) = 0$, so the sequence splits: $\mathcal{E} \cong \mathcal{O} \oplus \mathcal{E}_1$, and $\det \mathcal{E} \cong \det \mathcal{E}_1$. By induction $\mathcal{E}_1 \cong \mathcal{O}^{r-2} \oplus \det \mathcal{E}_1$. (Claim)

Failure for projective curves. On $C = \mathbf{P}^1$ the Euler sequence $0 \to \mathcal{O}(-1) \to \mathcal{O}^2 \to \mathcal{O}(1) \to 0$ gives $[\mathcal{O}(-1)] + [\mathcal{O}(1)] = 2[\mathcal{O}] = [\mathcal{O} \oplus \mathcal{O}]$ in $K(\mathbf{P}^1)$, i.e. $[\mathcal{O}(1) \oplus \mathcal{O}(-1)] = [\mathcal{O} \oplus \mathcal{O}]$ (the class of a direct sum is the sum of the classes, by the split exact sequence). These two sheaves have the same rank 2 and the same class, but they are not isomorphic: $H^0\big((\mathcal{O}(1) \oplus \mathcal{O}(-1)) \otimes \mathcal{O}(-1)\big) = H^0(\mathcal{O} \oplus \mathcal{O}(-2))$ has dimension 1, whereas $H^0(\mathcal{O}^2 \otimes \mathcal{O}(-1)) = 0$. □

Exercise 2.17

Problem.

(a) Let $\varphi\colon \mathbf{P}^1_k \to \mathbf{P}^3_k$ be the 3-uple embedding (I, Ex. 2.12). Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ. Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C, so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbf{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbf{Z}$. Determine l and m.

(b) Repeat part (a) for the embedding $\varphi\colon \mathbf{P}^1 \to \mathbf{P}^3$ given by $x_0 = t^4$, $x_1 = t^3u$, $x_2 = tu^3$, $x_3 = u^4$, whose image is a nonsingular rational quartic curve. [*Answer:* If $\operatorname{char} k \neq 2$, then $l = m = -7$; if $\operatorname{char} k = 2$, then $l, m = -6, -8$.]

(Starred in the book.)

Solution. Let $\varphi = (\varphi_0, \ldots, \varphi_3)\colon \mathbf{P}^1 \to \mathbf{P}^3$ be given by forms of degree d in t, u (no common zero), which is an embedding with image C. We treat (a) ($d = 3$, $\varphi = (t^3, t^2u, tu^2, u^3)$) and (b) ($d = 4$) simultaneously, in every characteristic. Put $\mathcal{Q} = \varphi^*(\mathcal{I}/\mathcal{I}^2)$; this is locally free of rank 2.

Step 1: $\mathcal{Q}$ as a kernel. Since C is a nonsingular subvariety of $\mathbf{P}^3$, (II, 8.17) gives an exact sequence $0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{P}^3}|_C \to \Omega_C \to 0$. Pull back by $\varphi\colon \mathbf{P}^1 \xrightarrow{\sim} C$:

$$0 \to \mathcal{Q} \to \varphi^*\Omega_{\mathbf{P}^3} \to \Omega_{\mathbf{P}^1} \to 0.$$

By the Euler sequences (II, 8.13), $\Omega_{\mathbf{P}^3} = \ker\big(\mathcal{O}(-1)^4 \xrightarrow{(x_i)} \mathcal{O}\big)$, so $\varphi^*\Omega_{\mathbf{P}^3} = \ker(\mathcal{O}(-d)^4 \xrightarrow{(\varphi_i)} \mathcal{O})$ (the pull-back of the surjection stays surjective and the kernel is locally free), and $\Omega_{\mathbf{P}^1} = \ker(\mathcal{O}(-1)^2 \xrightarrow{(t,u)} \mathcal{O})$. The map $\varphi^*\Omega_{\mathbf{P}^3} \to \Omega_{\mathbf{P}^1}$ is induced (by functoriality of the Euler sequence for the ring map $x_i \mapsto \varphi_i$, $dx_i \mapsto d\varphi_i$) by

$$\mathcal{O}(-d)^4 \to \mathcal{O}(-1)^2, \qquad (g_i) \mapsto \Big(\sum_i g_i \frac{\partial \varphi_i}{\partial t},\ \sum_i g_i \frac{\partial \varphi_i}{\partial u}\Big).$$

Since $\Omega_{\mathbf{P}^1} \subseteq \mathcal{O}(-1)^2$ we get

$$\mathcal{Q} = \Big\{(g_i) \in \mathcal{O}(-d)^4 :\ \sum g_i\varphi_i = 0,\ \sum g_i\varphi_{i,t} = 0,\ \sum g_i\varphi_{i,u} = 0\Big\}.$$

Its degree is $\deg \varphi^*\Omega_{\mathbf{P}^3} - \deg \Omega_{\mathbf{P}^1} = -4d+2$. By (2.14), $\mathcal{Q} \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$, $l \geq m$, with $l+m = -4d+2$. Left exactness of H^0 gives, for every s,

$$H^0(\mathcal{Q}(s)) = \Big\{(g_i) \in H^0(\mathcal{O}(s-d))^4 : \sum g_i\varphi_i = \sum g_i\varphi_{i,t} = \sum g_i\varphi_{i,u} = 0\Big\}. \tag{4}$$

(a) $d=3$: $\varphi = (t^3, t^2u, tu^2, u^3)$, $l+m=-10$. Take $s=4$, so $g_i = a_it+b_iu$ are linear forms. With $\varphi_{,t} = (3t^2, 2tu, u^2, 0)$, $\varphi_{,u} = (0, t^2, 2tu, 3u^2)$ the three equations in (4) become, coefficientwise:

$$\begin{aligned}
\textstyle\sum g_i\varphi_i = 0: &\quad a_0=0,\ \ b_0+a_1=0,\ \ b_1+a_2=0,\ \ b_2+a_3=0,\ \ b_3=0;\\
\textstyle\sum g_i\varphi_{i,t} = 0: &\quad 3a_0=0,\ \ 3b_0+2a_1=0,\ \ 2b_1+a_2=0,\ \ b_2=0;\\
\textstyle\sum g_i\varphi_{i,u} = 0: &\quad a_1=0,\ \ b_1+2a_2=0,\ \ 2b_2+3a_3=0,\ \ 3b_3=0.
\end{aligned}$$

From these (valid in every characteristic): $a_0=b_3=0$, $a_1=0$, $b_2=0$; then $b_0=-a_1=0$, $a_3=-b_2=0$; and $b_1=-a_2$ with $2b_1+a_2=0$ gives $b_1=0=a_2$. So all coefficients vanish and $H^0(\mathcal{Q}(4))=0$. Hence $l+4<0$, $l\leq -5$, and with $l+m=-10$, $l\geq m$: $l=m=-5$.

Answer (a): $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(-5)\oplus\mathcal{O}(-5)$ (in all characteristics), i.e. the normal bundle of the twisted cubic is $\mathcal{O}(5)\oplus\mathcal{O}(5)$.

(b) $d=4$: $\varphi = (t^4, t^3u, tu^3, u^4)$, $l+m=-14$. Here $\varphi_{,t} = (4t^3, 3t^2u, u^3, 0)$, $\varphi_{,u} = (0, t^3, 3tu^2, 4u^3)$. Take $s=6$, so $g_i = a_it^2 + b_itu + c_iu^2$ (quadrics). The equations (4) give, coefficientwise:

$$\begin{aligned}
\textstyle\sum g_i\varphi_i = 0: &\quad a_0=0,\ b_0+a_1=0,\ c_0+b_1=0,\ c_1+a_2=0,\ b_2+a_3=0,\ c_2+b_3=0,\ c_3=0;\\
\textstyle\sum g_i\varphi_{i,t} = 0: &\quad 4a_0=0,\ 4b_0+3a_1=0,\ 4c_0+3b_1=0,\ 3c_1+a_2=0,\ b_2=0,\ c_2=0;\\
\textstyle\sum g_i\varphi_{i,u} = 0: &\quad a_1=0,\ b_1=0,\ c_1+3a_2=0,\ 3b_2+4a_3=0,\ 3c_2+4b_3=0,\ 4c_3=0.
\end{aligned}$$

(For instance $g_0\cdot 4t^3 + g_1\cdot 3t^2u + g_2u^3 = 0$ has the coefficients $4a_0$ of t^5, $4b_0+3a_1$ of t^4u, $4c_0+3b_1$ of t^3u^2, $3c_1+a_2$ of t^2u^3, b_2 of tu^4, c_2 of u^5; and similarly for the others.) In every characteristic: $a_0=c_3=0$, $b_2=c_2=0$, $a_1=b_1=0$; hence $b_0=-a_1=0$, $c_0=-b_1=0$, $a_3=-b_2=0$, $b_3=-c_2=0$. What remains are c_1, a_2 with

$$c_1+a_2=0, \qquad 3c_1+a_2=0, \qquad c_1+3a_2=0,$$

i.e. $a_2=-c_1$ and $2c_1=0$.

- If $\operatorname{char} k \neq 2$: $c_1=a_2=0$, all coefficients vanish, $H^0(\mathcal{Q}(6))=0$. Then $l+6<0$, $l\leq -7$, and with $l+m=-14$, $l\geq m$: $l=m=-7$.
- If $\operatorname{char} k = 2$: $c_1=-a_2$ is free, so $\dim H^0(\mathcal{Q}(6)) = 1$ (spanned by $(0,u^2,t^2,0)$: indeed $u^2\cdot t^3u + t^2\cdot tu^3 = 2t^3u^3 = 0$ and the derivative conditions hold since $4=0$ and $3c_1+a_2 = c_1+a_2 = 0$). Since $h^0(\mathcal{O}(l+6)) + h^0(\mathcal{O}(m+6)) = 1$ with $l+m=-14$, $l\geq m$, we must have $l+6=0$ and $m+6=-2$: $l=-6$, $m=-8$.

Answer (b): $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(-7)\oplus\mathcal{O}(-7)$ if $\operatorname{char} k \neq 2$, and $\cong \mathcal{O}(-6)\oplus\mathcal{O}(-8)$ if $\operatorname{char} k = 2$. □

3. Monoidal transformations

Conventions. As throughout Chapter V, k is algebraically closed and a *surface* is a nonsingular projective surface over k. For a monoidal transformation $\pi : \widetilde{X} \to X$ with centre P I write $E = \pi^{-1}(P)$ for the exceptional curve. I use freely the results of the text: $E \cong \mathbf{P}^1$, $E^2 = -1$ (Prop. 3.1); $\pi^*C.\pi^*D = C.D$, $\pi^*C.E = 0$ (Prop. 3.2); $\pi^*C = \widetilde{C} + \mu_P(C)E$ for an effective divisor C (Prop. 3.6), hence $\widetilde{C}.E = \mu_P(C)$ and $p_a(\widetilde{C}) = p_a(C) - \frac{1}{2}r(r-1)$ for $r = \mu_P(C)$ (Cor. 3.7); and the notion of infinitely near points (Section 3, before Example 3.9.2). Standard facts of commutative algebra beyond the text are named where they are used.

Exercise 3.1

> **Problem.** Let X be a nonsingular projective variety of any dimension, let Y be a nonsingular subvariety, and let $\pi : \widetilde{X} \to X$ be obtained by blowing up Y. Show that $p_a(\widetilde{X}) = p_a(X)$.

Solution. Let $n = \dim X$ and $r = \operatorname{codim}(Y, X)$. Recall (III, Ex. 5.3) that $p_a(X) = (-1)^n(\chi(\mathcal{O}_X) - 1)$, so it suffices to show $H^i(\widetilde{X}, \mathcal{O}_{\widetilde{X}}) \cong H^i(X, \mathcal{O}_X)$ for all i. If $r = 1$ the ideal of Y is invertible and π is an isomorphism, so assume $r \geq 2$. By (II, 8.24), $\widetilde{X}$ is a nonsingular projective variety of dimension n, $E = \pi^{-1}(Y)$ is a nonsingular divisor, $f = \pi|_E : E \to Y$ is a projective bundle with fibres $\mathbf{P}^{r-1}$, and the ideal $\mathcal{J}$ of E satisfies $\mathcal{J}/\mathcal{J}^2 \cong \mathcal{O}_E(1)$ (the tautological sheaf of the projective bundle), so that the conormal sheaf of E in $\widetilde{X}$ is $\mathcal{O}_E(1)$.

Step 1: $\pi_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$. The sheaf $\pi_*\mathcal{O}_{\widetilde{X}}$ is a coherent sheaf of $\mathcal{O}_X$-algebras (II, 5.20), contained in the constant sheaf $K(\widetilde{X}) = K(X)$ because π is birational. Hence it is a finite $\mathcal{O}_X$-algebra consisting of elements of $K(X)$, so each of its sections is integral over $\mathcal{O}_X$. As X is nonsingular, hence normal, $\pi_*\mathcal{O}_{\widetilde{X}} \subseteq \mathcal{O}_X$, and the other inclusion is clear.

Step 2: $R^i\pi_*\mathcal{O}_{\widetilde{X}} = 0$ *for* $i > 0$. These are coherent sheaves supported on Y (over $X - Y$ the map π is an isomorphism), so it suffices to show that the stalk at every closed point $y \in Y$ vanishes. By the theorem on formal functions (III, 11.1),

$$\left(R^i\pi_*\mathcal{O}_{\widetilde{X}}\right)^\wedge_y \cong \varprojlim_m H^i(\widetilde{X}_m, \mathcal{O}_{\widetilde{X}_m}),$$

where $\widetilde{X}_m$ is the closed subscheme of $\widetilde{X}$ defined by $\mathcal{I}^m$, $\mathcal{I} = \mathfrak{m}_y\mathcal{O}_{\widetilde{X}}$. The subscheme defined by $\mathcal{I}$ is the scheme-theoretic fibre $F = \pi^{-1}(y) = f^{-1}(y) \cong \mathbf{P}^{r-1}$, a nonsingular subvariety of the nonsingular variety $\widetilde{X}$. Hence its ideal is locally generated by a regular sequence and $\mathcal{I}^a/\mathcal{I}^{a+1} \cong S^a(\mathcal{I}/\mathcal{I}^2)$, where $\mathcal{I}/\mathcal{I}^2$ is the conormal bundle of F in $\widetilde{X}$. For the chain $F \subset E \subset \widetilde{X}$ of nonsingular varieties we have an exact sequence of conormal bundles

$$0 \to (\mathcal{J}/\mathcal{J}^2)|_F \to \mathcal{I}/\mathcal{I}^2 \to \mathcal{N}^\vee_{F/E} \to 0.$$

Here $(\mathcal{J}/\mathcal{J}^2)|_F = \mathcal{O}_{\mathbf{P}^{r-1}}(1)$, and $\mathcal{N}^\vee_{F/E}$ is free of rank $n-r$, because F is the fibre of the smooth morphism f over the reduced point y, so its ideal in E is generated by the pull-backs of $n-r$ local parameters of Y at y. A symmetric power of an extension has a filtration whose graded pieces are $S^a(\mathcal{O}(1)) \otimes S^b(\mathcal{O}^{n-r}) = \mathcal{O}_{\mathbf{P}^{r-1}}(a)^{\oplus c_{a,b}}$ with $a, b \geq 0$, $a + b$ equal to the power. Since $H^i(\mathbf{P}^{r-1}, \mathcal{O}(a)) = 0$ for $i > 0$, $a \geq 0$, we get $H^i(F, \mathcal{I}^a/\mathcal{I}^{a+1}) = 0$ for all $i > 0$, $a \geq 0$. The exact sequences $0 \to \mathcal{I}^m/\mathcal{I}^{m+1} \to \mathcal{O}_{\widetilde{X}_{m+1}} \to \mathcal{O}_{\widetilde{X}_m} \to 0$ and induction on m ($\widetilde{X}_1 = F$ is covered

by the case $a = 0$) give $H^i(\widetilde{X}_m, \mathcal{O}_{\widetilde{X}_m}) = 0$ for all $i > 0$ and all m. So the completion of the stalk $(R^i\pi_*\mathcal{O}_{\widetilde{X}})_y$ is 0; as this stalk is a finitely generated module over the local ring $\mathcal{O}_{X,y}$ and completion is faithful (Krull), the stalk is 0.

Step 3. By Steps 1, 2 and (III, Ex. 8.1) (the Leray spectral sequence degenerates), $H^i(\widetilde{X}, \mathcal{O}_{\widetilde{X}}) \cong H^i(X, \pi_*\mathcal{O}_{\widetilde{X}}) = H^i(X, \mathcal{O}_X)$ for all $i \geq 0$. Hence $\chi(\mathcal{O}_{\widetilde{X}}) = \chi(\mathcal{O}_X)$ and $p_a(\widetilde{X}) = p_a(X)$. (For $n = 2$ and Y a point this is Prop. 3.4 and Cor. 3.5.) □

Exercise 3.2

> **Problem.** Let C and D be curves on a surface X, meeting at a point P. Let $\pi : \widetilde{X} \to X$ be the monoidal transformation with center P. Show that $\widetilde{C}.\widetilde{D} = C.D - \mu_P(C) \cdot \mu_P(D)$. Conclude that $C.D = \sum \mu_P(C) \cdot \mu_P(D)$, where the sum is taken over all intersection points of C and D, including infinitely near intersection points.

Solution. *The formula.* Put $r = \mu_P(C)$, $s = \mu_P(D)$. By Prop. 3.6, $\pi^*C = \widetilde{C} + rE$ and $\pi^*D = \widetilde{D} + sE$, so $\widetilde{C} = \pi^*C - rE$, $\widetilde{D} = \pi^*D - sE$. By Prop. 3.2, $\pi^*C.\pi^*D = C.D$, $\pi^*C.E = \pi^*D.E = 0$ and $E^2 = -1$, hence

$$\widetilde{C}.\widetilde{D} = (\pi^*C - rE).(\pi^*D - sE) = C.D + rs\,E^2 = C.D - rs.$$

(Equivalently, $\widetilde{C}.E = r$, $\widetilde{D}.E = s$ by Cor. 3.7 and $\pi^*C.\pi^*D = (\widetilde{C} + rE).(\widetilde{D} + sE) = \widetilde{C}.\widetilde{D} + rs + rs - rs$.)

The consequence. Let C, D have no common component (otherwise $C.D$ is not a sum over finitely many intersection points); then $C \cap D$ is finite. Blow up the points of $C \cap D$ one after another. A monoidal transformation with centre P does not change the strict transforms of C and D near the other points of $C \cap D$ (it is an isomorphism away from P), so applying the formula repeatedly we get, with $C^{(1)}, D^{(1)}$ the strict transforms on the resulting surface $X^{(1)}$,

$$C^{(1)}.D^{(1)} = C.D - \sum_{P \in C \cap D} \mu_P(C)\mu_P(D).$$

The strict transforms $C^{(1)}, D^{(1)}$ still have no common component, so $C^{(1)}.D^{(1)} \geq 0$ (Prop. 1.4), and every term $\mu_P(C)\mu_P(D)$ is ≥ 1 because $P \in C \cap D$ (Remark 3.5.2). Hence $C^{(1)}.D^{(1)} < C.D$ as soon as $C \cap D \neq \emptyset$. Repeating with $C^{(1)}, D^{(1)}$ (their common points lie on the exceptional curves; they are the intersection points of C and D that are infinitely near, of the first order, to points of $C \cap D$), the nonnegative integer $C^{(j)}.D^{(j)}$ strictly decreases until $C^{(m)} \cap D^{(m)} = \emptyset$, at which stage $C^{(m)}.D^{(m)} = 0$ by Prop. 1.4. Summing the identities gives

$$C.D = \sum_Q \mu_Q(C)\,\mu_Q(D),$$

the sum over all (ordinary and infinitely near) points Q lying on both (strict transforms of) C and D, where μ_Q denotes the multiplicity of the strict transform at Q (the process is finite, so the sum is). □

Exercise 3.3

Problem. Let $\pi : \widetilde{X} \to X$ be a monoidal transformation, and let D be a very ample divisor on X. Show that $2\pi^*D - E$ is ample on $\widetilde{X}$. [*Hint*: Use a suitable generalization of (I, Ex. 7.5) to curves in $\mathbf{P}^n$.]

Solution. I use Nakai's criterion (Thm. 1.10): put $H = 2\pi^*D - E$; we must show $H^2 > 0$ and $H.\Gamma > 0$ for every irreducible curve $\Gamma \subseteq \widetilde{X}$.

Self-intersection. By Prop. 3.2, $H^2 = 4D^2 - 4\pi^*D.E + E^2 = 4D^2 - 1$. Since D is very ample, $D.C > 0$ for every curve C (Remark 1.6.2), and applying this with C an effective divisor in $|D|$ gives $D^2 > 0$, so $D^2 \geq 1$ and $H^2 \geq 3 > 0$.

Curves. $H.E = 2\pi^*D.E - E^2 = 1 > 0$. Let $\Gamma \neq E$ be an irreducible curve, $C = \pi(\Gamma)$, so $\Gamma = \widetilde{C}$ and $\pi^*C = \widetilde{C} + rE$ with $r = \mu_P(C) \geq 0$. Then

$$H.\widetilde{C} = 2\pi^*D.(\pi^*C - rE) - E.(\pi^*C - rE) = 2D.C - r.$$

Claim: $D.C \geq r$. If $r = 0$ this is clear ($D.C > 0$ since D is ample), so let $r \geq 1$, i.e. $P \in C$. Embed $X \subseteq \mathbf{P}^N$ by $|D|$. Hyperplanes through P cut out divisors $D' \in |D|$ with $P \in D'$, and there is such a hyperplane not containing C (the hyperplanes through P have intersection $\{P\}$, and C is a curve). Then D' and C have no common component, $\mu_P(D') \geq 1$, and by Ex. 3.2 (the first formula), applied to C and D',

$$0 \leq \widetilde{C}.\widetilde{D'} = C.D' - \mu_P(C)\mu_P(D') \leq D.C - r,$$

where the first inequality uses Prop. 1.4 for the strict transforms (they have no common component) and $C.D' = C.D$ since $D' \sim D$. This proves the claim (it is the generalization of I, Ex. 7.5: the degree of a curve in projective space is at least its multiplicity at any point).

So $H.\widetilde{C} = 2D.C - r \geq 2r - r$ if $r \geq 1$, i.e. $H.\widetilde{C} \geq r \geq 1$, and $H.\widetilde{C} = 2D.C > 0$ if $r = 0$. Thus $H.\Gamma > 0$ for all irreducible curves and H is ample. □

Exercise 3.4

Problem. *Multiplicity of a Local Ring.* (See Nagata [7, Ch III, Section 23] or Zariski–Samuel [1, vol 2, Ch VIII, Section 10].) Let A be a noetherian local ring with maximal ideal $\mathfrak{m}$. For any $l > 0$, let $\psi(l) = \text{length}(A/\mathfrak{m}^l)$. We call ψ the Hilbert–Samuel function of A.

(a) Show that there is a polynomial $P_A(z) \in \mathbf{Q}[z]$ such that $P_A(l) = \psi(l)$ for all $l \gg 0$. This is the Hilbert–Samuel polynomial of A. [*Hint*: Consider the graded ring $\text{gr}_{\mathfrak{m}} A = \bigoplus_{d \geq 0} \mathfrak{m}^d/\mathfrak{m}^{d+1}$, and apply (I, 7.5).]

(b) Show that $\deg P_A = \dim A$.

(c) Let $n = \dim A$. Then we define the *multiplicity* of A, denoted $\mu(A)$, to be $(n!) \cdot$ (leading coefficient of P_A). If P is a point on a noetherian scheme X, we define the *multiplicity* of P on X, $\mu_P(X)$, to be $\mu(\mathcal{O}_{P,X})$.

(d) Show that for a point P on a curve C on a surface X, this definition of $\mu_P(C)$ coincides with the one in the text just before (3.5.2).

(e) If Y is a variety of degree d in $\mathbf{P}^n$, show that the vertex of the cone over Y is a point of multiplicity d.

Solution. (a) Let $\kappa = A/\mathfrak{m}$ and let $x_1, \dots, x_s$ generate $\mathfrak{m}$. Then $G = \operatorname{gr}_{\mathfrak{m}} A$ is a quotient of the polynomial ring $S = \kappa[X_1, \dots, X_s]$ (map $X_i \mapsto$ the class of x_i in $\mathfrak{m}/\mathfrak{m}^2$), so G is a finitely generated graded S-module, and each $G_d = \mathfrak{m}^d/\mathfrak{m}^{d+1}$ is a finite-dimensional κ-vector space whose length as an A-module is $\dim_\kappa G_d$. By the Hilbert polynomial theorem (I, 7.5; its proof works over any field κ) there is $Q(z) \in \mathbf{Q}[z]$ with $\dim_\kappa G_d = Q(d)$ for $d \gg 0$. Since $\operatorname{length}(\mathfrak{m}^d/\mathfrak{m}^{d+1}) = \dim_\kappa G_d$ and lengths add in the filtration $A \supset \mathfrak{m} \supset \cdots \supset \mathfrak{m}^l$,

$$\psi(l) = \sum_{d=0}^{l-1} \dim_\kappa G_d.$$

Choose $R(z) \in \mathbf{Q}[z]$ with $R(z) - R(z-1) = Q(z-1)$ (a polynomial in the binomial basis $\binom{z}{j}$ can be summed termwise; $\deg R = \deg Q + 1$). For $l \geq l_0$ large, $\psi(l) - \psi(l-1) = Q(l-1) = R(l) - R(l-1)$, so $\psi(l) - R(l)$ is a constant c for $l \geq l_0$, and $P_A = R + c \in \mathbf{Q}[z]$ works.

(b) This is the dimension theorem for noetherian local rings: $\deg P_A = d(A) = \dim A$ (Atiyah–Macdonald, Thm. 11.14; Matsumura, Thm. 13.4). (The proof there shows $d(A) \leq \delta(A)$ – the least number of generators of an $\mathfrak{m}$-primary ideal – by comparing $\mathfrak{m}^l$ with the powers of such an ideal, and then $\dim A \leq d(A)$ by induction on $\dim A$, using the exact sequence $0 \to A/(\mathfrak{q} : x) \to A/\mathfrak{q} \to A/(\mathfrak{q} + xA) \to 0$; Krull's theorem gives $\delta(A) \leq \dim A$.)

(c) This is a definition. Note that $\mu(A)$ is a positive integer: P_A is an integer-valued polynomial of degree n (it is a sum of binomial coefficients $\binom{z}{j}$ with integral coefficients), so its leading coefficient is $c/n!$ with $c \in \mathbf{Z}$, and $c > 0$ because $P_A(l) \to \infty$.

(d) Let $A = \mathcal{O}_{P,X}$, a regular local ring of dimension 2 (as X is a surface) with maximal ideal $\mathfrak{m}$ and residue field k, and let $f \in A$ be a local equation of C, so $\mathcal{O}_{P,C} = B = A/fA$ (dimension 1 for $P \in C$). Let r be the largest integer with $f \in \mathfrak{m}^r$ (the definition in the text). Because A is regular, $\operatorname{gr}_{\mathfrak{m}} A \cong k[X, Y]$ is a domain. For $g \in A$, $g \neq 0$, let j be the integer with $g \in \mathfrak{m}^j \setminus \mathfrak{m}^{j+1}$ (it exists by Krull's intersection theorem). Then the initial forms multiply: $\operatorname{in}(fg) = \operatorname{in}(f)\operatorname{in}(g) \neq 0$, so $fg \in \mathfrak{m}^{r+j} \setminus \mathfrak{m}^{r+j+1}$. Consequently

$$(\mathfrak{m}^l : f) = \mathfrak{m}^{l-r} \qquad (l \geq r).$$

Now $\operatorname{length}(B/\mathfrak{m}_B^l) = \operatorname{length}(A/(\mathfrak{m}^l + fA)) = \operatorname{length}(A/\mathfrak{m}^l) - \operatorname{length}((fA + \mathfrak{m}^l)/\mathfrak{m}^l)$, and $(fA + \mathfrak{m}^l)/\mathfrak{m}^l \cong fA/(fA \cap \mathfrak{m}^l) \cong A/(\mathfrak{m}^l : f) = A/\mathfrak{m}^{l-r}$. Since $P_A(l) = \sum_{d<l}(d+1) = \binom{l+1}{2}$, we get for $l \geq r$

$$\psi_B(l) = \binom{l+1}{2} - \binom{l-r+1}{2} = rl - \binom{r}{2}.$$

This is a polynomial of degree $1 = \dim B$ with leading coefficient r, so $\mu(B) = 1! \cdot r = r$, as claimed.

(e) Let $Y \subseteq \mathbf{P}^n$ be a variety of dimension t and degree d, with homogeneous coordinate ring $R = S(Y) = k[x_0, \dots, x_n]/I(Y)$. Embed $\mathbf{P}^n$ as a hyperplane in $\mathbf{P}^{n+1}$ and let $V = (0 : \cdots : 0 : 1)$. The cone over Y with vertex V is the subscheme of $\mathbf{P}^{n+1}$ with homogeneous ideal $I(Y)k[x_0, \dots, x_{n+1}]$; in the affine chart $x_{n+1} = 1$ it is $\operatorname{Spec} R$ with V the origin, so $\mathcal{O}_V = A = R_{\mathfrak{m}}$, $\mathfrak{m} = R_+ = (x_0, \dots, x_n)$. As R is graded and generated by R_1 over $k = R_0$, we have $\mathfrak{m}^j/\mathfrak{m}^{j+1} = R_j$, and since $R/\mathfrak{m}^l$ has finite length and is supported at the origin, $\operatorname{length}(A/\mathfrak{m}^l A) = \dim_k R/\mathfrak{m}^l = \sum_{j<l} \dim_k R_j = \sum_{j<l} \varphi_Y(j)$. By (I, 7.5 and the definition of degree), $\varphi_Y(j) = P_Y(j)$ for $j \gg 0$, where P_Y has degree t and leading coefficient $d/t!$. Therefore $P_A(l) = \sum_{j<l} P_Y(j) + \text{const}$ has

degree $t+1 = \dim A$ and leading coefficient $\dfrac{d}{t!\,(t+1)} = \dfrac{d}{(t+1)!}$, so $\mu(A) = (t+1)! \cdot \dfrac{d}{(t+1)!} = d$.
□

Exercise 3.5

> **Problem.** Let $a_1, \ldots, a_r$, $r \geq 5$, be distinct elements of k, and let C be the curve in $\mathbf{P}^2$ given by the (affine) equation $y^2 = \prod_{i=1}^{r}(x - a_i)$. Show that the point P at infinity on the y-axis is a singular point. Compute δ_P, and $g(\widetilde{Y})$, where $\widetilde{Y}$ is the normalization of Y. Show in this way that one obtains hyperelliptic curves of every genus $g \geq 2$.

Solution. *Hypothesis.* I assume $\operatorname{char} k \neq 2$ (as is implicit: in characteristic 2 the function field $k(x, y)$, $y^2 = f(x)$, is purely inseparable over $k(x)$ and the curve has genus 0). Write $Y = C$ and $m = r - 2 \geq 3$.

The curve. Homogenizing, $C:\ y^2 z^{r-2} = \prod_{i=1}^{r}(x - a_i z)$, a plane curve of degree r; it is irreducible, since $f(x) = \prod(x - a_i)$ is squarefree of positive degree, hence not a square in $k(x)$, so $y^2 - f$ is irreducible in $k(x)[y]$. Thus $p_a(C) = \frac{1}{2}(r-1)(r-2)$ (Ex. 1.5.1). Its affine points are nonsingular: $\partial/\partial y = 2y$ and $\partial/\partial x = -f'(x)$ vanish simultaneously with $y^2 = f(x)$ only if $f(x) = f'(x) = 0$, impossible since f has simple roots. The only point at infinity is $P = (0 : 1 : 0)$ ($z = 0$ forces $x = 0$).

The point P. In the chart $y = 1$, with coordinates $(u, w) = (x, z)$, P is the origin and C is $F(u, w) = w^m - \prod_{i=1}^{r}(u - a_i w) = 0$. The second term is homogeneous of degree r and w^m has degree $m = r - 2 < r$, so $\mu_P(C) = m = r - 2 \geq 3 \geq 2$: P is a singular point, with tangent cone w^m.

Embedded resolution. Blow up P. In the chart $w = tu$ we have $F = u^m\big(t^m - u^2 U(t)\big)$ where $U(t) = \prod(1 - a_i t)$, $U(0) = 1$. In the other chart $u = sw$ one gets $w^m(1 - w^2 \prod(s - a_i))$, whose strict transform $1 - w^2 \prod(s - a_i)$ does not meet the exceptional curve $w = 0$. So the strict transform C_1 is $G_1 = t^m - u^2 U(t)$ near $Q_1 = (t, u) = (0, 0)$ and has multiplicity $\min(m, 2) = 2$ there (as $m \geq 3$), with tangent cone u^2. Blow up Q_1, in the chart $u = vt$ (the tangent direction): $G_1 = t^2(t^{m-2} - v^2 U(t))$; in the other chart $t = su$ one gets $u^2(s^m u^{m-2} - U(su))$, whose strict transform takes the value -1 on $u = 0$, so does not meet the exceptional curve. Inductively, the strict transform C_j is, near the only point $Q_j = (0, 0)$ of the new exceptional curve through which it passes with multiplicity ≥ 1, given by

$$G_j = t^{m_j} - v^2 U(t), \qquad m_j = m - 2(j - 1),$$

and its multiplicity at Q_j is $\min(m_j, 2)$ as long as $m_j \geq 1$ (with the same argument for the other chart). If $m_j \geq 3$ the tangent cone is v^2 and blowing up gives $G_{j+1} = t^{m_j - 2} - v^2 U$; if $m_j = 2$ then $G_j = t^2 - v^2 U(t)$ has two distinct tangents $t = \pm v$ (as $U(0) = 1$, char $\neq 2$) and one more blow-up separates the two branches into two nonsingular points; if $m_j = 1$ then G_j is nonsingular. Hence the singular points of C and its infinitely near points are P (multiplicity $r - 2$) and $Q_1, \ldots, Q_N$, each of multiplicity 2, where N is the number of $j \geq 1$ with $m - 2(j - 1) \geq 2$, i.e. $N = \lfloor m/2 \rfloor$. By Example 3.9.3 (and IV, Ex. 1.8)

$$\delta_P = \tfrac{1}{2} m(m-1) + \left\lfloor \tfrac{m}{2} \right\rfloor = \tfrac{1}{2}(r-2)(r-3) + \left\lfloor \frac{r-2}{2} \right\rfloor.$$

The genus. C has no other singular points, so by Example 3.9.2 the normalization $\widetilde{Y}$ (the strict transform in the last blown-up surface, a nonsingular projective curve birational to C) has

$$g(\widetilde{Y}) = \tfrac{1}{2}(r-1)(r-2) - \tfrac{1}{2}(r-2)(r-3) - \left\lfloor \frac{r-2}{2} \right\rfloor = (r-2) - \left\lfloor \frac{r-2}{2} \right\rfloor = \left\lfloor \frac{r-1}{2} \right\rfloor.$$

Hyperelliptic curves. $K(\widetilde{Y}) = k(x,y)$ is a quadratic extension of $k(x)$ (as $y^2 = f(x)$ is irreducible), so x defines a finite morphism $\widetilde{Y} \to \mathbf{P}^1$ of degree 2. For $r = 2g+1$ ($g \geq 2$, so $r \geq 5$) we obtain $g(\widetilde{Y}) = g$. Hence $\widetilde{Y}$ is hyperelliptic of genus g for every $g \geq 2$. (E.g. $r = 5, 6$ give $g = 2$; $r = 7, 8$ give $g = 3$.) □

Exercise 3.6

> **Problem.** Show that analytically isomorphic curve singularities (I, 5.6.1) are equivalent in the sense of (3.9.4), but not conversely.

Solution. *Conventions.* By (I, 5.6.1) points $P \in C \subset X$ and $P' \in C' \subset X'$ (curves on surfaces) are analytically isomorphic if $\widehat{\mathcal{O}}_{P,C} \cong \widehat{\mathcal{O}}_{P',C'}$ as k-algebras. For equivalence in the sense of (3.9.4) I use the following reading: the sequences of monoidal transformations resolving C and C' blow up, at each stage, corresponding singular points of the reduced total transforms, and the correspondence of components and singular points preserves incidence and multiplicities; in particular the multiplicity of the strict transform of C at each blown-up point, and the incidence graph of the total transform, are the same. (If both singular points are nonsingular points of the curves they are trivially equivalent, so assume P, P' singular.)

Analytically isomorphic ⇒ equivalent. Choose isomorphisms $\widehat{\mathcal{O}}_{P,X} \cong k[[x,y]]$, $\widehat{\mathcal{O}}_{P',X'} \cong k[[x',y']]$ (X, X' nonsingular surfaces, k algebraically closed, Cohen structure theorem); then $\widehat{\mathcal{O}}_{P,C} = k[[x,y]]/(f)$, $\widehat{\mathcal{O}}_{P',C'} = k[[x',y']]/(f')$ for local equations f, f' of C, C', and since P, P' are singular, $f, f' \in \mathfrak{m}^2$. Let $\theta : k[[x,y]]/(f) \to k[[x',y']]/(f')$ be an isomorphism. Lift $\theta(\bar{x}), \theta(\bar{y})$ to elements $\alpha(x), \alpha(y) \in \mathfrak{m}'$, giving a continuous local k-algebra homomorphism $\alpha : k[[x,y]] \to k[[x',y']]$ inducing θ. On $\mathfrak{m}/\mathfrak{m}^2 \to \mathfrak{m}'/\mathfrak{m}'^2$ it is an isomorphism ($f, f' \in \mathfrak{m}^2$, so these spaces are the cotangent spaces of the curves, and θ is an isomorphism), hence α is an isomorphism (formal inverse function theorem). The kernel of $k[[x,y]] \xrightarrow{\alpha} k[[x',y']] \to k[[x',y']]/(f')$ is (f), because this composite induces the injective map θ on $k[[x,y]]/(f)$ and kills f. So $\alpha^{-1}((f')) = (f)$, i.e. $\alpha(f) = u\,f'$ with u a unit.

Now the data entering the definition of equivalence are computed on the completions. Blowing up the maximal ideal $\mathfrak{m}$ of $k[[x,y]]$ gives, over the closed points Q of the exceptional curve $E = \mathbf{P}(\mathfrak{m}/\mathfrak{m}^2)$, local rings $\widehat{\mathcal{O}}_{\widetilde{X},Q} \cong k[[x_1,t_1]]$ described by the formulas $y = x_1 t_1$ or $x = y_1 t_1'$ (in terms of the generators x, y), the exceptional curve being $x_1 = 0$ and the strict transform of C being given by $f(x_1, x_1 t_1)/x_1^r$, $r =$ multiplicity. The isomorphism α induces a bijection $E \to E'$ ($\mathbf{P}(\mathfrak{m}/\mathfrak{m}^2) \cong \mathbf{P}(\mathfrak{m}'/\mathfrak{m}'^2)$) and, for corresponding points $Q \leftrightarrow Q'$, an isomorphism of the completed local rings of the blown-up surfaces carrying the ideal of E to that of E' and the ideal of the strict transform $\widetilde{C}$ to that of $\widetilde{C}'$ (the strict transform is the closure of the complement of E in the total transform $\alpha(f) = uf'$ and the exceptional divisor is divided out; both are intrinsic to the pair $(\widehat{\mathcal{O}}, (f))$). This gives equal multiplicities $\mu_P(C) = \mu_{P'}(C')$, a bijection of the points $Q \in \widetilde{C} \cap E$, and for every such point an isomorphism of the completions of the pairs (surface; total reduced transform). Repeating at each singular point of the reduced total transform, we

obtain by induction that the multiplicities of the strict transforms at the infinitely near points, the incidence of the components and the position of all singular points of the reduced total transforms correspond at every stage. In particular a resolution $U_n \to \cdots \to U_0$ of C by blowing up successively the singular points of the total transform induces the corresponding sequence $U'_n \to \cdots \to U'_0$ for C', which is also an embedded resolution (normal crossings is a property of the completed local rings), and the required correspondence exists. So $P \in C$ and $P' \in C'$ are equivalent.

Not conversely. Let $\lambda \in k$, $\lambda \neq 0, 1$, and let $C_\lambda \subset \mathbf{A}^2$ be the union of four lines through the origin, $f_\lambda = xy(x-y)(x-\lambda y) = 0$ (an ordinary quadruple point).

(i) *All C_λ are equivalent at* 0. Blow up the origin ($\mu = 4$): the strict transform consists of four disjoint smooth branches meeting the exceptional curve E transversally in four distinct points (the four tangent directions), so the reduced total transform has only nodes (normal crossings), and this single blow-up is an embedded resolution for every λ. Any bijection between the four branches (and the four intersection points with E) is a correspondence preserving incidence and multiplicities. Hence all C_λ are equivalent.

(ii) *C_λ and C_μ are analytically isomorphic only if μ lies in the S_3-orbit* $\{\lambda, \lambda^{-1}, 1-\lambda, (1-\lambda)^{-1}, \lambda(\lambda-1)^{-1}, (\lambda-1)\lambda^{-1}\}$ *of* λ. The ideal (f_λ) is homogeneous, so $\operatorname{gr}_{\mathfrak{m}} \widehat{\mathcal{O}}_{0,C_\lambda} = k[x,y]/(f_\lambda)$. An isomorphism $\widehat{\mathcal{O}}_{0,C_\lambda} \cong \widehat{\mathcal{O}}_{0,C_\mu}$ induces an isomorphism of these graded k-algebras, which are generated in degree 1; so it is given by an invertible linear substitution T of (x,y) with $T(f_\lambda) = c\, f_\mu$, $c \in k^*$. Thus the projective transformation of $\mathbf{P}^1$ induced by T carries the four zeros $\{0, \infty, 1, \lambda\}$ of f_λ to the four zeros $\{0, \infty, 1, \mu\}$ of f_μ, so the two sets of four points have the same cross-ratio class, i.e. μ lies in the S_3-orbit of λ. Since k is infinite and orbits have at most six elements, we can choose μ outside the orbit of λ; then C_λ, C_μ are equivalent by (i) but not analytically isomorphic. □

Exercise 3.7

Problem. For each of the following singularities at $(0,0)$ in the plane, give an embedded resolution, compute δ_P, and decide which ones are equivalent.

(a) $x^3 + y^5 = 0$.
(b) $x^3 + x^4 + y^5 = 0$.
(c) $x^3 + y^4 + y^5 = 0$.
(d) $x^3 + y^5 + y^6 = 0$.
(e) $x^3 + xy^3 + y^5 = 0$.

Solution. Notation: P_0 is the origin, E_i the exceptional curve of the i-th blow-up and $\widetilde{E}_i$ its strict transform at later stages, C_i the strict transform of the curve on the i-th surface; $\delta_P = \sum \frac{1}{2} r_Q(r_Q - 1)$ over P_0 and the infinitely near singular points Q of the curve with multiplicity r_Q (Example 3.9.3). I describe each step in the chart containing the relevant point, computing with the substitutions $x = vy$ (the tangent cone is x^3 in all five cases, so the tangent direction is $x = 0$) and the subsequent ones; in each case I checked the other chart: it contains no point of the strict transform of the curve. No assumption on the characteristic is needed.

(a), (b), (d) together. Write $f = x^3 A(x) + y^5 B(y)$ with A, B polynomials, $A(0) = B(0) = 1$: (a) $A = B = 1$; (b) $A = 1 + x$, $B = 1$; (d) $A = 1$, $B = 1 + y$.

Step 1 (centre P_0, multiplicity 3). With $x = vy$: $f = y^3\big(v^3A(vy)+y^2B(y)\big)$. The strict transform $C_1:\ v^3A(vy) + y^2B(y) = 0$ has multiplicity 2 at $Q_1 = (v, y) = (0, 0)$ with tangent cone y^2 (the line $y = 0$ is the tangent line of E_1). The only singular point of the reduced total transform $C_1 + E_1$ is Q_1 (mult. 3).

Step 2 (centre Q_1, mult. 2). With $y = sv$: $v^3A(sv^2) + s^2v^2B(sv) = v^2\big[vA(sv^2) + s^2B(sv)\big]$. So $C_2:\ G = vA(sv^2) + s^2B(sv) = 0$ is nonsingular at $Q_2 = (0, 0)$ with tangent line $v = 0$; $E_2:\ v = 0$, $\widetilde{E}_1:\ s = 0$. So C_2 is tangent to E_2 (since $G|_{v=0} = s^2B(0) = s^2$: contact order exactly 2) and transversal to $\widetilde{E}_1$ ($G|_{s=0} = v$). The only singular point of the reduced total transform is Q_2, where three smooth branches $C_2, E_2, \widetilde{E}_1$ meet, C_2, E_2 being tangent (mult. 3).

Step 3 (centre Q_2, mult. 1 of C_2). With $v = as$: $G = s\,[aA(as^3) + sB(as^2)]$. So $C_3:\ aA(as^3) + sB(as^2) = 0$ is smooth with tangent line $a + s = 0$ at $Q_3 = (a, s) = (0, 0)$; $E_3:\ s = 0$, $\widetilde{E}_2:\ a = 0$; $\widetilde{E}_1$ meets E_3 elsewhere. At Q_3 the three curves $C_3, E_3, \widetilde{E}_2$ are smooth with three different tangents (a triple point of the reduced total transform).

Step 4 (centre Q_3, mult. 1). Blowing up Q_3 separates the three branches (different tangents). The reduced total transform now consists of $C_4, \widetilde{E}_1, \dots, \widetilde{E}_3, E_4$ with only nodes: E_4 meets $C_4, \widetilde{E}_2, \widetilde{E}_3$ at three distinct points, and $\widetilde{E}_1$ meets $\widetilde{E}_3$ transversally. So this is an embedded resolution, by four blow-ups with multiplicities $3, 2, 1, 1$ of C at the centres. The singular infinitely near points of C are P_0 (3), Q_1 (2), hence

$$\delta_P = \tfrac{1}{2}\cdot 3\cdot 2 + \tfrac{1}{2}\cdot 2\cdot 1 = 4.$$

(For (a) this agrees with $\delta = (3-1)(5-1)/2 = 4$ for x^3+y^5, a cusp with semigroup $\langle 3, 5\rangle$.) The data – multiplicities $3, 2, 1, 1$ of C at the successive centres, the incidence of the total transforms at each stage, the fact that the two new singular points at each stage are as described – are identical for (a), (b), (d); so **(a), (b), (d) are equivalent.**

(c) $x^3 + y^4 + y^5$. Step 1: $x = vy$ gives $y^3(v^3 + y + y^2)$; $C_1:\ v^3 + y(1 + y) = 0$ is nonsingular at Q_1, tangent to $E_1:\ y = 0$ with contact order 3 ($C_1|_{y=0} = v^3$); so P_0 has mult. 3 and the total transform $C_1 + E_1$ has a singular point Q_1 (mult. 2, tangent branches). Step 2 (centre Q_1, mult. 1 of C_1), $y = sv$: $v(v^2 + s(1 + sv))$; $C_2:\ v^2 + s + s^2v = 0$ is smooth at $Q_2 = (0, 0)$, tangent to $\widetilde{E}_1:\ s = 0$ (contact 2), transversal to $E_2:\ v = 0$. Step 3 (centre Q_2), $s = av$: $v(v + a + a^2v^2)$; $C_3:\ v + a + a^2v^2 = 0$ is smooth, and C_3, E_3 ($v = 0$), $\widetilde{E}_1$ ($a = 0$) pass through $Q_3 = (0, 0)$ with three distinct tangents. Step 4 (centre Q_3) separates them. Thus an embedded resolution needs four blow-ups, with multiplicities $3, 1, 1, 1$ of C at the centres, and $\delta_P = \frac{1}{2}\cdot 3\cdot 2 = 3$ (equivalent to the cusp of type E_6, $x^3 + y^4$, with $\delta = (3 - 1)(4 - 1)/2 = 3$).

(e) $x^3+xy^3+y^5$. Step 1: $x = vy$ gives $y^3(v^3+vy+y^2)$; $C_1:\ v^3+vy+y^2 = 0$ has multiplicity 2 at $Q_1 = (0, 0)$ with tangent cone $y(v+y)$: two *distinct* tangents, so Q_1 is a node of C_1. Step 2 (centre Q_1), $y = sv$: $v^2(v + s + s^2)$; $C_2:\ v + s + s^2 = 0$ is smooth and meets $E_2:\ v = 0$ transversally in the two points $s = 0$ and $s = -1$. At $Q_2 = (0, 0)$ the three curves C_2 ($v + s = 0$), E_2 ($v = 0$), $\widetilde{E}_1$ ($s = 0$) have three distinct tangents; at $(0, -1)$ only C_2 and E_2 meet, transversally. Step 3 (centre Q_2, mult. 1) separates the three branches and gives normal crossings. So the embedded resolution has three blow-ups with multiplicities $3, 2, 1$, and

$$\delta_P = \tfrac{1}{2}\cdot 3\cdot 2 + \tfrac{1}{2}\cdot 2\cdot 1 = 4.$$

(In fact $f = x^3 + xy^3 + y^5$ has two branches, a cusp and a smooth branch tangent to it, $\delta = 1 + 0 + 3 = 4$.)

Which are equivalent. $\delta_P = 4$ for (a), (b), (d), (e) and $\delta_P = 3$ for (c). Equivalent singularities have the same multiplicities of the strict transforms at corresponding centres (these multiplicities, and hence δ_P, are determined by the correspondence in (3.9.4), cf. the discussion after (3.9.3)); (c) has multiplicity 1 at the second centre against 2 for the others, so (c) is equivalent to none of the others. (e) has the same multiplicities $3, 2$ as (a), but after the second blow-up the reduced total transform has two singular points in (e) (Q_2 and the node $(0, -1)$) and only one in (a),(b),(d) (the point Q_2); a correspondence between the singular points of the total transforms at this stage (which is required by the definition) cannot exist, so (e) is not equivalent to (a). (Even more simply, (e) has two branches at the origin, as noted above, whereas (a), (b), (d) have one, so already the components of the reduced total transform at the first stage cannot correspond.) Hence the equivalence classes are

$$\{\text{(a),(b),(d)}\}, \qquad \{\text{(c)}\}, \qquad \{\text{(e)}\}.$$

Only (a) $\sim$ (b) $\sim$ (d) is an instance of "same δ_P and equivalent"; (a) and (e) have the same δ_P (and the same multiplicities $3, 2$ of the first two singular points) but are not equivalent. □

Exercise 3.8

> **Problem.** Show that the following two singularities have the same multiplicity, and the same configuration of infinitely near singular points with the same multiplicities, hence the same δ_P, but are not equivalent.
>
> (a) $x^4 - xy^4 = 0$.
> (b) $x^4 - x^2y^3 - x^2y^5 + y^8 = 0$.

Solution. *This exercise is wrong as printed.* I re-read the equations on the scan at high resolution (the exponents and signs are as in the restatement above) and computed the blow-up sequences of both singularities, by hand and by computer. The two curves do have the same multiplicity 4 at the origin, but *not* the same configuration of infinitely near points, and they do not have the same δ_P.

(a) $f = x^4 - xy^4 = x(x^3 - y^4)$ is a line $x = 0$ together with the curve $x^3 = y^4$. Blow up the origin ($\mu = 4$, tangent cone x^4): in the chart $x = vy$, $f = y^4(v^4 - vy)$, so $C_1 : v(v^3 - y) = 0$ has a node at $Q_1 = (0, 0)$ (two smooth branches $v = 0$ and $y = v^3$, with distinct tangents) and nothing else on E_1 (the other chart gives $1 - u^4x$, which does not vanish on $x = 0$). Blowing up Q_1 separates the branches into two nonsingular points. So the infinitely near singular points are P_0 (mult. 4) and Q_1 (mult. 2), and $\delta_P = 6+1 = 7$ (check: $\delta = \delta(x^3 = y^4)+\delta(\text{line})+(x\cdot(x^3-y^4))_0 = 3+0+4 = 7$).

(b) $f = x^4 - x^2y^3 - x^2y^5 + y^8 = (x^2 - y^3)(x^2 - y^5)$, a cusp $(2, 3)$ and a cusp $(2, 5)$. Blow up the origin ($\mu = 4$): in the chart $x = vy$, $f = y^4\big(v^4 - v^2y - v^2y^3 + y^4\big) = y^4(v^2 - y)(v^2 - y^3)$, whose strict transform has multiplicity 3 at $Q_1 = (0, 0)$ (lowest term $-v^2y$, tangent cone v^2y), not 2. Blowing up Q_1 (charts $v = wy$ and $y = tv$) gives nonsingular curves, so the singular infinitely near points are P_0 (mult. 4) and Q_1 (mult. 3), and $\delta_P = 6 + 3 = 9$ (check: $1 + 2 + (x^2 - y^3, x^2 - y^5)_0 = 1 + 2 + 6 = 9$).

So (a) has multiplicity sequence $(4, 2)$ and $\delta_P = 7$, (b) has $(4, 3)$ and $\delta_P = 9$; the singularities are in particular not equivalent, but the stronger claim of the exercise does not hold. I was not able to reconstruct the intended equations.

A correct instance of the phenomenon. Consider (a′) $y(y-x^2)=0$ (a tacnode, two smooth branches tangent to order 2) and (b′) $y^2=x^5$ (a cusp of type A_4).

Same configuration, same multiplicities, same δ_P. For (a′): blow up the origin ($\mu=2$), $y=ux$: $x^2u(u-x)$; $C_1:\ u(u-x)=0$ has a node at $Q_1=(0,0)$ ($\mu=2$), and one more blow-up separates the branches. For (b′): $y=ux$: $x^2(u^2-x^3)$; $C_1:\ u^2=x^3$ is a cusp at Q_1 ($\mu=2$), and after one more blow-up ($u=wx$: $x^2(w^2-x)$) the strict transform is nonsingular. So both have infinitely near singular points P_0, Q_1 with multiplicities $(2,2)$, and $\delta_P=1+1=2$ for both.

Not equivalent. After the second blow-up the reduced total transform for (a′) is $C_2'+C_2''+\widetilde{E}_1+E_2$ with E_2 meeting the three other curves at three distinct points (C_1 has two branches with tangents $u=0$, $u=x$ and $E_1:\ x=0$ has a third tangent at Q_1): normal crossings, three singular points. For (b′) the total transform is $C_2+\widetilde{E}_1+E_2$ with E_2 meeting $\widetilde{E}_1$ transversally and C_2 tangent to E_2 ($w^2=x$, $E_2:\ x=0$): so it has a point where C_2 is tangent to E_2, which is not a normal crossing, and the reduced total transform has only two singular points (and one more blow-up is needed). Also the curve $y(y-x^2)$ has two irreducible components through 0 and $y^2=x^5$ only one. No bijection of components and singular points compatible with the blow-ups exists (already at the first stage the reduced total transform, namely the curve itself, has two components for (a′) and one for (b′)), so (a′) and (b′) are not equivalent in the sense of (3.9.4), although they have the same multiplicities, the same infinitely near singular points and the same δ_P. □

4. The cubic surface in projective 3-space

Conventions and notation. As in the text, k is algebraically closed and a surface is a nonsingular projective surface. X always denotes the cubic surface of (4.7.3): $\pi: X\to\mathbf{P}^2$ is the blow-up of six points $P_1,\dots,P_6$, no three collinear and not all six on a conic, with exceptional curves E_i; l, e_i are the classes of π^*(line) and E_i, so $l^2=1$, $e_i^2=-1$, $l.e_i=e_i.e_j=0$; $h=3l-\sum e_i$ is the hyperplane class, $K=-h$ (4.8), and the 27 lines are E_i, $F_{ij}\sim l-e_i-e_j$, $G_j\sim 2l-\sum_{i\neq j}e_i$ (4.9). I use freely (4.1)–(4.13), Thm. 1.9 (Hodge index), Thm. 1.10 (Nakai–Moishezon), the adjunction formula (1.5) and Riemann–Roch (1.6), and, where the exercise comes from Section 5, Cor. 5.4 and Thm. 5.7. "Bertini" means (II, 8.18) together with the connectedness of hyperplane sections (III, 7.9.1): on a surface a very ample divisor class contains an irreducible nonsingular curve, and a general member of a base-point-free subsystem of a very ample system avoids any given finite set of points.

Exercise 4.1

> **Problem.** The linear system of conics in $\mathbf{P}^2$ with two assigned base points P_1 and P_2 (4.1) determines a morphism ψ of X' (which is $\mathbf{P}^2$ with P_1 and P_2 blown up) to a nonsingular quadric surface Y in $\mathbf{P}^3$, and furthermore X' is isomorphic via ψ to Y with one point blown up.

Solution. Take $P_1=(1:0:0)$, $P_2=(0:1:0)$. A conic through P_1,P_2 has no x_0^2, x_1^2 terms, so the system $\mathfrak{d}$ of such conics is spanned by $x_0x_1,\ x_0x_2,\ x_1x_2,\ x_2^2$ (dimension 3, as in 4.2(a)). By (4.1) ($r=2\le 4$) $\mathfrak{d}$ has no unassigned base points, so $\mathfrak{d}'=|2l-e_1-e_2|$ is base point free on X'

and defines a morphism $\psi : X' \to \mathbf{P}^3$, $\psi^*\mathcal{O}(1) = \mathcal{O}(D')$, $D' = 2l - e_1 - e_2$, given on $\mathbf{P}^2$ by

$$\psi(x_0 : x_1 : x_2) = (y_0 : y_1 : y_2 : y_3) = (x_0x_1 : x_0x_2 : x_1x_2 : x_2^2).$$

Since $y_0y_3 = x_0x_1x_2^2 = y_1y_2$, the image lies in the quadric $Y : \ y_0y_3 - y_1y_2 = 0$, which is nonsingular (a quadratic form of rank 4); $Y \cong \mathbf{P}^1 \times \mathbf{P}^1$ via the Segre embedding $(a : b), (c : d) \mapsto (ac : ad : bc : bd)$.

ψ is birational onto Y. Projection from $R = (1 : 0 : 0 : 0) \in Y$ is a rational map $Y \dashrightarrow \mathbf{P}^2$, $y \mapsto (y_1 : y_2 : y_3)$, and $(y_1 : y_2 : y_3) \circ \psi = (x_0x_2 : x_1x_2 : x_2^2) = (x_0 : x_1 : x_2)$. So ψ is generically injective, $\psi(X')$ is a closed irreducible surface contained in the irreducible surface Y, hence $\psi(X') = Y$, and $K(\mathbf{P}^2) \subseteq K(Y) \subseteq K(X') = K(\mathbf{P}^2)$, so ψ is birational.

Exactly one curve is contracted. An irreducible curve $\Gamma \subseteq X'$ is mapped to a point iff $D'.\Gamma = 0$. Let L be the line through P_1, P_2, $\widetilde{L} \sim l - e_1 - e_2$ its strict transform; then $D'.\widetilde{L} = 2 + e_1^2 + e_2^2 = 0$, and indeed $\psi(\widetilde{L}) = \psi(\{x_2 = 0\}) = (1 : 0 : 0 : 0) = R$. Also $D'.E_i = -e_i^2 = 1 > 0$. If Γ is the strict transform of an irreducible plane curve $C \neq L$ of degree d with multiplicities m_1, m_2 at P_1, P_2, then $\Gamma \sim dl - m_1e_1 - m_2e_2$ (Prop. 3.6) and $D'.\Gamma = 2d - m_1 - m_2$. If $d = 1$, $C \neq L$ passes through at most one of P_1, P_2, so $D'.\Gamma \geq 1$; if $d \geq 2$ then $m_i \leq d - 1$ (a line through P_i and a further point of C meets C in at least $m_i + 1$ points, so $m_i + 1 \leq d$ by Bezout, C being irreducible of degree ≥ 2), so $D'.\Gamma \geq 2$. Hence $\widetilde{L}$ is the only irreducible curve contracted by ψ.

By Cor. 5.4, a birational morphism of surfaces contracting exactly n curves is a composition of exactly n monoidal transformations. Here $n = 1$, so $\psi : X' \to Y$ is the monoidal transformation with centre the point $\psi(\widetilde{L}) = R$: X' is isomorphic via ψ to Y with the point R blown up. (Its exceptional curve is $\widetilde{L}$; the two rulings of $Y \cong \mathbf{P}^1 \times \mathbf{P}^1$ pull back to the pencils $|l - e_2|$, $|l - e_1|$, since $(y_0 : y_2) = (x_0 : x_2)$, $(y_0 : y_1) = (x_1 : x_2)$.) □

Exercise 4.2

> **Problem.** Let φ be the quadratic transformation of $\mathbf{P}^2$, centered at P_1, P_2, P_3. If C is an irreducible curve of degree d in $\mathbf{P}^2$, with points of multiplicity r_1, r_2, r_3 at P_1, P_2, P_3, then the strict transform C' of C by φ has degree $d' = 2d - r_1 - r_2 - r_3$, and has points of multiplicity $d - r_2 - r_3$ at Q_1, $d - r_1 - r_3$ at Q_2, and $d - r_1 - r_2$ at Q_3. The curve C may have arbitrary singularities. [*Hint*: Use (Ex. 3.2).]

Solution. (The strict transform of a line L_{jk} through two of the centres is a point, so I assume, as the statement does implicitly, that C is not one of the three lines L_{12}, L_{13}, L_{23}.) Use the notation of (4.2.3): X is the blow-up of $\mathbf{P}^2$ at P_1, P_2, P_3 via π, and of the second plane at Q_1, Q_2, Q_3 via ψ, $\varphi = \psi \circ \pi^{-1}$; ψ is given by the base-point-free system of conics through P_1, P_2, P_3, so $l' := \psi^*(\text{line}) \sim 2l - e_1 - e_2 - e_3$; and $E_i' := \psi^{-1}(Q_i) = \widetilde{L}_{jk}$, so $e_i' \sim l - e_j - e_k$ ($\{i, j, k\} = \{1, 2, 3\}$), by Prop. 3.6 applied to L_{jk}.

Let $\widetilde{C} \subseteq X$ be the strict transform of C under π; $\widetilde{C} \sim dl - r_1e_1 - r_2e_2 - r_3e_3$ (Prop. 3.6). It is an irreducible curve different from E_i and from the $\widetilde{L}_{jk}$, the only curves contracted by ψ (hypothesis on C). So $C' = \psi(\widetilde{C})$ is a curve, $\widetilde{C}$ is its strict transform under the monoidal transformations ψ (ψ is an isomorphism outside the E_i', and $\widetilde{C}$ is irreducible, not one of the E_i'), and $\psi|_{\widetilde{C}} : \widetilde{C} \to C'$ is birational. Let $d' = \deg C'$ and $r_i' = \mu_{Q_i}(C')$. By the projection

formula (Prop. 3.2 (b), (d) for the three monoidal transformations), $d' = C'.(\text{line}) = \widetilde{C}.l'$ and, by Cor. 3.7, $r'_i = \widetilde{C}.E'_i$. Therefore

$$d' = (dl - \sum r_m e_m).(2l - \sum e_m) = 2d - r_1 - r_2 - r_3,$$

since $l^2 = 1$, $e_m^2 = -1$ give $(-r_m e_m).(-e_m) = r_m e_m^2 = -r_m$; and for $\{i,j,k\} = \{1,2,3\}$,

$$r'_i = (dl - \sum r_m e_m).(l - e_j - e_k) = d - r_j - r_k.$$

This is the claim; no assumption on the singularities of C was used. (Equivalently: apply Ex. 3.2 to C and a general line, resp. to C and L_{jk}.) □

Exercise 4.3

> **Problem.** Let C be an irreducible curve in $\mathbf{P}^2$. Then there exists a finite sequence of quadratic transformations, centered at suitable triples of points, so that the strict transform C' of C has only ordinary singularities, i.e., multiple points with all distinct tangent directions (I, Ex. 5.14). Use (3.8).

Solution. *Hypothesis.* I assume $\operatorname{char} k = 0$. The only place where this is used is the transversality of a general line through a point to a curve (Step 3); the argument works verbatim whenever the projection of the curve from the relevant point is separable. (The hypothesis is needed for this proof: in characteristic p the irreducible curve $x^{2p+1} = y^{p+1}$, parametrized by (t^{p+1}, t^{2p+1}), has a non-ordinary singular point at 0 and all its tangent lines pass through 0, so no line through 0 meets it transversally elsewhere.)

Step 1: ordinary points. Let P be a point of multiplicity r of a curve C on a surface, $\pi : \widetilde{X} \to X$ the blow-up of P, E the exceptional curve. In the notation of the proof of (3.6), $\widetilde{C} \cap E = \{f_r(1,u) = 0\}$ in the chart $y = ux$, where f_r is the tangent cone, and the local intersection multiplicity of $\widetilde{C}$ with E at a point equals the multiplicity of the corresponding root of $f_r(1,u)$. Hence *P is ordinary iff $\widetilde{C} \cap E$ consists of r distinct points* (recall $\widetilde{C}.E = r$).

Step 2: the tree of a singular point. For a singular point P of C let $T(P)$ be the finite rooted tree whose vertices are the singular points of C infinitely near to P (including P): the children of a vertex Q are the singular points of the strict transform of C lying on the exceptional curve obtained by blowing up Q. It is finite: along a chain the arithmetic genus drops by at least 1 at each step (3.7), $p_a \geq 0$, and each vertex has finitely many children. Let $n(P) = |T(P)|$, and for a plane curve C put

$$N(C) = \sum_{P \text{ singular, not ordinary}} n(P).$$

Then $N(C) = 0$ iff all singular points of C are ordinary. Note $n(P) = 1 + \sum_{S \text{ child of } P} n(S)$, and (germs being intrinsic) $T(S)$ for a child S is the subtree of $T(P)$ at S.

Step 3: the move. Let C have degree $d \geq 2$ and let P_1 be a non-ordinary singular point, of multiplicity $r \geq 2$ (so $r \leq d-1$, C being irreducible and not a line). Choose P_2, P_3 as follows. The tangent lines to C at its nonsingular points form (the image of the Gauss map) a set contained in a proper closed subset of the dual plane, since C is not a line; so there are lines L_{23} not tangent to C at any point, not through P_1, not through any singular point of C, and

meeting C in d distinct points (transversally, at nonsingular points). Fix such an L_{23}. For a point $P_2 \in L_{23}$ the line $L_{12} = P_1P_2$ runs over all lines through P_1 as P_2 runs over L_{23}. In characteristic 0 the projection from P_1 restricted to the strict transform of C is separable, so the general line through P_1 meets C outside P_1 in distinct points, transversally, and at nonsingular points; it is not in the tangent cone at P_1 (finitely many exceptions) and does not pass through the other singular points. So we can choose $P_2, P_3 \in L_{23} \setminus C$, distinct, such that L_{12} and L_{13} have these properties. The points P_1, P_2, P_3 are not collinear ($P_1 \notin L_{23}$). Let φ be the quadratic transformation centred at P_1, P_2, P_3 and C' the strict transform; with the notation of Ex. 4.2, C' has degree $2d - r$ and multiplicities d at Q_1, $d - r$ at Q_2, Q_3 (as $r_2 = r_3 = 0$).

Claim: Q_1, Q_2, Q_3 are ordinary points (or nonsingular if $d - r = 1$). Indeed $E_1' = \widetilde{L}_{23}$ and $\widetilde{C} \cap \widetilde{L}_{23} = C \cap L_{23}$ is d distinct points, so Q_1 is ordinary by Step 1 applied to ψ. C meets L_{13} at P_1 with intersection multiplicity exactly r (not a tangent) and in $d - r$ further distinct points, so $\widetilde{C} \cap \widetilde{L}_{13}$ consists of $d - r$ distinct points (the point of E_1 in the direction L_{13} is not on $\widetilde{C}$), all transversal; thus Q_2 is ordinary, and likewise Q_3.

Claim: the other singular points of C' are the images of the singular points of C other than P_1 and of the singular points of $\widetilde{C}$ on E_1, with the same trees. Indeed ψ is an isomorphism over $\mathbf{P}^2 - \{Q_1, Q_2, Q_3\}$, and the singular points of $\widetilde{C} \subseteq X$ are the singular points of C other than P_1 (none of them is on a line L_{jk}, and $P_2, P_3 \notin C$) and the singular points of $\widetilde{C}$ on E_1 (not on $\widetilde{L}_{12}, \widetilde{L}_{13}$ by the choice of directions, nor on $\widetilde{L}_{23}$). The germs of $(X, \widetilde{C})$ and of $(\mathbf{P}^2, C')$ at corresponding points are isomorphic, and ordinariness and $T(\cdot)$ depend only on the germ.

Consequently

$$N(C') = N(C) - n(P_1) + \sum_{S \text{ child of } P_1,\ S \text{ not ordinary}} n(S) \ \leq \ N(C) - 1,$$

because $n(P_1) = 1 + \sum_S n(S)$ and the points Q_i contribute 0.

Conclusion. The strict transform of an irreducible curve is irreducible, N is a nonnegative integer and decreases by at least 1 at each move; after at most $N(C)$ quadratic transformations (each centred at three non-collinear points of the current plane) we reach a curve with $N = 0$, i.e. with only ordinary singularities. (A curve of degree ≤ 2 has $N = 0$ trivially.) □

Exercise 4.4

> **Problem.** (a) Use (4.5) to prove the following lemma on cubics: If C is an irreducible plane cubic, if L is a line meeting C in points P, Q, R, and L' is a line meeting C in P', Q', R', and if P'' is the third intersection of the line PP' with C, and Q'', R'' are defined similarly, then P'', Q'', R'' are collinear.
>
> (b) Let P_0 be an inflection point of C, and define the group operation on the set of regular points of C by the geometric recipe "let the line PQ meet C at R, and let P_0R meet C at T, then $P + Q = T$" as in (II, 6.10.2) and (II, 6.11.4). Use (a) to show that this operation is associative.

Solution. *Remarks on the statement.* In (a) the lines PP', QQ', RR' must be defined, and the statement is about the generic situation of nine *distinct* points $P, Q, R, P', Q', R', P'', Q'', R''$ on C (otherwise one needs the versions of (4.4), (4.5) with infinitely near points). I prove (a) in

that case, which is the situation to which (4.5) applies directly, and in (b) I deduce associativity for all triples from the generic ones by a closedness argument; for (b) I assume $\operatorname{char} k \neq 2, 3$ (needed to see that the generic nine points are distinct; in characteristic 2 and 3 there are cubics all of whose tangent lines pass through one point, and then the coincidences $S_1 = T_1$ below can occur identically).

(a) Let $\mathfrak{d}$ be the system of cubics through the eight points $P, Q, R, P', Q', R', P'', Q''$. These lie on the irreducible cubic C; a line not contained in C meets C in at most 3 points and a conic (having no common component with C) in at most 6, so no 4 of them are collinear and no 7 lie on a conic. By Cor. 4.4(b) (with $r = 8$), $\dim \mathfrak{d} = 1$: $\mathfrak{d}$ is a pencil. Let $\ell_P = PP'P''$, $\ell_Q = QQ'Q''$, $\ell_R = RR'R''$ be the three lines. The cubic $\mathfrak{c}_1 = \ell_P + \ell_Q + \ell_R$ contains all nine points, so $\mathfrak{c}_1 \in \mathfrak{d}$, and $\mathfrak{c}_1 \neq C$ since C is irreducible. Hence $\mathfrak{d} = \{\lambda C + \mu \mathfrak{c}_1\}$ and the base locus of $\mathfrak{d}$ is $C \cap \mathfrak{c}_1$. By Bezout this has nine points counted with multiplicity and it contains the nine distinct points above, so it consists exactly of them. In particular every member of $\mathfrak{d}$ passes through R'' (the ninth point, cf. Cor. 4.5).

Now let M be the line through P'' and Q'' (they are distinct) and consider the cubic $\mathfrak{c}_2 = L + L' + M$. It contains $P, Q, R \in L$, $P', Q', R' \in L'$ and $P'', Q'' \in M$, so it contains the eight points, $\mathfrak{c}_2 \in \mathfrak{d}$, hence $R'' \in \mathfrak{c}_2$. But $R'' \notin L$ (since $L \cap C = \{P, Q, R\}$ and R'' is different from P, Q, R), and likewise $R'' \notin L'$. So $R'' \in M$: P'', Q'', R'' are collinear. $\square$(a)

(b) Write $X * Y$ for the third intersection point of the line XY (the tangent line if $X = Y$) with C, for regular points X, Y; $X * Y$ is again a regular point, because a singular point on a line meeting C at two regular points (or doubly at one) would give intersection multiplicity $\geq 4 > 3$ (the line is not in C). The operation is $X + Y = P_0 * (X * Y)$ and $*$ is symmetric. (It has P_0 as neutral element and $P_0 * P_0 = P_0$ because P_0 is a flex; only associativity is asked.)

Generic case. Let P, Q, R be regular points and put $S_1 = Q * R$, $S_2 = P * Q$, $T_1 = P_0 * S_1 = Q + R$, $T_2 = P_0 * S_2 = P + Q$, $W = R * T_2$. Suppose the nine points

$$Q,\ R,\ S_1,\ S_2,\ T_2,\ P_0,\ P,\ W,\ T_1$$

are pairwise distinct. Apply (a) with the lines $L = QRS_1$ and $L' = P_0 S_2 T_2$: the third intersection of the line through Q and S_2 (it is the line PQ) is P; that of the line RT_2 is W; that of the line $S_1 P_0$ is T_1. By (a), P, W, T_1 are collinear, i.e. $W = P * T_1$. Hence

$$(P + Q) + R = P_0 * (T_2 * R) = P_0 * W = P_0 * (P * T_1) = P + (Q + R).$$

All triples. Let C_{reg} be the (irreducible, infinite) variety of regular points. The map $* : C_{\mathrm{reg}} \times C_{\mathrm{reg}} \to C_{\mathrm{reg}}$ is a morphism. Let F be the cubic form. For $X \neq Y$ (vectors representing the points) the points of the line are $\lambda X + \mu Y$ and $F(\lambda X + \mu Y) = \lambda\mu(\lambda A + \mu B)$ with $A = \nabla F(X) \cdot Y$, $B = \nabla F(Y) \cdot X$ (not both zero, as the line is not in C), so $X * Y = BX - AY$ off the diagonal. Near the diagonal use a divided difference. Let $U \subseteq C_{\mathrm{reg}}$ be an affine open set with a regular lift $\gamma : U \to k^3 \setminus 0$ of the embedding and a function $t : U \to \mathbf{A}^1$ which is a local parameter everywhere on U (so t is etale). On a neighbourhood W of the diagonal of $U \times U$ the ideal of the diagonal is generated by $\delta = t(q) - t(p)$, so for $X = \gamma(p)$, $Y = \gamma(q)$ we can write $Y = X + \delta\Gamma$ with Γ regular on W, and $\Gamma = d\gamma/dt$ on the diagonal; X, Γ are linearly independent everywhere (off the diagonal because $Y \not\sim X$, on it because Γ is a tangent vector of the smooth curve C_{reg}), and span the line XY (the tangent line on the diagonal). Since $F(X) = 0$,

$$F(\lambda X + \mu\Gamma) = \mu\,(a\lambda^2 + b\lambda\mu + c\mu^2), \qquad a = \nabla F(X) \cdot \Gamma,\ c = F(\Gamma),$$

with b regular. As $\nabla F(X)\cdot X = 3F(X) = 0$, $a = \delta^{-1}\nabla F(X)\cdot Y$ vanishes on the diagonal ($\nabla F(X)\cdot\gamma' = \frac{d}{dt}F(\gamma) = 0$ there), so $a = \delta a'$ with a' regular. The quadratic form has the root $(1,\delta)$ (the point Y), whence $a + b\delta + c\delta^2 = 0$, $b = -a' - c\delta$ (an identity of regular functions on the irreducible W), and

$$F(\lambda X + \mu\Gamma) = \mu(\mu - \delta\lambda)(c\mu - a'\lambda).$$

The roots $\mu = 0$, $\mu = \delta\lambda$ are X and Y, so $X * Y = cX + a'\Gamma$; this is nonzero because F does not vanish identically on the line, it is regular on W (also on the diagonal), and agrees with $BX - AY$ off the diagonal. Gluing, $*$ is a morphism. Hence $(P,Q,R) \mapsto (P+Q)+R$ and $(P,Q,R) \mapsto P + (Q+R)$ are morphisms $C^3_{\mathrm{reg}} \to C$, and since the target is separated they agree everywhere as soon as they agree on a nonempty open subset. Distinctness of the nine points is an open condition; it remains to see that it is nonempty. The nine points, as functions of (P,Q,R), depend respectively on $\{Q\},\{R\},\{Q,R\},\{P,Q\},\{P,Q\},\emptyset,\{P\},\{P,Q,R\},\{Q,R\}$ (e.g. $W = R * T_2$: for fixed Q,R the map $P \mapsto P * Q$ is nonconstant, hence so is $P \mapsto T_2$ and $P \mapsto R * T_2$, because $Y \mapsto R * Y$ is finite-to-one). Two functions with different supports are not identically equal and a nonconstant function is not constant; so the only possible identical coincidences are $S_1 \equiv T_1$ and $S_2 \equiv T_2$, i.e. $P_0 * Y = Y$ for all Y in an infinite set, i.e. the tangent line at infinitely many $Y \in C_{\mathrm{reg}}$ passes through P_0. These are the points of C on the first polar of P_0, a quadric $\sum_i (P_0)_i \partial_i F = 0$; as C is an irreducible cubic it is not contained in a quadric, so the polar vanishes identically, i.e. $\partial F/\partial y \equiv 0$ in coordinates where $P_0 = (0:1:0)$, and since $\operatorname{char} k \neq 2,3$ the form F (of degree 3) does not involve y, so C is a union of three lines through P_0, a contradiction. Hence the nine points are distinct for (P,Q,R) in a nonempty open set, and the identity of the two morphisms there gives associativity for all triples. □

Exercise 4.5

> **Problem.** Prove Pascal's theorem: if A,B,C,A',B',C' are any six points on a conic, then the points $P = AB'.A'B$, $Q = AC'.A'C$, $R = BC'.B'C$ are collinear (Fig. 22).

Solution. I take the conic Γ nonsingular and the six points distinct, so that the lines in the statement are defined (no three of the six points are collinear, as a line meets Γ in at most 2 points). Let $\mathfrak{X} = AB' + BC' + CA'$ and $\mathfrak{Y} = A'B + B'C + C'A$ be the two cubics, each a union of three lines. Then A,B,C,A',B',C' lie on both, and so do P ($\in AB'$ and $A'B$), R ($\in BC'$ and $B'C$), Q ($\in CA'$ and $C'A$). The cubics $\mathfrak{X},\mathfrak{Y}$ are not proportional (the line AB' is none of $A'B, B'C, C'A$, since it would contain at least three of the six points).

The points P,Q,R are not on Γ: e.g. $R \in BC' \cap \Gamma = \{B,C'\}$ and $R \in B'C \cap \Gamma = \{B',C\}$ is impossible as the six points are distinct.

Choose a seventh point $S \in \Gamma$ different from the six, and let $\mathfrak{Z} = \lambda\mathfrak{X} + \mu\mathfrak{Y}$, $(\lambda,\mu) \neq (0,0)$, be a member of the pencil with $\mathfrak{Z}(S) = 0$ (one linear condition). Then $\mathfrak{Z}$ meets the conic Γ in at least 7 points A,B,C,A',B',C',S, while a cubic not containing Γ meets Γ in at most 6 points (Bezout, I, 7.7); so $\mathfrak{Z}$ contains Γ, and $\mathfrak{Z} = \Gamma + \ell$ for a line ℓ. Now P,Q,R are base points of the pencil, so lie on $\mathfrak{Z} = \Gamma \cup \ell$, and not on Γ; hence $P,Q,R \in \ell$ are collinear. □

(This is the Cayley–Bacharach argument behind (4.5); here the pencil through the nine points $\mathfrak{X} \cap \mathfrak{Y}$ is used directly, which avoids checking the hypotheses of (4.5) for the eight points $A,\dots,C',P,Q$.)

Singular conics. The theorem also holds for six distinct points on a singular conic (two lines, or a double line), whenever P, Q, R are defined, i.e. the three pairs of lines are pairs of distinct lines. In homogeneous coordinates $P = (A \times B') \times (A' \times B)$, $Q = (A \times C') \times (A' \times C)$, $R = (B \times C') \times (B' \times C)$ (cross products of vectors), so $G = \det(P, Q, R)$ is a polynomial in the coordinates of the six points, and the proof above shows $G = 0$ on every six-tuple of distinct points of a nonsingular conic. Let $\mathcal{T} \subseteq \mathbf{P}^5 \times (\mathbf{P}^2)^6$ be the set of pairs (conic, six points on it). It is cut out by six equations in a space of dimension 17, so every component of $\mathcal{T}$ has dimension ≥ 11, whereas the part of $\mathcal{T}$ over the singular conics (a hypersurface of dimension 4 in $\mathbf{P}^5$, with fibres Γ^6 of dimension 6) has dimension ≤ 10. Hence every component of $\mathcal{T}$ meets the part over nonsingular conics, so $\mathcal{T}$ is the closure of the six-tuples of distinct points on nonsingular conics, and G vanishes on all of $\mathcal{T}$: if P, Q, R are defined, they are collinear.

Exercise 4.6

> **Problem.** Generalize (4.5) as follows: given 13 points $P_1, \ldots, P_{13}$ in the plane, there are three additional determined points P_{14}, P_{15}, P_{16}, such that all quartic curves through $P_1, \ldots, P_{13}$ also pass through P_{14}, P_{15}, P_{16}. What hypotheses are necessary on $P_1, \ldots, P_{13}$ for this to be true?

Solution. This exercise is open-ended; here is what I can prove. Let $P_1, \ldots, P_{13}$ be distinct ordinary points of $\mathbf{P}^2$, $\pi : X' \to \mathbf{P}^2$ the blow-up at them, $\mathfrak{d}$ the system of quartics through them, i.e. $\mathfrak{d}' = |D'|$, $D' = 4l - \sum_{i=1}^{13} e_i$, on X'. Quartics form a projective space of dimension 14 and each point imposes one linear condition, so always $\dim \mathfrak{d} \geq 1$.

Hypotheses. (H1) the 13 points impose independent conditions on quartics, i.e. $\dim \mathfrak{d} = 1$ ($\mathfrak{d}$ is a pencil); (H2) the pencil $\mathfrak{d}$ has no fixed component.

Claim. Under (H1) and (H2) the pencil has exactly three further base points P_{14}, P_{15}, P_{16} counted with multiplicity (they may be infinitely near to each other or to the P_i, or coincide), every quartic through $P_1, \ldots, P_{13}$ passes through them (as the problem asserts), and they are determined by $P_1, \ldots, P_{13}$.

Proof. Let C_1', C_2' be two distinct members of $\mathfrak{d}'$. By (H2) they have no common component. By Prop. 3.2 (rules (a)–(c)), $C_1'.C_2' = D'^2 = 16 - 13 = 3$. Hence $C_1' \cap C_2'$ is a finite set and their intersection scheme has length 3 (Prop. 1.4); every member of the pencil $\mathfrak{d}'$ contains this scheme because it is generated by C_1', C_2'. Its points are the unassigned base points of $\mathfrak{d}$ (in the language of Section 4): a point of $C_1' \cap C_2'$ off the exceptional curves is an ordinary point $P_{14}, \ldots,$ a point on some E_i is a point infinitely near P_i. By (H1) the quartics through $P_1, \ldots, P_{13}$ are exactly the members of $\mathfrak{d}$, so they all pass through the length-3 scheme $\{P_{14}, P_{15}, P_{16}\}$. □

Necessity. If (H1) fails, the statement fails in general: e.g. if $P_1, \ldots, P_6$ are collinear, every quartic through them contains the line L (it meets L in $\geq 6 > 4$ points), so the quartics through $P_1, \ldots, P_{13}$ are $L+$ (cubics through $P_7, \ldots, P_{13}$), a net ($\dim = 9 - 7 = 2$) whose base locus is L and the seven points: there are no additional isolated base points. Similarly if all 13 points lie on an irreducible cubic E, the quartics $E + L'$ (L' any line) form a system of dimension ≥ 2 contained in the system, and the base locus is E only. If (H1) holds but (H2) fails, the base locus of the pencil contains a curve, and the quantity $D'^2 = 3$ no longer counts isolated points; again no three extra points are determined as base points. In the same way, ten of the points on a conic make every quartic through them contain the conic, and then (H1) fails. On the

other hand (H1) holds for a general choice of the points: choosing them one by one outside the base locus of the system of quartics through the previous ones, the dimension drops by exactly one each time (as in 4.2(a)), so that $\dim \mathfrak{d} = 14 - 13 = 1$ at the end. I have not worked out a complete list of geometric conditions on the P_i equivalent to (H1) and (H2) (this would be the quartic analogue of the conditions "no 4 collinear, no 7 on a conic" of (4.3), (4.4), and is more complicated), and I do not discuss (H2) for general points.

Remark (Cayley–Bacharach). If C_1, C_2 are quartics meeting in 16 distinct points and P_{14}, P_{15}, P_{16} are any three of them, the other 13 satisfy the conclusion exactly when P_{14}, P_{15}, P_{16} are *not collinear*; so for 13 points that are among the intersection points of two quartics (a typical configuration of the situation above) the extra hypothesis is that the three residual points are not collinear. I only state this as a remark, since it uses the Cayley–Bacharach theorem (Eisenbud–Green–Harris), not proved here. □

Exercise 4.7

Problem. If D is any divisor of degree d on the cubic surface (4.7.3), show that

$$p_a(D) \le \begin{cases} \dfrac{1}{6}(d-1)(d-2) & \text{if } d \equiv 1, 2 \pmod 3, \\ \dfrac{1}{6}(d-1)(d-2) + \dfrac{2}{3} & \text{if } d \equiv 0 \pmod 3. \end{cases}$$

Show furthermore that for every $d > 0$, this maximum is achieved by some irreducible nonsingular curve.

Solution. For a divisor class D put $p_a(D) = \frac{1}{2}D.(D+K) + 1$ (for an effective divisor this is its arithmetic genus, by Ex. 1.3 / Prop. 4.8(g)); the degree of D is $d = D.h$. Since $K = -h$, $D.K = -d$ and

$$p_a(D) = \tfrac{1}{2}(D^2 - d) + 1.$$

Two constraints on D^2. (i) By the Hodge index theorem (Thm. 1.9; h is ample) applied to $D' = 3D - d\,h$, which satisfies $D'.h = 3d - 3d = 0$ (recall $h^2 = 3$), we get $D'^2 \le 0$, i.e. $9D^2 - 6d^2 + 3d^2 \le 0$, i.e. $3D^2 \le d^2$. (ii) By Riemann–Roch, $\chi(\mathcal{L}(D)) = \frac{1}{2}D.(D-K)+1+p_a(X) \in \mathbf{Z}$ with $p_a(X) = 0$ (X is a rational surface), so $D^2 + d$ is even, i.e. $D^2 \equiv d \pmod 2$.

The bound. If $3 \mid d$, then $D^2 \le d^2/3$ and $p_a(D) \le \frac{1}{2}(\frac{d^2}{3} - d) + 1 = \frac{d^2-3d+6}{6} = \frac{1}{6}(d-1)(d-2) + \frac{2}{3}$. If $3 \nmid d$ then $d^2 \equiv 1 \pmod 3$ and (i) gives $D^2 \le \frac{d^2-1}{3}$. Write $d = 3q+1$ or $3q+2$: then $\frac{d^2-1}{3} = 3q^2 + 2q$ resp. $3q^2 + 4q + 1$, which has the parity of q resp. $q+1$, whereas d has the parity of $q+1$ resp. q; so by (ii) $D^2 \ne \frac{d^2-1}{3}$ and hence $D^2 \le \frac{d^2-1}{3} - 1 = \frac{d^2-4}{3}$. Then $p_a(D) \le \frac{1}{2}(\frac{d^2-4}{3} - d) + 1 = \frac{d^2-3d+2}{6} = \frac{1}{6}(d-1)(d-2)$.

The maximum is attained. By Ex. 4.8 below (whose proof does not use this exercise) a divisor class D contains an irreducible nonsingular curve if $D.L \ge 0$ for all 27 lines and $D^2 > 0$, or D is a line, or D has degree 2 with $D^2 = 0$. For a curve of class D the genus is $p_a(D)$. Take:

- $d = 1$: $D = e_1$ (a line, $p_a = 0 = \frac{1}{6} \cdot 0 \cdot (-1)$). $d = 2$: $D = l - e_1$ (a conic class, $D^2 = 0$, $p_a = 0$).
- $d = 3q$, $q \ge 1$: $D = qh$; $D^2 = 3q^2 = d^2/3$, $D.L = q > 0$.

- $d = 3q+1$, $q \geq 1$: $D = qh+e_1$; $D^2 = 3q^2+2q-1 = \frac{d^2-4}{3} > 0$, and $D.L = q+e_1.L \geq q-1 \geq 0$ ($e_1.L \geq -1$ for each line).
- $d = 3q+2$, $q \geq 1$: $D = qh + (l - e_1)$ (degree $3q+2$); $D^2 = 3q^2+4q = \frac{d^2-4}{3}$, $D.L \geq q > 0$ since $l - e_1$ is base point free, hence $(l - e_1).L \geq 0$ for every curve L.

In each case the curve has genus $\frac{1}{2}(D^2 - d) + 1$ equal to the bound. □

Exercise 4.8

> **Problem.** Show that a divisor class D on the cubic surface contains an irreducible curve $\Leftrightarrow$ it contains an irreducible nonsingular curve $\Leftrightarrow$ it is either (a) one of the 27 lines, or (b) a conic (meaning a curve of degree 2) with $D^2 = 0$, or (c) $D.L \geq 0$ for every line L, and $D^2 > 0$. [*Hint*: Generalize (4.11) to the surfaces obtained by blowing up $2, 3, 4$, or 5 points of $\mathbf{P}^2$, and combine with our earlier results about curves on $\mathbf{P}^1 \times \mathbf{P}^1$ and the rational ruled surface X_1 (2.18).] (Starred in the book.)

Solution. I read (b) as: D has degree $D.h = 2$ and $D^2 = 0$. We prove: irreducible nonsingular $\Rightarrow$ irreducible $\Rightarrow$ [(a), (b) or (c)] $\Rightarrow$ irreducible nonsingular curve. The first implication is trivial.

Irreducible $\Rightarrow$ (a), (b) or (c). Let C be an irreducible curve, of degree d. If $d = 1$ it is a line, (a). Let $d \geq 2$. For a line $L \neq C$ we have $C.L \geq 0$. By (1.5) and (4.8) $C^2 = 2p_a(C) - 2 + d$, and $p_a(C) \geq 0$ for an integral projective curve ($h^0(\mathcal{O}_C) = 1$). So $C^2 \geq d - 2 \geq 0$, with $C^2 = 0$ iff $d = 2$, $p_a = 0$ (a conic, (b)); otherwise $C^2 > 0$ and (c) holds.

Preparations for the converse.

Lemma A (normal form). Let D be a class with $D.L \geq 0$ for all 27 lines. After a suitable choice of the blow-down structure $X \to \mathbf{P}^2$ (i.e. of six skew lines E'_i, Prop. 4.10) we can write $D \sim al - \sum b_i e_i$ with $b_1 \geq \cdots \geq b_6 \geq 0$ and $a \geq b_1 + b_2 + b_3$.

Proof. Exactly as in the proof of (4.11): choose E'_6 with $D.E'_6$ minimal among all lines; E'_5 with $D.E'_5$ minimal among the lines skew to E'_6; E'_4 minimal among lines skew to E'_6, E'_5; E'_3 minimal among lines skew to E'_6, E'_5, E'_4. Then three lines remain which are skew to $E'_3, \ldots, E'_6$: two skew lines E'_1, E'_2 (numbered so that $D.E'_1 \geq D.E'_2$) and a third meeting both (F'_{12}). By Prop. 4.10 these six skew lines are the exceptional curves of a blow-down $X \to \mathbf{P}^2$, and with respect to it $b_i = D.E'_i$ (so $b_i \geq 0$), $D.F'_{12} = a - b_1 - b_2$. By construction $b_6 \leq b_5 \leq b_4 \leq b_3 \leq b_2 \leq b_1$ (E'_1, E'_2 were candidates for E'_3), and F'_{12}, being skew to E'_4, E'_5, E'_6, was a candidate for E'_3, so $a - b_1 - b_2 \geq b_3$. □

For classes (and with the convention that only the indices $\leq s$ occur) let $D_0 = l$, $D_1 = l - e_1$, $D_2 = 2l - e_1 - e_2$, $D_3 = 2l - e_1 - e_2 - e_3$, $D_4 = 2l - e_1 - \cdots - e_4$, $D_5 = 3l - e_1 - \cdots - e_5$, $D_6 = 3l - e_1 - \cdots - e_6 = h$ (as in the proof of Lemma 4.12). The linear systems $|D_0|, \ldots, |D_5|$ are base point free by (4.1) and (4.3) (and on the blow-up at fewer points they are the pull-backs), and D_6 is very ample (4.6). If a', b'_i are integers with $b'_1 \geq \cdots \geq b'_6 \geq 0$ and $a' \geq b'_1 + b'_2 + b'_5$, then

$$a'l - \sum_i b'_i e_i = c_0 D_0 + \cdots + c_6 D_6, \quad c_0 = a' - b'_1 - b'_2 - b'_5,\ c_i = b'_i - b'_{i+1}\ (1 \leq i \leq 5),\ c_6 = b'_6\ (\geq 0) \tag{$*$}$$

(compare coefficients of e_i and of l).

(a), (b), (c) $\Rightarrow$ irreducible nonsingular.

(a) A line is irreducible and nonsingular.

(b) Let $D.h = 2$, $D^2 = 0$. By Riemann–Roch $l(D) \geq \frac{1}{2}D.(D-K)+1 = \frac{1}{2}\cdot 2+1 = 2$ ($l(K-D) = 0$ since $\deg(K-D) < 0$), so $\dim |D| \geq 1$ and D is effective. Members of $|D|$ have degree 2, so they are either irreducible plane conics or sums of (at most) two lines. There are only finitely many (at most 27^2) sums of two lines, while a pencil has infinitely many members; so some member is an irreducible conic, which is a nonsingular plane conic.

(c) Let $D.L \geq 0$ for all lines and $D^2 > 0$. By Lemma A, $D \sim al - \sum b_i e_i$ with $b_1 \geq \cdots \geq b_6 \geq 0$, $a \geq b_1 + b_2 + b_3$, in suitable coordinates. Let s be the number of nonzero b_i (so $b_i = 0$ for $i > s$), let X_s be $\mathbf{P}^2$ blown up at $P_1, \ldots, P_s$, and $p : X \to X_s$ the blow-down of $E_{s+1}, \ldots, E_6$; then $D = p^*\overline{D}$ with $\overline{D} = al - \sum_{i\leq s} b_i e_i$ on X_s (not to be confused with the classes $D_0, \ldots, D_6$ above). It suffices to find an irreducible nonsingular curve in $|\overline{D}|$ avoiding the points $P_{s+1}, \ldots, P_6 \in X_s$: its pull-back is isomorphic to it and lies in $|D|$.

Case $s \geq 3$. Put $b'_i = b_i - 1$ for $i \leq s$ and $b'_i = 0$ for $i > s$, $a' = a - 3$. Then $b'_1 \geq \cdots \geq b'_6 \geq 0$ and $b'_1 + b'_2 + b'_5 \leq b'_1 + b'_2 + b'_3 = b_1 + b_2 + b_3 - 3 \leq a - 3$, so $(*)$ applies, with $c_i = 0$ for $i > s$. Hence $N := \overline{D} - (-K_{X_s}) = (a-3)l - \sum_{i\leq s}(b_i - 1)e_i$ is a nonnegative combination of $D_0, \ldots, D_s$, hence base point free on X_s, while $-K_{X_s} = 3l - \sum_{i\leq s} e_i$ is very ample (4.6, $r = s \leq 6$). So $\overline{D}$ is very ample (II, Ex. 7.5), and Bertini gives the required curve.

Case $s = 0$. $D = al$ with $a \geq 1$ (since $D^2 = a^2 > 0$ and $D.F_{12} = a \geq 0$); a general plane curve of degree a is irreducible, nonsingular and misses $P_1, \ldots, P_6$ (Bertini on $\mathbf{P}^2$).

Case $s = 2$. $\overline{D} = al - b_1 e_1 - b_2 e_2$, $b_1 \geq b_2 \geq 1$, $\gamma := D.F_{12} = a - b_1 - b_2 \geq 0$. If $\gamma \geq 1$: $\overline{D} = -K_{X_2} + N$ with $N = (a-3)l - (b_1 - 1)e_1 - (b_2 - 1)e_2 = (\gamma - 1)D_0 + (b_1 - b_2)D_1 + (b_2 - 1)D_2$, a nonnegative combination of base point free classes, and $-K_{X_2}$ is very ample (4.6): $\overline{D}$ is very ample. If $\gamma = 0$: $\overline{D} = b_1(l - e_1) + b_2(l - e_2)$. By Ex. 4.1, X_2 is $\mathbf{P}^1 \times \mathbf{P}^1$ blown up at one point, the classes $l - e_1, l - e_2$ being the pull-backs of the two rulings; so $\overline{D}$ is the pull-back of the divisor class of type (b_1, b_2) (or (b_2, b_1)), which is very ample on $\mathbf{P}^1 \times \mathbf{P}^1$ since $b_1, b_2 \geq 1$. A general member of that system avoids the centre of the blow-up and the images of $P_3, \ldots, P_6$; it is irreducible and nonsingular (Bertini), and its pull-back is the required curve.

Case $s = 1$. $\overline{D} = al - b_1 e_1$, $b_1 \geq 1$; $D^2 > 0$ and $D.F_{12} = a - b_1 \geq 0$ give $a > b_1$. On X_1 (the rational ruled surface of invariant $e = 1$ with $C_0 = e_1$, $f = l - e_1$, (2.11.5)): $al - b_1 e_1 = (a - b_1)C_0 + af$. By Cor. 2.18(a) it is very ample, since $a - b_1 > 0$ and $a > (a - b_1)e = a - b_1$. Bertini gives the curve.

So in every case D contains an irreducible nonsingular curve. □

Exercise 4.9

Problem. If C is an irreducible non-singular curve of degree d on the cubic surface, and if the genus $g > 0$, then

$$g \geq \begin{cases} \frac{1}{2}(d-6) & \text{if } d \text{ is even, } d \geq 8, \\ \frac{1}{2}(d-5) & \text{if } d \text{ is odd, } d \geq 13, \end{cases}$$

and this minimum value of $g > 0$ is achieved for each d in the range given.

Solution. Let $C \sim al - \sum b_i e_i$. Since $g > 0$, C is not rational, so not a line, and $C.L \geq 0$ for all lines L. By Lemma A (in Ex. 4.8) we may assume $b_1 \geq \cdots \geq b_6 \geq 0$ and $a \geq b_1+b_2+b_3$. We have $d = 3a-\sum b_i$ and $g = \frac{1}{2}(a-1)(a-2)-\frac{1}{2}\sum b_i(b_i-1)$ (4.8). Put $x = b_1$, $m = a-x$ $(\geq b_2+b_3 \geq 0)$, $S = \sum_{i\geq 2} b_i$, $T = \sum_{i\geq 2} b_i(b_i-1)$. A direct computation gives

$$2g = (m-1)(2x+m-2) - T, \qquad d = 2x+3m-S,$$

and therefore

$$E := 2g-(d-6) = (m-2)(2x+m-4) + \sum_{i\geq 2} b_i(2-b_i). \tag{$**$}$$

$m \geq 2$. If $m \leq 1$ then $2g = -T \leq 0$ for $m = 1$, and for $m = 0$ we get $b_2 = b_3 = 0$, $T = 0$, $2g = 2-2x$, so $g > 0$ forces $x = 0$, $a = 0$, impossible. So $m \geq 2$ and $b_2+b_3 \leq m$, $b_i \leq b_3 \leq m/2$ for $i \geq 3$.

Claim: $E \geq 0$ unless $(m,x) = (3,0)$, i.e. $C \sim 3l$, $d = 9$.

$m = 2$: then $b_2 \leq 2$ and $b_2 = 2$ forces $b_3 = \cdots = 0$; every term $b_i(2-b_i) \geq 0$ (as $b_i \leq 2$), so $E \geq 0$. *$m = 3$:* $E = (2x-1)+\sum b_i(2-b_i)$. If $x = 0$ all $b_i = 0$, $a = 3$ (the exceptional case). If $x \geq 1$: for $b_2 = 3$ (then $b_3 = \cdots = 0$, $x \geq 3$): $E \geq 5-3 = 2$; for $b_2 = 2$ (then $b_3 \leq 1$): the sum is ≥ 0 and $E \geq 2x-1 \geq 3$; for $b_2 \leq 1$: sum ≥ 0, $E \geq 1$. *$m \geq 4$:* then $E \geq (m-2)(2b_2+m-4)-\sum_{i\geq 2} f(b_i)$ with $f(t) = \max(0, t(t-2))$, nondecreasing in $t \geq 0$, so $\sum_{i\geq 2} f(b_i) \leq f(u)+4f(v)$ with $u = b_2$, $v = b_3$ $(x \geq u \geq v,\ u+v \leq m)$. Put $\Phi = (m-2)(2u+m-4)$. If $v \leq 2$, then $f(v) = 0$ and $f(u) \leq u(m-2) \leq \Phi$ (as $u \leq m$, $m \geq 4$). If $v \geq 3$, then $m \geq 6$ and the bound is $u(u-2)+4v(v-2)$ with $3 \leq v \leq \min(u, m-u)$. For $u \leq m/2$ it is $\leq 5u(u-2)$ and $g_1(u) = \Phi - 5u(u-2)$ is concave in u with $g_1(3) = m^2-19 > 0$, $g_1(m/2) = \frac{3}{4}m^2-3m+8 > 0$, so $g_1 > 0$ on $[3, m/2]$. For $u \geq m/2$ the bound is increasing in v, hence $\leq u(u-2)+4(m-u)(m-u-2)$ (for $u \leq m-3$), a convex function of u, while Φ is linear in u; at $u = m/2$ the difference is $g_1(m/2) > 0$ and at $u = m-3$ it is $(m-2)(3m-10)-(m^2-8m+27) = 2m^2-8m-7 > 0$; by concavity of the difference it is positive on $[m/2, m-3]$. So $E \geq 0$ in all cases.

Conclusion. If d is even, $2g \geq d-6$, i.e. $g \geq \frac{1}{2}(d-6)$ (this holds for all even d with $g > 0$). If d is odd and $d \neq 9$, then $2g$ is even and $d-6$ is odd, so $E = 2g-(d-6)$ is odd and ≥ 0, hence ≥ 1 and $2g \geq d-5$, $g \geq \frac{1}{2}(d-5)$. (For $d = 9$ the plane-cubic class $3l$ gives $g = 1 < 2$; $d = 9$ is not in the range of the exercise, and neither is $d = 11$ although the inequality holds there too.)

The minimum is attained. For $d = 2a-2+2 = 2a+2$ even, $a = \frac{d-2}{2} \geq 3$: $D = al-(a-2)e_1$ has $d = 3a-(a-2) = 2a+2$, $D^2 = a^2-(a-2)^2 = 4a-4 > 0$, and $D.L \geq 0$ for all lines: $D.E_1 = a-2$, $D.E_i = 0$ $(i \geq 2)$, $D.F_{1j} = 2$, $D.F_{ij} = a$ $(1 \notin \{i,j\})$, $D.G_1 = 2a$, $D.G_j = a+2$ $(j \geq 2)$. By Ex. 4.8(c) it contains an irreducible nonsingular curve, of genus $\frac{1}{2}(4a-4-2a-2)+1 = a-2 = \frac{1}{2}(d-6)$. For $d = 2a+1$ odd, $a = \frac{d-1}{2} \geq 3$: $D = al-(a-2)e_1-e_2$ has degree $3a-(a-2)-1 = 2a+1$, $D^2 = a^2-(a-2)^2-1 = 4a-5 > 0$, and $D.E_1 = a-2$, $D.E_2 = 1$, $D.E_i = 0$ $(i \geq 3)$, $D.F_{12} = 1$, $D.F_{1j} = 2$ $(j \geq 3)$, $D.F_{2j} = a-1$, $D.F_{ij} = a$ $(i,j \geq 3)$, $D.G_1 = 2a-1$, $D.G_2 = a+2$, $D.G_j = a+1$ $(j \geq 3)$, all ≥ 0; so by Ex. 4.8 there is an irreducible nonsingular curve in $|D|$, of genus $\frac{1}{2}(4a-5-2a-1)+1 = a-2 = \frac{1}{2}(d-5)$. These are the plane curves of degree a with an $(a-2)$-fold point (and, for d odd, one more simple point). □

Exercise 4.10

Problem. A curious consequence of the implication (iv)⇒(iii) of (4.11) is the following numerical fact: Given integers $a, b_1, \ldots, b_6$ such that $b_i > 0$ for each i, $a - b_i - b_j > 0$ for each i, j, and $2a - \sum_{i \neq j} b_i > 0$ for each j, we must necessarily have $a^2 - \sum b_i^2 > 0$. Prove this directly (for $a, b_1, \ldots, b_6 \in \mathbf{R}$) using methods of freshman calculus.

Solution. I read the second condition for $i \neq j$ (this is the line condition $D.F_{ij} > 0$); requiring it also for $i = j$ would only strengthen the hypothesis. The hypotheses are symmetric in $b_1, \ldots, b_6$, so we may assume $b_1 \geq b_2 \geq \cdots \geq b_6 > 0$. Then the hypotheses give $u := a - b_1 - b_2 > 0$ and (taking $j = 6$) $v := 2a - (b_1 + \cdots + b_5) > 0$. Put $x_i = b_i - b_{i+1} \geq 0$ for $1 \leq i \leq 5$ and $x_6 = b_6 > 0$. Expanding both sides one checks the identity

$$\begin{aligned} a^2 - \sum_{i=1}^{6} b_i^2 = {} & \tfrac{1}{2}\, uv + \tfrac{3}{2}\, ux_1 + ux_2 + \tfrac{7}{2}\, ux_3 + \tfrac{1}{2}\, ux_5 + \tfrac{1}{2}\, ux_6 \\ & + vx_2 + 2vx_4 + 2vx_5 + 2vx_6 \\ & + x_1x_2 + 2x_1x_3 + 3x_2x_3 + x_2x_5 + x_2x_6 + x_3^2 + 2x_4x_5 + 2x_4x_6 + x_5^2 + 2x_5x_6. \end{aligned}$$

(The identity is a polynomial identity in the seven independent variables $u, x_1, \ldots, x_6$, after substituting $b_i = x_i + x_{i+1} + \cdots + x_6$, $a = u + b_1 + b_2$ and $v = 2a - b_1 - \cdots - b_5$, i.e. $v = 2u + b_1 + b_2 - b_3 - b_4 - b_5$. It is verified by expanding both sides; I also checked it by computer algebra.) All terms on the right are ≥ 0 (as $u, v, x_i \geq 0$) and the first term $\frac{1}{2}uv$ is > 0. Hence $a^2 - \sum b_i^2 > 0$. □

Exercise 4.11

Problem. *The Weyl Groups.* Given any diagram consisting of points and line segments joining some of them, we define an abstract group, given by generators and relations, as follows: each point represents a generator x_i. The relations are $x_i^2 = 1$ for each i; $(x_i x_j)^2 = 1$ if i and j are not joined by a line segment, and $(x_i x_j)^3 = 1$ if i and j are joined by a line segment.

(a) The Weyl group $\mathbf{A}_n$ is defined using the diagram of $n-1$ points, each joined to the next. Show that it is isomorphic to the symmetric group Σ_n as follows: map the generators of $\mathbf{A}_n$ to the elements $(12), (23), \ldots, (n-1, n)$ of Σ_n, to get a surjective homomorphism $\mathbf{A}_n \to \Sigma_n$. Then estimate the number of elements of $\mathbf{A}_n$ to show in fact it is an isomorphism.

(b) The Weyl group $\mathbf{E}_6$ is defined using the diagram of five points $x_1, \ldots, x_5$ in a row, joined to the next, together with a sixth point y joined to the middle point x_3. Call the generators $x_1, \ldots, x_5$ and y. Show that one obtains a surjective homomorphism $\mathbf{E}_6 \to G$, the group of automorphisms of the configuration of 27 lines (4.10.1), by sending $x_1, \ldots, x_5$ to the permutations $(12), (23), \ldots, (56)$ of the E_i, respectively, and y to the element associated with the quadratic transformation based at P_1, P_2, P_3.

∗(c) Estimate the number of elements in $\mathbf{E}_6$, and thus conclude that $\mathbf{E}_6 \cong G$.

Note: See Manin [3, Sections 25, 26] for more about Weyl groups, root systems, and exceptional curves. (Starred in the book: part (c).)

Solution. (a) The elements $s_i = (i\ i+1)$ of Σ_n satisfy $s_i^2 = 1$, $(s_i s_j)^2 = 1$ if $|i - j| \geq 2$ (disjoint transpositions) and $(s_i s_{i+1})^3 = 1$ (a 3-cycle has order 3). So $x_i \mapsto s_i$ defines a homomorphism

$\mathbf{A}_n \to \Sigma_n$, surjective because adjacent transpositions generate Σ_n. To show $|\mathbf{A}_n| \le n!$ we use induction on n ($\mathbf{A}_1$ is trivial, $\mathbf{A}_2$ is generated by one involution, so $|\mathbf{A}_2| \le 2$). Let $H \subseteq \mathbf{A}_n$ be the subgroup generated by $x_1, \dots, x_{n-2}$; it is a quotient of $\mathbf{A}_{n-1}$ (the defining relations of $\mathbf{A}_{n-1}$ hold), so $|H| \le (n-1)!$. Put $c_n = 1$ and $c_j = x_{n-1}x_{n-2}\cdots x_j$ $(1 \le j \le n-1)$. I claim $\bigcup_{j=1}^{n} Hc_j$ is stable under right multiplication by each generator x_i; since it contains 1 and the x_i are involutions this implies that it is all of $\mathbf{A}_n$, so $[\mathbf{A}_n : H] \le n$ and $|\mathbf{A}_n| \le n\,(n-1)! = n!$. The verification uses only the relations: for c_j $(1 \le j \le n-1)$, if $i = j$ then $c_j x_j = c_{j+1}$; if $i = j-1$ then $c_j x_{j-1} = c_{j-1}$; if $i < j-1$ then x_i commutes with $x_j, \dots, x_{n-1}$ and $c_j x_i = x_i c_j$; if $i = j+1$ then

$$c_j x_{j+1} = x_{n-1}\cdots x_{j+2}(x_{j+1}x_j x_{j+1}) = x_{n-1}\cdots x_{j+2}(x_j x_{j+1} x_j) = x_j c_j$$

(using $x_j x_{j+1} x_j = x_{j+1} x_j x_{j+1}$, from $(x_j x_{j+1})^3 = 1$, and that x_j commutes with $x_{j+2}, \dots, x_{n-1}$); if $i \ge j+2$ then x_i commutes with $x_{i-2}, \dots, x_j$ and

$$c_j x_i = x_{n-1}\cdots x_{i+1}(x_i x_{i-1} x_i)x_{i-2}\cdots x_j = x_{n-1}\cdots x_{i+1}(x_{i-1} x_i x_{i-1})x_{i-2}\cdots x_j = x_{i-1} c_j.$$

In all cases the result lies in $Hc_{j'}$ for some j' (the elements x_i, x_j, x_{i-1} occurring as left factors are in H because their indices are $\le n-2$). For $c_n = 1$: $x_i \in H$ for $i \le n-2$ and $x_{n-1} = c_{n-1}$. Hence $|\mathbf{A}_n| \le n! = |\Sigma_n|$, and the surjection $\mathbf{A}_n \to \Sigma_n$ is an isomorphism. □(a)

(b) *The group G and Pic.* By (4.10.1) G acts simply transitively on ordered sextuples of mutually skew lines. Let N be the 27×27 matrix of intersection numbers of the lines. The classes of the lines span $\operatorname{Pic} X$ (e.g. $e_i = [E_i]$ and $l = [F_{12}] + e_1 + e_2$) and the intersection form is nondegenerate (unimodular) on $\operatorname{Pic} X \cong \mathbf{Z}^7$. An element $g \in G$ preserves the incidence relations, hence N (intersection numbers of lines are -1 on the diagonal and 0 or 1 off it, determined by incidence). If $\sum n_L[L] = 0$ in $\operatorname{Pic} X$ then $Nn = 0$ (intersect with each line), so $Nn' = 0$ for $n' = n \circ g^{-1}$, so $\sum n'_L[L]$ is orthogonal to all lines, hence 0. Therefore $[L] \mapsto [gL]$ extends to a linear map T_g of $\operatorname{Pic} X$, an isometry; $g \mapsto T_g$ is an injective group homomorphism $G \to O(\operatorname{Pic} X)$ (injective because T_g determines the permutation of the lines).

The generators. Let $\sigma_i \in G$ be the automorphism with $E_i \leftrightarrow E_{i+1}$ and the other E_m fixed (4.10.1), and $\tau \in G$ the automorphism with $E_1, \dots, E_6 \mapsto F_{23}, F_{13}, F_{12}, E_4, E_5, E_6$ (these six lines are mutually skew); this is the automorphism associated with the quadratic transformation based at P_1, P_2, P_3 (Case 2 in the proof of (4.10)): the new blow-down π' has $l' = 2l - e_1 - e_2 - e_3$ and $e'_1 = l - e_2 - e_3$, $e'_2 = l - e_1 - e_3$, $e'_3 = l - e_1 - e_2$, $e'_j = e_j$ $(j \ge 4)$ (Ex. 4.2). Put $\alpha_i = e_i - e_{i+1}$ $(1 \le i \le 5)$, $\alpha_0 = l - e_1 - e_2 - e_3$; for $\alpha \in \operatorname{Pic} X$ with $\alpha^2 = -2$, $\alpha.K = 0$ let $s_\alpha(z) = z + (z.\alpha)\alpha$, an isometric involution with $s_\alpha(K) = K$. Then s_{α_i} swaps e_i, e_{i+1} and fixes the other e_m and l; s_{α_0} is $l \mapsto l'$, $e_i \mapsto e'_i$ (e.g. $s_{\alpha_0}(e_1) = e_1 + (e_1.\alpha_0)\alpha_0 = e_1 + \alpha_0 = l - e_2 - e_3$). *Identification of T_{σ_i} and T_τ.* For $m \ne i, i+1$, $T_{\sigma_i}(e_m) = e_m$, and T_{σ_i} swaps e_i, e_{i+1}. Moreover $l = [F_{12}] + e_1 + e_2$, and $\sigma_i F_{12}$ is the unique line meeting $\sigma_i E_1, \sigma_i E_2$ and no other E_m (incidence is preserved), i.e. the line $F_{\sigma(1)\sigma(2)}$ for the transposition $\sigma = (i\ i+1)$, of class $l - e_{\sigma(1)} - e_{\sigma(2)}$ (4.10.1); so $T_{\sigma_i}(l) = l - e_{\sigma(1)} - e_{\sigma(2)} + e_{\sigma(1)} + e_{\sigma(2)} = l$ and $T_{\sigma_i} = s_{\alpha_i}$ (they agree on the basis $l, e_1, \dots, e_6$). Similarly $T_\tau(e_m) = e'_m$, and τF_{12} is the unique line meeting $E'_1 = F_{23}$, $E'_2 = F_{13}$ and no other E'_m; for the blow-down structure π' this is the strict transform of the line through the first two new points, of class $l' - e'_1 - e'_2$. So $T_\tau(l) = T_\tau(F_{12}) + e'_1 + e'_2 = l' = 2l - e_1 - e_2 - e_3$, and $T_\tau = s_{\alpha_0}$.

Relations. For α, β as above, $\alpha.\beta = 0$ gives $s_\alpha s_\beta = s_\beta s_\alpha$; and $\alpha.\beta = 1$ gives $s_\alpha(\beta) = \beta + \alpha = s_\beta(\alpha)$, so $s_\alpha s_\beta s_\alpha = s_{s_\alpha(\beta)} = s_{\alpha+\beta} = s_\beta s_\alpha s_\beta$, whence $(s_\alpha s_\beta)^3 = 1$. Now $\alpha_i.\alpha_{i+1} = (e_i -$

$e_{i+1}).(e_{i+1}-e_{i+2}) = -e_{i+1}^2 = 1$, $\alpha_i.\alpha_j = 0$ for $|i-j| \geq 2$, $\alpha_3.\alpha_0 = (e_3-e_4).(l-e_1-e_2-e_3) = -e_3^2 = 1$ and $\alpha_i.\alpha_0 = 0$ for $i = 1,2,4,5$ (e.g. $(e_1-e_2).\alpha_0 = 1-1 = 0$). So the reflections $s_{\alpha_1},\dots,s_{\alpha_5},s_{\alpha_0}$ satisfy the defining relations of $\mathbf{E}_6$ for the diagram $\alpha_1-\alpha_2-\alpha_3-\alpha_4-\alpha_5$ with α_0 attached to α_3. As $G \to O(\operatorname{Pic} X)$ is injective and $T_{\sigma_i} = s_{\alpha_i}$, $T_\tau = s_{\alpha_0}$, the elements $\sigma_1,\dots,\sigma_5,\tau$ satisfy the same relations in G. So $x_i \mapsto \sigma_i$, $y \mapsto \tau$ defines a homomorphism $\rho : \mathbf{E}_6 \to G$.

Surjectivity. Let $H = \rho(\mathbf{E}_6)$; it contains the σ_i, hence all permutations of $E_1,\dots,E_6$, and τ, hence (conjugating by permutations) the elements associated with the quadratic transformations based at any three of the P_i. The proof of (4.10) transforms an arbitrary ordered sextuple of skew lines into $(E_1,\dots,E_6)$ by a finite sequence of operations of exactly these two kinds (relabelling $E_2',\dots,E_5'$ and quadratic transformations based at three points of the current blow-down); each operation replaces the sextuple by its image under an element of H. Hence H acts transitively on ordered skew sextuples. Since G acts simply transitively on them (4.10.1), for $g \in G$ there is $h \in H$ with $h(E_1,\dots,E_6) = g(E_1,\dots,E_6)$, hence $g = h$. So ρ is surjective. □(b)

(c) By (4.10.1), $|G| = 27\cdot16\cdot10\cdot6\cdot2\cdot1 = 51840$. I show $|\mathbf{E}_6| \leq 51840$; then the surjection ρ is an isomorphism. Let $\mathbf{D}$ be the abstract group of the diagram $x_2-x_3-x_4-x_5$ with y attached to x_3 (type D_5); its subgroup $\mathbf{B} = \langle x_2,x_3,x_4,x_5\rangle$ is a quotient of the abstract group of the diagram of 4 points in a row, which has $120 = 5!$ elements by (a) (with $n = 5$). A Todd–Coxeter coset enumeration of $\mathbf{B}$ in $\mathbf{D}$ closes with 16 cosets, so $|\mathbf{D}| \leq 16\cdot120 = 1920$. Next, a Todd–Coxeter enumeration of the cosets of $\langle x_2,\dots,x_5,y\rangle$ in the abstract group $\mathbf{E}_6$ closes with 27 cosets, so this subgroup has index 27 in $\mathbf{E}_6$ (I carried out both enumerations by computer, with the coset enumeration of sympy). The subgroup $\langle x_2,\dots,x_5,y\rangle$ of $\mathbf{E}_6$ is a quotient of $\mathbf{D}$ (the relations of $\mathbf{D}$ hold in it), so it has at most 1920 elements, and

$$|\mathbf{E}_6| \leq 27\cdot1920 = 51840 = |G|.$$

Together with the surjection $\rho : \mathbf{E}_6 \to G$ this gives $\mathbf{E}_6 \cong G$. (Independent conceptual proof: by the computation in (b), the action of $\mathbf{E}_6$ on $K^\perp \otimes \mathbf{R}$ through s_{α_i} is the geometric representation of the Coxeter group with the form $B = -\frac{1}{2}(\ ,\ )$, since $B(\alpha,\alpha) = 1$ and $B(\alpha_i,\alpha_j) = -\frac{1}{2}$ exactly for joined points. By Tits' theorem this representation is faithful, so $\mathbf{E}_6 \to G \subseteq O(\operatorname{Pic} X)$ is injective, hence an isomorphism.) □

Exercise 4.12

> **Problem.** Use (4.11) to show that if D is any ample divisor on the cubic surface X, then $H^1(X,\mathcal{O}_X(-D)) = 0$. This is Kodaira's vanishing theorem for the cubic surface (III, 7.15).

Solution. Lemma. Let Y be a surface with $H^1(Y,\mathcal{O}_Y) = 0$ and let D be very ample on Y. Then $H^1(Y,\mathcal{O}_Y(-D)) = 0$.

Proof. By Bertini there is an irreducible nonsingular curve $C \in |D|$ (hyperplane sections of a nonsingular projective surface are connected, III, 7.9.1, and a connected nonsingular curve is irreducible). The exact sequence $0 \to \mathcal{O}_Y(-D) \to \mathcal{O}_Y \to \mathcal{O}_C \to 0$ gives

$$H^0(\mathcal{O}_Y) \to H^0(\mathcal{O}_C) \to H^1(\mathcal{O}_Y(-D)) \to H^1(\mathcal{O}_Y) = 0.$$

Both $H^0(\mathcal{O}_Y)$ and $H^0(\mathcal{O}_C)$ equal k (integral projective schemes over $k = \bar{k}$), and the first map is the identity of k; so $H^1(\mathcal{O}_Y(-D)) = 0$. □

Now X is the blow-up of $\mathbf{P}^2$ at six points, so by Prop. 3.4 (applied six times) $H^1(X, \mathcal{O}_X) \cong H^1(\mathbf{P}^2, \mathcal{O}) = 0$ (III, 5.1). If D is ample on X, then D is very ample by Thm. 4.11 ((ii)⇒(i)), and the Lemma gives $H^1(X, \mathcal{O}_X(-D)) = 0$. □

Exercise 4.13

Problem. Let X be the Del Pezzo surface of degree 4 in $\mathbf{P}^4$ obtained by blowing up 5 points of $\mathbf{P}^2$ (4.7).

(a) Show that X contains 16 lines.

(b) Show that X is a complete intersection of two quadric hypersurfaces in $\mathbf{P}^4$ (the converse follows from (4.7.1)).

Solution. Here $P_1, \dots, P_5$ are five points of $\mathbf{P}^2$, no three collinear, $\pi : X \to \mathbf{P}^2$ the blow-up with exceptional curves E_i, and $X \subseteq \mathbf{P}^4$ is embedded by $|h|$, $h = 3l - \sum_{i\le 5} e_i = -K$ (Cor. 4.7, $r = 5$); $h^2 = 9 - 5 = 4$.

(a) A line of $\mathbf{P}^4$ contained in X is an irreducible curve $\Gamma \subseteq X$ with $h.\Gamma = 1$, $p_a(\Gamma) = 0$, so $\Gamma^2 = 2p_a - 2 - K.\Gamma = -2 + 1 = -1$. Conversely an irreducible curve with $h.\Gamma = 1$ is a line. So we look for irreducible curves Γ with $-K.\Gamma = 1$, $\Gamma^2 = -1$. If $\Gamma \neq E_i$, then $\Gamma \sim al - \sum b_i e_i$ with $a = \Gamma.l > 0$ and $b_i = \Gamma.E_i \geq 0$ (Prop. 3.6; $a = 0$ would mean Γ is contracted, i.e. $\Gamma = E_i$), and

$$3a - \sum_{i=1}^{5} b_i = 1, \qquad a^2 - \sum b_i^2 = -1.$$

By Schwarz, $(\sum b_i)^2 \leq 5 \sum b_i^2$: $(3a-1)^2 \leq 5(a^2+1)$, i.e. $(2a+1)(a-2) \leq 0$, so $a \leq 2$. If $a = 1$: $\sum b_i = 2$, $\sum b_i^2 = 2$, so two b_i equal 1: $\Gamma \sim l - e_i - e_j$. If $a = 2$: $\sum b_i = 5 = \sum b_i^2$, so all $b_i = 1$: $\Gamma \sim 2l - \sum_{i\le 5} e_i$. Conversely each of these classes is represented by a line: E_i (5 of them, a curve $\cong \mathbf{P}^1$ of degree $h.e_i = 1$); the strict transform of the line P_iP_j ($\binom{5}{2} = 10$ of them; it contains no third point since no three P_i are collinear, degree $3 - 2 = 1$); and the strict transform of the unique conic through the five points (irreducible since no three are collinear, degree $6 - 5 = 1$). In total $5 + 10 + 1 = 16$ lines.

(b) *Cohomology.* The Lemma of Ex. 4.12 applies to X (here $H^1(\mathcal{O}_X) = 0$, X being a blow-up of $\mathbf{P}^2$). By Riemann–Roch (1.6), with $p_a(X) = 0$: $\chi(\mathcal{O}_X(nh)) = \frac{1}{2}nh.(nh - K) + 1 = \frac{1}{2}n(n+1)h^2 + 1 = 2n(n+1) + 1$, so $\chi(\mathcal{O}(h)) = 5$ and $\chi(\mathcal{O}(2h)) = 13$. By Serre duality and $K = -h$: $H^2(\mathcal{O}_X(nh)) \cong H^0(\mathcal{O}_X(-(n+1)h))^\vee = 0$ and $H^1(\mathcal{O}_X(nh)) \cong H^1(\mathcal{O}_X(-(n+1)h))^\vee = 0$ for $n \geq 1$ by the Lemma ($(n+1)h$ is very ample). Hence $h^0(\mathcal{O}_X(h)) = 5$ and $h^0(\mathcal{O}_X(2h)) = 13$.

Quadrics. Since $X \subseteq \mathbf{P}^4$ is embedded by a complete linear system with $h^0 = 5$, X is not contained in a hyperplane. The restriction $H^0(\mathbf{P}^4, \mathcal{O}(2)) \to H^0(X, \mathcal{O}_X(2))$ goes from a 15-dimensional space to a 13-dimensional one, so there are two linearly independent quadratic forms q_1, q_2 vanishing on X. Each quadric $Q_i = V(q_i)$ is irreducible (a reducible quadric is a union of two hyperplanes, and the irreducible X would lie in one), and $Q_1 \neq Q_2$. So $Y = Q_1 \cap Q_2$ has pure dimension 2 (each component has dimension $\geq 4 - 2$, and < 3), is a complete intersection, hence Cohen–Macaulay, and by Bezout (I, 7.7) $\sum_j i(Q_1, Q_2; Z_j) \deg Z_j = 4$, the sum over the irreducible components Z_j of Y. Now X is one of the Z_j with $\deg X = 4$ and $i \geq 1$, so $Y = X$ as sets, $i = 1$, and Y is generically reduced; a Cohen–Macaulay scheme has no embedded

components, so $Y = X$ as schemes: $X = Q_1 \cap Q_2$ is a complete intersection (and, as (q_1, q_2) is a saturated ideal being a complete intersection ideal, $I(X) = (q_1, q_2)$). □

Exercise 4.14

Problem. Using the method of (4.13.1), verify that there are nonsingular curves in $\mathbf{P}^3$ with $d = 8$, $g = 6, 7$; $d = 9$, $g = 7, 8, 9$; $d = 10$, $g = 8, 9, 10, 11$. Combining with (IV, Section 6), this completes the determination of all possible g for curves of degree $d \leq 10$ in $\mathbf{P}^3$.

Solution. By Cor. 4.13(a), a class $D \sim al - \sum b_i e_i$ on the cubic surface is ample (equivalently very ample) iff $b_i > 0$ for all i, $a > b_i + b_j$ for $i \neq j$, and $2a > \sum_{i \neq j} b_i$ for each j; then by Cor. 4.13(b) it contains an irreducible nonsingular curve C, a nonsingular curve in $\mathbf{P}^3$, of degree $d = 3a - \sum b_i$ and genus $g = \frac{1}{2}(a-1)(a-2) - \frac{1}{2}\sum b_i(b_i - 1)$ (4.8). The following classes work. In each row, the conditions say: $\max(b_i + b_j)$ $(i \neq j) < a$, and $2a - (\sum b_i - \min b_j) > 0$.

(d, g)	$(a; b_1, \ldots, b_6)$	$\sum b_i$	$\max_{i \neq j}(b_i + b_j)$	$2a - (\sum b_i - \min b)$	g
$(8, 6)$	$(6; 2, 2, 2, 2, 1, 1)$	10	$4 < 6$	$12 - 9 = 3 > 0$	$10 - 4 = 6$
$(8, 7)$	$(7; 3, 2, 2, 2, 2, 2)$	13	$5 < 7$	$14 - 11 = 3 > 0$	$15 - 8 = 7$
$(9, 7)$	$(6; 2, 2, 2, 1, 1, 1)$	9	$4 < 6$	$12 - 8 = 4 > 0$	$10 - 3 = 7$
$(9, 8)$	$(7; 3, 2, 2, 2, 2, 1)$	12	$5 < 7$	$14 - 11 = 3 > 0$	$15 - 7 = 8$
$(9, 9)$	$(7; 2, 2, 2, 2, 2, 2)$	12	$4 < 7$	$14 - 10 = 4 > 0$	$15 - 6 = 9$
$(10, 8)$	$(6; 2, 2, 1, 1, 1, 1)$	8	$4 < 6$	$12 - 7 = 5 > 0$	$10 - 2 = 8$
$(10, 9)$	$(7; 3, 2, 2, 2, 1, 1)$	11	$5 < 7$	$14 - 10 = 4 > 0$	$15 - 6 = 9$
$(10, 10)$	$(7; 2, 2, 2, 2, 2, 1)$	11	$4 < 7$	$14 - 10 = 4 > 0$	$15 - 5 = 10$
$(10, 11)$	$(8; 3, 3, 2, 2, 2, 2)$	14	$6 < 8$	$16 - 12 = 4 > 0$	$21 - 10 = 11$

The degrees are $d = 3a - \sum b_i$: $18 - 10 = 8$, $21 - 13 = 8$, $18 - 9 = 9$, $21 - 12 = 9$, $21 - 12 = 9$, $18 - 8 = 10$, $21 - 11 = 10$, $21 - 11 = 10$, $24 - 14 = 10$. (In the genus column, $\frac{1}{2}(a-1)(a-2)$ is $10, 15, 10, 15, 15, 10, 15, 15, 21$ and $\frac{1}{2}\sum b_i(b_i - 1)$ is the number of the $b_i = 2$ plus 3 for each $b_i = 3$.) So there exist nonsingular curves in $\mathbf{P}^3$ with $(d, g) = (8, 6), (8, 7), (9, 7), (9, 8), (9, 9), (10, 8), (10, 9), (10, 10), (10, 11)$, as required. (That the remaining values of g are excluded is the content of (IV, Section 6), as stated in the exercise.) □

Exercise 4.15

Problem. Let $P_1, \ldots, P_r$ be a finite set of (ordinary) points of $\mathbf{P}^2$, no 3 collinear. We define an *admissible transformation* to be a quadratic transformation (4.2.3) centered at some three of the P_i (call them P_1, P_2, P_3). This gives a new $\mathbf{P}^2$, and a new set of r points, namely Q_1, Q_2, Q_3, and the images of $P_4, \ldots, P_r$. We say that $P_1, \ldots, P_r$ are *in general position* if no three are collinear, and furthermore after any finite sequence of admissible transformations, the new set of r points also has no three collinear.

(a) A set of 6 points is in general position if and only if no three are collinear and not all six lie on a conic.

(b) If $P_1, \ldots, P_r$ are in general position, then the r points obtained by any finite sequence of admissible transformations are also in general position.

(c) Assume the ground field k is uncountable. Then given $P_1, \dots, P_r$ in general position, there is a dense subset $V \subseteq \mathbf{P}^2$ such that for any $P_{r+1} \in V$, $P_1, \dots, P_{r+1}$ will be in general position. [*Hint*: Prove a lemma that when k is uncountable, a variety cannot be equal to the union of a countable family of proper closed subsets.]

(d) Now take $P_1, \dots, P_r \in \mathbf{P}^2$ in general position, and let X be the surface obtained by blowing up $P_1, \dots, P_r$. If $r = 7$, show that X has exactly 56 irreducible nonsingular curves C with $g = 0$, $C^2 = -1$, and that these are the only irreducible curves with negative self-intersection. Ditto for $r = 8$, the number being 240.

∗(e) For $r = 9$, show that the surface X defined in (d) has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$. [*Hint*: Let L be the line joining P_1 and P_2. Show that there exist finite sequences of admissible transformations such that the strict transform of L becomes a plane curve of arbitrarily high degree.]

This example is apparently due to Kodaira — see Nagata [5, II, p. 283]. (Starred in the book: part (e).)

Solution. *Preliminaries.* Let P_1, P_2, P_3 be three of the points (non-collinear). The admissible transformation based at them is the quadratic transformation of (4.2.3): $X_3 = \mathrm{Bl}_{P_1,P_2,P_3}\mathbf{P}^2$ has two blow-downs, $\pi : X_3 \to \mathbf{P}^2$ (classes l, e_i) and $\psi : X_3 \to \mathbf{P}^2$ (classes l', e_i', blow-up of the new points Q_1, Q_2, Q_3), where $l' = 2l - e_1 - e_2 - e_3$ and $e_i' = l - e_j - e_k$ for $\{i, j, k\} = \{1, 2, 3\}$; ψ contracts $\widetilde{L}_{jk}$ (of class e_i') to Q_i and maps E_i onto the line Q_jQ_k. On the blow-up X of all r points the same transformation changes the blow-down structure: $(l; e_1, \dots, e_r)$ is replaced by $(l'; e_1', e_2', e_3', e_4, \dots, e_r)$, the exceptional curves of the new structure being $\widetilde{L}_{23}, \widetilde{L}_{13}, \widetilde{L}_{12}, E_4, \dots, E_r$ and the centres the new set of r points. By Ex. 4.2 a plane curve of degree a with multiplicities b_i at the P_i becomes a plane curve of degree $a' = 2a - b_1 - b_2 - b_3$ with multiplicities $b_1' = a - b_2 - b_3$, $b_2' = a - b_1 - b_3$, $b_3' = a - b_1 - b_2$, $b_m' = b_m$ $(m \geq 4)$, provided its strict transform is not contracted by ψ.

(a) Let $S = \{P_1, \dots, P_6\}$ with no three collinear. *If the six points lie on a conic* Γ *then* S *is not in general position*: Γ is irreducible (no three collinear), of degree 2 with multiplicity 1 at P_1, P_2, P_3; its strict transform under the admissible transformation based at P_1, P_2, P_3 has degree $2 \cdot 2 - 3 = 1$ (Ex. 4.2) and passes through the images of P_4, P_5, P_6: three collinear points.

Conversely let S satisfy $\mathcal{P}$: "no three collinear and not all six on a conic". I show that an admissible transformation, based on P_1, P_2, P_3 say, produces a set $S' = \{Q_1, Q_2, Q_3, P_4', P_5', P_6'\}$ with $\mathcal{P}$; then by induction every finite sequence leaves $\mathcal{P}$ (in particular "no three collinear") invariant, so S is in general position. Let $X = X_3$ as above; $P_a' = \psi(\widetilde{P}_a)$ for $a \geq 4$, where $\widetilde{P}_a = \pi^{-1}(P_a)$ is not on any E_i and (since no three of the P's are collinear) not on any $\widetilde{L}_{jk}$.

No three of S' *collinear.* Let ℓ' be a line of the new plane containing three points of S'; let $I = \{i : Q_i \in \ell'\}$ ($|I| \leq 2$, the Q_i being non-collinear). Its strict transform $\widetilde{\ell'}$ in X satisfies $\psi^*\ell' = \widetilde{\ell'} + \sum_{i \in I} E_i'$, so $\widetilde{\ell'} \sim l' - \sum_{i \in I} e_i'$. If $I = \emptyset$: $\widetilde{\ell'} \sim 2l - e_1 - e_2 - e_3$, an irreducible conic (the image of $\widetilde{\ell'}$ in the old plane, which has degree $\widetilde{\ell'}.l = 2$ and multiplicity $\widetilde{\ell'}.e_i = 1$ at P_1, P_2, P_3) through P_1, P_2, P_3 which contains the three points P_4, P_5, P_6 (as $\widetilde{P}_a \in \widetilde{\ell'}$): six points on a conic, contradicting $\mathcal{P}$. If $I = \{i\}$: $\widetilde{\ell'} \sim l - e_i$, a line through P_i, which contains two of P_4, P_5, P_6: three collinear points, contradiction. If $I = \{i, j\}$: $\widetilde{\ell'} \sim e_k$, so $\widetilde{\ell'} = E_k$, and the third point of S' on ℓ' is some P_a'; but $\psi^{-1}(\ell') = E_k \cup \widetilde{L}_{jk} \cup \widetilde{L}_{ik}$ does not contain $\widetilde{P}_a$ (a point of $\widetilde{L}_{ik}$ would make P_a, P_i, P_k collinear). Contradiction.

Not all six of S' *on a conic.* Let Γ' be a conic through S'; it is irreducible (no three of S'

collinear). Then $\psi^*\Gamma' = \widetilde{\Gamma}' + E'_1 + E'_2 + E'_3$ and $\widetilde{\Gamma}' \sim 2l' - e'_1 - e'_2 - e'_3 = l$ (a short computation), an irreducible curve of class l, i.e. the pull-back of a line of the old plane, which contains $\widetilde{P}_4, \widetilde{P}_5, \widetilde{P}_6$: contradiction with $\mathcal{P}$. Hence S' satisfies $\mathcal{P}$, which proves (a). $\square$(a)

(b) This is immediate from the definition: let S be in general position and S^σ obtained from S by a finite sequence σ of admissible transformations. A finite sequence σ' of admissible transformations starting from S^σ gives, composed with σ, a finite sequence from S, so the set obtained has no three collinear, and S^σ itself has no three collinear ($\sigma' = \emptyset$). $\square$(b)

(c) *Lemma.* If k is uncountable, the plane $\mathbf{P}^2$ (and any nonempty open subset of it) is not a countable union of proper closed subsets. Proof: a proper closed subset is a finite union of points and irreducible curves. Suppose $\mathbf{P}^2 = \bigcup_n Z_n$ with Z_n proper closed. Each Z_n contains only finitely many lines, so the countably many Z_n contain only countably many lines altogether, while $\mathbf{P}^2$ has uncountably many lines; choose a line ℓ not contained in any Z_n. Then each $\ell \cap Z_n$ is finite, so $\ell = \bigcup(\ell \cap Z_n)$ would be countable, but $\ell \cong \mathbf{P}^1(k)$ is uncountable. The same holds for an open subset U (apply the argument to a line meeting U; $U \cap \ell$ is cofinite in ℓ, hence uncountable). The same is true for $(\mathbf{P}^2)^m$ and its nonempty open subsets, by induction on m: if Z_n are proper closed subsets of $(\mathbf{P}^2)^{m-1} \times \mathbf{P}^2$, the set Y_n of points x with $\{x\} \times \mathbf{P}^2 \subseteq Z_n$ is a proper closed subset of $(\mathbf{P}^2)^{m-1}$, so by induction there is $x \notin \bigcup Y_n$, and then the sets $Z_n \cap (\{x\} \times \mathbf{P}^2)$ are proper closed subsets of $\mathbf{P}^2$, whose union is not $\mathbf{P}^2$. $\square$

Let $m \geq 1$. Given a finite sequence $\sigma = (t_1, \ldots, t_n)$ of triples of indices (a "sequence of centres") and a triple τ of indices, let $B_{\sigma,\tau} \subseteq (\mathbf{P}^2)^m$ be the set of m-tuples S such that σ is an admissible sequence for S (each t_s is, at its stage, a non-collinear triple) and the points of τ are collinear in the final set; let $B_\emptyset$ be the set of tuples with three collinear points. By definition, S is in general position iff $S \notin \bigcup B_{\sigma,\tau} \cup B_\emptyset$, a countable union. Each quadratic transformation with centres of index t (after normalizing by the projectivity taking the three centres and a fourth point of the set to $(1:0:0), (0:1:0), (0:0:1), (1:1:1)$, i.e. on the dense open set of tuples where this fourth point is in general position with respect to the centres) is the restriction of the Cremona map $(x:y:z) \mapsto (yz:xz:xy)$, so defines a rational map $\Psi_t : (\mathbf{P}^2)^m \dashrightarrow (\mathbf{P}^2)^m$ which is an involution (the quadratic transformation based at Q_1, Q_2, Q_3 undoes the original one), hence *birational*. Collinearity is invariant under projectivities, so $B_{\sigma,\tau} = U_\sigma \cap \Psi_\sigma^{-1}(\mathrm{Coll}_\tau)$, where $\Psi_\sigma = \Psi_{t_n} \circ \cdots \circ \Psi_{t_1}$ is a birational self-map defined on a dense open set U_σ (a composition of birational maps), and Coll_τ is the proper closed subset of tuples whose points indexed by τ are collinear. Since Ψ_σ is dominant, $B_{\sigma,\tau}$ is a proper closed subset of U_σ (it is not all of U_σ), hence contained in a proper closed subset of $(\mathbf{P}^2)^m$ (the complement of U_σ together with its closure). Therefore:

(i) for every m, the set of m-tuples in general position contains the complement of a countable union of proper closed subsets $Z_1, Z_2, \ldots$ of $(\mathbf{P}^2)^m$; in particular it is nonempty and dense (by the Lemma for $(\mathbf{P}^2)^m$ and its nonempty open subsets).

(ii) the fibre statement. Apply (i) to $m = r+1$ and let Z_n be the proper closed subsets. For each n the set $W_n = \{S_0 \in (\mathbf{P}^2)^r : \{S_0\} \times \mathbf{P}^2 \subseteq Z_n\}$ is closed and proper (if W_n were everything, Z_n would be everything). For $S_0 \notin \bigcup_n W_n$ and every n, $Z_n \cap (\{S_0\} \times \mathbf{P}^2)$ is a proper closed subset of the fibre. By the Lemma, the set V of points P of $\mathbf{P}^2$ with $(S_0, P) \notin \bigcup Z_n$ is the complement of a countable union of proper closed subsets, hence dense, and for $P \in V$ the $(r+1)$-tuple (S_0, P) is in general position. This proves (c) for $S_0 = (P_1, \ldots, P_r)$ outside the countable union of proper closed subsets $\bigcup_n W_n \subseteq (\mathbf{P}^2)^r$ ("very general" r-tuples); in particular, starting from a

general-position tuple and choosing the points inductively (each time very general with respect to the previous ones) one produces general-position tuples of every length r, which is what (d) and (e) need.

What holds for every given S_0. Let $S_0 = (P_1, \dots, P_r)$ be any tuple in general position and consider the conditions (σ, τ) in which $P_{r+1} = P$ is never a centre of σ. The tuples $S_0^{(s)}$ obtained from S_0 after the first s steps are in general position by (b), so every step is admissible for every P, and P is carried along by a birational map $\Phi_\sigma : \mathbf{P}^2 \dashrightarrow \mathbf{P}^2$ (a composite of the Cremona maps $(x : y : z) \mapsto (yz : xz : xy)$ and projectivities, all depending only on S_0) to a point $P' = \Phi_\sigma(P)$ on the final plane. If τ does not contain the index $r+1$, its points are not collinear, since S_0^σ is in general position; if it does, collinearity means that P' lies on a line through two distinct points of S_0^σ, or that P is a point where Φ_σ is not defined or (at some step) lies on a side of the triangle of centres. As Φ_σ is dominant, these P lie in a proper closed subset of $\mathbf{P}^2$. So, for every S_0 in general position, all conditions in which P_{r+1} is not a centre exclude only a countable union of proper closed subsets of $\mathbf{P}^2$. What is left are the conditions in which P_{r+1} is itself a centre at some stage; for these I only have the argument of (ii).

Honest remark. For an arbitrary given r-tuple in general position the argument above shows only that each "bad" set for P_{r+1} is closed in an open set (of definition); that none of them contains a whole open piece of the fibre follows from the dominance argument only for S_0 outside the countable union $\bigcup W_n$. I do not have a proof that the countable union $\bigcup W_n$ is disjoint from the set of general-position tuples. So (c) is proved here for very general $P_1, \dots, P_r$ (including the existence of general-position tuples of every length), not for every given general-position tuple. $\square$(c)

(d) Let $r \in \{7, 8\}$, $P_1, \dots, P_r$ in general position, X the blow-up, $K = -3l + \sum e_i$.

Step 1: $-K.\Gamma > 0$ for every irreducible curve $\Gamma \subseteq X$. We prove this by induction on $a = \Gamma.l$, for all configurations in general position simultaneously ($-K$ is intrinsic to X). If $\Gamma = E_j$, $-K.\Gamma = 1$. Otherwise $a > 0$ and $b_i = \Gamma.E_i \geq 0$, $\Gamma \sim al - \sum b_i e_i$. Suppose first there are three indices, say $1, 2, 3$ after renumbering, with $b_1 + b_2 + b_3 > a$. Apply the admissible transformation based at P_1, P_2, P_3: we get a new blow-down structure $X \to \mathbf{P}^2$ with centres in general position (by (b); its centres are the new set of r points), in which Γ has degree $a' = 2a - b_1 - b_2 - b_3 < a$ (if $a' = 0$, Γ is contracted, so Γ is an exceptional curve of the new structure and $-K.\Gamma = 1$). Since $-K.\Gamma$ does not depend on the blow-down structure, induction applies. Otherwise $b_i + b_j + b_k \leq a$ for all triples; ordering $b_1 \geq \cdots \geq b_r$, with $s = b_1 + b_2 + b_3 \leq a$ we have $b_i \leq b_3 \leq s/3$ for $i \geq 4$ and

$$\sum_{i=1}^{r} b_i \leq s + (r-3)\frac{s}{3} = \frac{r}{3}s \leq \frac{r}{3}a < 3a \quad (r \leq 8,\ a > 0),$$

so $-K.\Gamma = 3a - \sum b_i > 0$. This proves Step 1.

Step 2. For an irreducible curve Γ, (1.5) gives $\Gamma^2 = 2p_a(\Gamma) - 2 - K.\Gamma = 2p_a(\Gamma) - 2 + (-K.\Gamma) \geq -2 + 1 = -1$ since $p_a(\Gamma) \geq 0$ and $-K.\Gamma \geq 1$. So the irreducible curves with negative self-intersection are exactly those with $\Gamma^2 = -1$, $p_a = 0$, $-K.\Gamma = 1$, i.e. the irreducible nonsingular curves with $g = 0$ and $C^2 = -1$ (nonsingular rational curves, $\cong \mathbf{P}^1$). Two distinct irreducible curves $C \neq C'$ with the same class would give $C.C' \geq 0$, but $C.C' = C^2 = -1$: so each class contains at most one irreducible curve, and the number of such curves is the number of classes which contain one.

Step 3: the possible classes. A class $(a; b_1, \dots, b_r)$ of such a curve $\Gamma \neq E_i$ satisfies $a \geq 1$, $b_i \geq 0$ and

$$\sum b_i = 3a - 1, \qquad \sum b_i^2 = a^2 + 1, \qquad \text{hence } \sum b_i(b_i - 1) = (a-1)(a-2).$$

By Schwarz, $(3a-1)^2 \leq r(a^2+1)$; for $r = 7$ this gives $a \leq 3$ and for $r = 8$, $a \leq 7$, with equality $a = 7$ excluded since it would force all $b_i = \frac{20}{8} \notin \mathbf{Z}$. The solutions are found as follows (write $b_i = c_i + t$ with t chosen to make $\sum c_i$ small):

- $a = 1$: $\sum b = 2, \sum b^2 = 2$: two b_i equal 1: $\binom{r}{2}$ classes (F_{ij}).
- $a = 2$: $\sum b = \sum b^2 = 5$, so $b_i \in \{0, 1\}$ with five ones: $\binom{r}{5}$ classes (conics through five points).
- $a = 3$: $\sum b = 8$, $\sum b^2 = 10$; with $c_i = b_i - 1$: $\sum c_i = 8 - r$, $\sum c_i^2 = 10 - 2 \cdot 8 + r$. For $r = 7$: $\sum c = 1$, $\sum c^2 = 1$: one $b_i = 2$, the others 1: 7 classes. For $r = 8$: $\sum c = 0$, $\sum c^2 = 2$: one $b_i = 2$, one $b_i = 0$, six 1's: $8 \cdot 7 = 56$ classes.
- ($r = 8$ only) $a = 4$: $\sum b = 11, \sum b^2 = 17$, $c_i = b_i - 1$: $\sum c = 3 = \sum c^2$, so $\sum c(c-1) = 0$, $c_i \in \{0, 1\}$: three $b_i = 2$, five $b_i = 1$: $\binom{8}{3} = 56$ classes. $a = 5$: $\sum b = 14$, $\sum b^2 = 26$, $\sum c = 6 = \sum c^2$: six $b_i = 2$, two $b_i = 1$: $\binom{8}{2} = 28$ classes. $a = 6$: $\sum b = 17$, $\sum b^2 = 37$, $c_i = b_i - 2$: $\sum c = 1$, $\sum c^2 = 1$: one $b_i = 3$, seven $b_i = 2$: 8 classes.

Adding the r classes e_i: for $r = 7$, 7+21+21+7 = 56; for $r = 8$, 8+28+56+56+56+28+8 = 240.

Step 4: each such class contains an irreducible curve. Induct on a: the classes e_i ($a = 0$) are the E_i. For $a \geq 1$ choose three indices with $b_i + b_j + b_k > a$: for $(1; 1, 1)$ any triple containing both 1's; for $(2; 1^5)$ three of the five 1's ($3 > 2$); for $(3; 2, 1^6[, 0])$ the 2 and two 1's ($4 > 3$); for $(4; 2^3, 1^5)$ the three 2's ($6 > 4$); for $(5; 2^6, 1, 1)$ three 2's ($6 > 5$); for $(6; 3, 2^7)$ the 3 and two 2's ($7 > 6$). The admissible transformation at these three points gives a new blow-down structure (centres in general position by (b)), in which, by the formulas of the Preliminaries, the class has the new coordinates: $(1; 1, 1, 0, \dots) \mapsto (0; 0, 0, -1, \dots) = e'_k$; $(2; 1^5) \mapsto (1; 1, 1, 0, \dots)$; $(3; 2, 1^6) \mapsto (2; 1^5)$; $(4; 2^3, 1^5) \mapsto (2; 1^5)$; $(5; 2^6, 1, 1) \mapsto (4; 2^3, 1^5)$; $(6; 3, 2^7) \mapsto (5; 2^6, 1, 1)$ (up to renumbering). By induction the class is the class of an irreducible curve in the new structure, hence in X (being the class of an irreducible curve does not depend on the blow-down structure). So each of the 56 (resp. 240) classes contains exactly one irreducible curve, which has $g = 0$, $C^2 = -1$.

Together with Steps 1, 2: X has exactly 56 (resp. 240) irreducible nonsingular curves with $g = 0$, $C^2 = -1$, and every irreducible curve with negative self-intersection is one of them. □(d)

(e) Let $r = 9$, $P_1, \dots, P_9$ in general position, L the line P_1P_2, $\widetilde{L} \sim l - e_1 - e_2$ its strict transform. All admissible transformations may be applied indefinitely, since every set obtained has no three collinear (def. of general position and (b)). *Claim: there is a sequence of admissible transformations after which $\widetilde{L}$ has degree a_n with $a_n \to \infty$.* Suppose $\widetilde{L}$ is, in the current blow-down structure, the strict transform of a plane curve of degree $a \geq 1$ with multiplicities $b_1, \dots, b_9 \geq 0$ (we shall see these stay ≥ 0); then $-K.\widetilde{L} = 3a - \sum b_i = 1$. Choose the three indices with the smallest b_i; then $b_i + b_j + b_k \leq \frac{1}{3}\sum b_m = \frac{3a-1}{3} < a$, so $b_i + b_j + b_k \leq a - 1$. The admissible transformation at these three points gives degree $a' = 2a - (b_i + b_j + b_k) \geq a + 1$ and new multiplicities $b'_i = a - b_j - b_k = b_i + (a - b_i - b_j - b_k) > b_i \geq 0$ (and similarly for j, k), the others unchanged; so $\widetilde{L}$ is again the strict transform of a plane curve (not contracted, since $a' > 0$) with nonnegative multiplicities, and $a' > a$. Starting with $a = 1$, $b = (1, 1, 0, \dots, 0)$ and

iterating, we obtain structures $\pi_n : X \to \mathbf{P}^2$ with $a_n = \widetilde{L}.l_n \geq n+1 \to \infty$, where l_n is the pulled-back line class and $E_1^{(n)}, \ldots, E_9^{(n)}$ the exceptional curves of π_n.

Now each $E_i^{(n)}$ is an irreducible nonsingular curve with $g = 0$, $(E_i^{(n)})^2 = -1$ (Prop. 3.1). Suppose there were only finitely many irreducible nonsingular curves C with $g = 0$, $C^2 = -1$ on X, say forming the finite set $\mathcal{N}$. Then all $E_i^{(n)} \in \mathcal{N}$, so $b_i^{(n)} = \widetilde{L}.E_i^{(n)} \leq M := \max\{\widetilde{L}.C : C \in \mathcal{N}\}$ (if $E_i^{(n)} = \widetilde{L}$ this number is -1). Hence $a_n = \frac{1}{3}(1 + \sum_i b_i^{(n)}) \leq \frac{1}{3}(1 + 9M)$ for all n, contradicting $a_n \to \infty$. So X has infinitely many irreducible nonsingular curves C with $g = 0$ and $C^2 = -1$. □

Exercise 4.16

> **Problem.** For the Fermat cubic $x_0^3 + x_1^3 + x_2^3 + x_3^3 = 0$, find the equations of the 27 lines explicitly, and verify their incidence relations. What is the group of automorphisms of this surface?

Solution. *Hypothesis.* Assume $\operatorname{char} k \neq 3$ (in characteristic 3 the form is $(x_0 + x_1 + x_2 + x_3)^3$, a triple plane, not a nonsingular cubic surface). The surface $X : F = \sum x_i^3 = 0$ is then nonsingular ($\partial F/\partial x_i = 3x_i^2$ vanish only at $x = 0$), so (4.9) applies. For the group in the last part I also assume $\operatorname{char} k \neq 2$ (see the end).

The equation $z^3 = -1$ has three distinct roots $\omega_0, \omega_1, \omega_2$ (namely $-1, -\omega, -\omega^2$ with ω a primitive cube root of unity; they are distinct because $\operatorname{char} k \neq 3$).

The 27 lines. For each of the three partitions of $\{0,1,2,3\}$ into two pairs, $\{01|23\}, \{02|13\}, \{03|12\}$, and for each choice of roots $\alpha, \beta \in \{\omega_0, \omega_1, \omega_2\}$ we have the line

$$L_{\alpha\beta}^{(1)} : \ x_0 = \alpha x_1, \ x_2 = \beta x_3; \qquad L_{\gamma\delta}^{(2)} : \ x_0 = \gamma x_2, \ x_1 = \delta x_3; \qquad L_{\mu\nu}^{(3)} : \ x_0 = \mu x_3, \ x_1 = \nu x_2.$$

Each lies on X: e.g. on $L_{\alpha\beta}^{(1)}$, $x_0^3 + x_1^3 = (\alpha^3 + 1)x_1^3 = 0$ and $x_2^3 + x_3^3 = (\beta^3 + 1)x_3^3 = 0$. There are $3 \cdot 3 \cdot 3 = 27$ distinct lines, so by Thm. 4.9 these are *all* the lines on X.

Incidence relations. Two of these lines meet iff the four linear equations have a common solution, i.e. a one-dimensional solution space; by direct computation:

- Lines of the same partition, e.g. $L_{\alpha\beta}^{(1)}$ and $L_{\alpha'\beta'}^{(1)}$: if $\alpha \neq \alpha'$ and $\beta \neq \beta'$ the equations force $x_0 = x_1 = x_2 = x_3 = 0$: skew. If $\alpha = \alpha'$, $\beta \neq \beta'$ they meet in the point $(\alpha : 1 : 0 : 0)$; if $\alpha \neq \alpha'$, $\beta = \beta'$ in $(0 : 0 : \beta : 1)$. So two lines of the same partition meet iff they agree in exactly one of the two parameters. (Same for the other partitions.)
- $L_{\alpha\beta}^{(1)}$ and $L_{\gamma\delta}^{(2)}$ meet iff $\alpha\delta = \beta\gamma$ (then in the point $(\alpha\delta : \delta : \beta : 1)$); $L_{\alpha\beta}^{(1)}$ and $L_{\mu\nu}^{(3)}$ meet iff $\mu = \alpha\beta\nu$; $L_{\gamma\delta}^{(2)}$ and $L_{\mu\nu}^{(3)}$ meet iff $\mu\nu = \gamma\delta$.

(I also verified these rules against the rank of the system for all $\binom{27}{2}$ pairs by computer algebra.) Consequently each line meets exactly 10 others: 4 of the same partition (two agreeing in the first parameter, two in the second) and for each of the other two partitions 3 (for the first line fixed, the condition $\alpha\delta = \beta\gamma$ determines γ from δ; three solutions, and $\gamma^3 = -1$ automatically).

This agrees with (4.9): E_i meets F_{ij} (5 lines) and G_j (5 lines), 10 in all. The 10 lines meeting a given line fall into 5 pairs of lines which meet each other (the five tritangent planes through the line), so there are $27 \cdot 5/3 = 45$ tritangent planes. *Check against the standard configuration.* The six lines $L^{(1)}_{\omega_0\omega_0}, L^{(1)}_{\omega_1\omega_1}, L^{(1)}_{\omega_2\omega_2}$ and $L^{(2)}_{\omega_0\omega_1}, L^{(2)}_{\omega_1\omega_2}, L^{(2)}_{\omega_2\omega_0}$ are mutually skew by the rules above (for lines of the same partition the two parameters differ; for $L^{(1)}_{\omega_i\omega_i}$ and $L^{(2)}_{\gamma\delta}$ the condition $\omega_i\delta = \omega_i\gamma$ fails since $\gamma \neq \delta$). Taking these as $E_1, \dots, E_6$, the remaining 21 lines are exactly the lines F_{ij} (the unique line meeting E_i, E_j and no other E_m) and G_j (the unique line meeting all E_i, $i \neq j$), in accordance with (4.10.1) (by which the remaining 21 lines are determined by $E_1, \dots, E_6$).

The group of automorphisms (assume $\operatorname{char} k \neq 2, 3$). By (4.7) $\omega_X \cong \mathcal{O}_X(-1)$, so every automorphism of X preserves the class $h = -K$, hence $\mathcal{O}_X(1)$, and acts on $H^0(X, \mathcal{O}_X(1)) = H^0(\mathbf{P}^3, \mathcal{O}(1))$ (by Riemann–Roch and the vanishing used in Ex. 4.13(b), $h^0(\mathcal{O}_X(h)) = h^2 + 1 = 4$; so the linear system $|h|$ is the complete one); so every automorphism of X is induced by a projective linear transformation $g \in PGL_4(k)$ with $g(X) = X$, i.e. with $F \circ g = cF$ for a constant c (the ideal of X is (F)). Conversely each such g gives an automorphism, and $g \neq 1$ acts nontrivially. Differentiating twice, $g^T \operatorname{Hess}(F)(gx)\, g = c \operatorname{Hess}(F)(x)$, and $\operatorname{Hess}(F) = 6 \operatorname{diag}(x_0, x_1, x_2, x_3)$, so

$$\det(g)^2 \cdot x_0' x_1' x_2' x_3' = c^4\, x_0 x_1 x_2 x_3, \qquad x' = gx.$$

By unique factorization the four linear forms $(gx)_i$ are proportional to the four coordinates in some order: g is a *monomial* matrix, $g = DP$ with P a permutation matrix and $D = \operatorname{diag}(d_0, \dots, d_3)$. Then $F \circ g = \sum d_i^3 x_{\pi(i)}^3 = cF$ forces $d_i^3 = c$ for all i, i.e. $d_i = c^{1/3}\varepsilon_i$ with $\varepsilon_i^3 = 1$. Modulo scalars we obtain

$$\operatorname{Aut}(X) \cong \left(\mu_3^4/\mu_3\right) \rtimes S_4 \cong (\mathbf{Z}/3)^3 \rtimes S_4, \qquad |\operatorname{Aut}(X)| = 27 \cdot 24 = 648,$$

the group generated by the permutations of the coordinates and by $x_i \mapsto \varepsilon x_i$ ($\varepsilon^3 = 1$). It acts on the 27 lines (faithfully: an automorphism fixing all 27 lines fixes six skew lines, so descends to a projectivity of $\mathbf{P}^2$ fixing six points no three collinear, hence is the identity; thus it is a subgroup of $G \cong W(E_6)$, of index 80 in G, with the three partitions permuted by $S_4 \to S_3$) with the lines of one partition forming one orbit under μ_3^3. (In characteristic 2 the Hessian argument is not available, and indeed the Fermat cubic is then the Hermitian surface over $\mathbf{F}_4$, whose automorphism group is the much larger group $PGU_4(2)$ of order 25920; I do not prove this here.) □

5. Birational transformations

Conventions. Throughout, k is algebraically closed, a *surface* is a nonsingular projective surface over k, a *curve* on a surface is an effective divisor, and "point" means closed point unless said otherwise. For a divisor D on a surface, $\mathcal{L}(D)$ is the corresponding invertible sheaf. We use freely that for an irreducible nonsingular curve C on a surface and any divisor D one has $D.C = \deg_C(\mathcal{L}(D) \otimes \mathcal{O}_C)$ (Lemma 1.3 and Example 1.4.1), in particular $C^2 = \deg$ of the normal bundle $N_C = \mathcal{L}(C) \otimes \mathcal{O}_C$; and that for two curves C, D without common component $C.D = \sum_P (C.D)_P \geq 0$ with every term positive at a common point (Prop. 1.4).

Exercise 5.1

Problem. Let f be a rational function on the surface X. Show that it is possible to "resolve the singularities of f" in the following sense: there is a birational morphism $g : X' \to X$ so that f induces a morphism of X' to $\mathbf{P}^1$. [*Hints*: Write the divisor of f as $(f) = \sum n_i C_i$. Then apply embedded resolution (3.9) to the curve $Y = \bigcup C_i$. Then blow up further as necessary whenever a curve of zeros meets a curve of poles until the zeros and poles of f are disjoint.]

Solution. If $(f) = 0$ then f is a nowhere vanishing regular function on the projective variety X, hence constant, and $g = \mathrm{id}$ works. Otherwise let $Y = \bigcup C_i$ be the support of (f). By Theorem 3.9 there is a composition $g_0 : X' \to X$ of monoidal transformations such that the total inverse image $g_0^{-1}(Y)$ is a divisor with normal crossings. We write f also for $f \circ g_0 \in K(X') = K(X)$. Since g_0 is an isomorphism over $X - Y$, where f has neither zeros nor poles, the divisor (f) on X' is supported on $g_0^{-1}(Y)$:

$$(f) = \sum_j a_j D_j, \qquad D_j \text{ the irreducible components of } g_0^{-1}(Y),\ a_j \in \mathbf{Z}.$$

Normal crossings on a surface means: every D_j is nonsingular, through a point pass at most two of the D_j (their local equations are part of a regular system of parameters, which has 2 elements), and if two pass they meet transversally there. Call D_j a *zero curve* if $a_j > 0$ and a *pole curve* if $a_j < 0$; let Z and P be the unions of the zero curves and of the pole curves.

Claim A. The rational map $f : X' \to \mathbf{P}^1$ is a morphism at every point $x \notin Z \cap P$. Indeed, if $x \notin P$, then $\mathrm{ord}_D(f) \geq 0$ for every irreducible curve D through x, and since $\mathcal{O}_{x,X'}$ is a regular, hence integrally closed, noetherian domain, it is the intersection of its localizations at height one primes, so $f \in \mathcal{O}_{x,X'}$ and f is a morphism to $\mathbf{A}^1 \subseteq \mathbf{P}^1$ near x. If $x \notin Z$, apply this to $1/f$.

So it remains to remove the finitely many points of $Z \cap P$. Consider the following property $(*)$ of a pair (surface X', rational function f): the irreducible components of the support of (f) are nonsingular, and through each point $Q \in Z \cap P$ pass exactly one zero curve C_x and one pole curve C_y, no other component of $\mathrm{Supp}(f)$, and C_x, C_y meet transversally at Q. After the application of (3.9), $(*)$ holds (through a point of $Z \cap P$ there are exactly two of the D_j, which must be one zero and one pole curve, as both have $a_j \neq 0$). Then, if x, y are local equations of C_x, C_y at Q, they are local parameters at Q, and $fx^{-a}y^b$ (with $a > 0$, $b > 0$ the orders of the zero and pole curve at Q) has divisor 0 near Q, hence is a unit ε in $\mathcal{O}_Q$ (a regular local ring is a UFD, so a function without zeros and poles near Q is a unit):

$$f = \varepsilon\, x^a y^{-b} \quad \text{near } Q, \qquad \varepsilon \in \mathcal{O}_Q^*.$$

Put $\mu(X', f) = \sum_{Q \in Z \cap P} (a_Q + b_Q)$. We show by induction on μ that one can reach $\mu = 0$, i.e. $Z \cap P = \emptyset$, by blowing up points while keeping $(*)$. If $\mu = 0$, we are done by Claim A. If $\mu > 0$ pick $Q \in Z \cap P$ with data a, b, x, y, ε as above, and let $\pi : X'' \to X'$ be the monoidal transformation with center Q, E its exceptional curve. By the description of the blowing up in the proof of (3.6), over a small neighbourhood V of Q the surface $\pi^{-1}(V)$ is covered by two charts with coordinates (s, t) and (s', t') in which

$$\text{chart 1 : } x = s,\ y = st; \qquad \text{chart 2 : } y = s',\ x = s't',$$

s, t (resp. s', t') are local parameters at every point of the chart, $E = \{s = 0\} = \{s' = 0\}$, the strict transform $\widetilde{C}_y$ of $C_y = \{y = 0\}$ is $\{t = 0\}$ in chart 1 and does not meet chart 2, and

the strict transform $\widetilde{C}_x$ of C_x is $\{t' = 0\}$ in chart 2 and does not meet chart 1. In particular $\widetilde{C}_x \cap \widetilde{C}_y = \emptyset$, and E meets each of them transversally in one point. Substituting,

$$f = \varepsilon\, s^{a-b}t^{-b} \quad \text{(chart 1)}, \qquad f = \varepsilon\, s'^{a-b}t'^{a} \quad \text{(chart 2)},$$

with ε still a unit. Hence over V the divisor of f is $a\widetilde{C}_x - b\widetilde{C}_y + (a-b)E$, and:

- if $a = b$, then E is not in the support, $\widetilde{C}_x$ is a zero curve, $\widetilde{C}_y$ a pole curve and they are disjoint: no point of $Z \cap P$ lies over Q;
- if $a > b$, then E is a zero curve of order $a - b$; the only point of $Z \cap P$ over Q is $E \cap \widetilde{C}_y$ (the origin of chart 1), where $f = \varepsilon s^{a-b}t^{-b}$, a point with data $(a-b, b)$;
- if $a < b$, then E is a pole curve of order $b - a$; the only point of $Z \cap P$ over Q is $E \cap \widetilde{C}_x$ (origin of chart 2), where $f = \varepsilon t'^{a} s'^{-(b-a)}$, with data $(a, b-a)$.

In all cases $(*)$ holds again (all curves are nonsingular, s, t resp. s', t' are local parameters, the two curves meet transversally), the other points of $Z \cap P$ are unchanged since π is an isomorphism away from E, and Q is replaced by at most one point whose weight is a or b, which is $< a + b$. So μ strictly decreases, and after finitely many blowing-ups $Z \cap P = \emptyset$. Let $g : X''' \to X$ be the composite of g_0 with all these monoidal transformations. By Claim A, $f \circ g$ defines a morphism $X''' \to \mathbf{P}^1$, and g is a birational morphism. □

Exercise 5.2

> **Problem.** Let $Y \cong \mathbf{P}^1$ be a curve in a surface X, with $Y^2 < 0$. Show that Y is contractible (5.7.2) to a point on a projective variety X_0 (in general singular).

Solution. Recall (Remark 5.7.2) that Y is contractible if there is a birational morphism $f : X \to X_0$ to a projective variety with $f(Y)$ a point P and $f : X - Y \to X_0 - P$ an isomorphism. We follow the proof of Theorem 5.7, replacing $Y^2 = -1$ by $Y^2 = -n$, $n \geq 1$.

Choice of H. Let H_0 be a very ample divisor and $h = H_0.Y > 0$. For $N \gg 0$ divisible by n the divisor $H = NH_0$ is very ample and $H^1(X, \mathcal{L}(H)) = 0$ (III, 5.2). Put $\lambda = H.Y = Nh$ and $m = \lambda/n = Nh/n$, a positive integer. Let $\mathcal{M} = \mathcal{L}(H + mY)$; then $\mathcal{M}.Y = \lambda + mY^2 = nm - mn = 0$.

Step 1: $H^1(X, \mathcal{L}(H + iY)) = 0$ for $0 \leq i \leq m$. For $i = 0$ this is the choice of H. For $i \geq 1$ use the exact sequence

$$0 \to \mathcal{L}(H + (i-1)Y) \to \mathcal{L}(H + iY) \to \mathcal{O}_Y \otimes \mathcal{L}(H + iY) \to 0.$$

Since $Y \cong \mathbf{P}^1$ and $(H+iY).Y = \lambda - in = n(m-i) \geq 0$ we have $\mathcal{O}_Y \otimes \mathcal{L}(H+iY) \cong \mathcal{O}_{\mathbf{P}^1}(n(m-i))$, which has $H^1 = 0$. The cohomology sequence and induction give the claim.

Step 2: $\mathcal{M}$ is generated by global sections. Off Y: H is very ample, so for $x \notin Y$ there is $\sigma \in H^0(\mathcal{L}(H))$ with $\sigma(x) \neq 0$; multiplying by the canonical section s_Y^m of $\mathcal{O}(mY)$, which does not vanish at x, gives a section of $\mathcal{M}$ not vanishing at x. On Y: $\mathcal{M} \otimes \mathcal{O}_Y$ has degree 0 on $Y \cong \mathbf{P}^1$, so it is $\cong \mathcal{O}_Y$ and is generated by the global section 1. The sequence $0 \to \mathcal{L}(H + (m-1)Y) \to \mathcal{M} \to \mathcal{M} \otimes \mathcal{O}_Y \to 0$ and $H^1(\mathcal{L}(H + (m-1)Y)) = 0$ (Step 1, $m \geq 1$) show

that $H^0(X, \mathcal{M}) \to H^0(Y, \mathcal{M} \otimes \mathcal{O}_Y)$ is surjective. So there is $s \in H^0(X, \mathcal{M})$ whose restriction to Y is nowhere zero; it generates $\mathcal{M}$ at every point of Y.

Step 3. Hence $\mathcal{M}$ defines a morphism $f_1 : X \to \mathbf{P}^N$; let $X_1 = f_1(X)$ (a closed subvariety of dimension 2). Since $f_1^*\mathcal{O}(1) = \mathcal{M}$ has degree 0 on Y, $f_1|_Y$ is constant (a nonconstant morphism $\mathbf{P}^1 \to \mathbf{P}^N$ pulls $\mathcal{O}(1)$ back to a sheaf of positive degree); say $f_1(Y) = P_1$. Now let $p \in X - Y$ and $q \in X$, $q \neq p$. If $q \notin Y$, very ampleness of H gives $\sigma \in H^0(\mathcal{L}(H))$ with $\sigma(p) \neq 0 = \sigma(q)$; if $q \in Y$ any σ with $\sigma(p) \neq 0$ will do, since s_Y^m vanishes at q. Then $s_Y^m\sigma$ is a section of $\mathcal{M}$ vanishing at q but not at p, so $f_1(p) \neq f_1(q)$. In particular $f_1^{-1}(P_1) = Y$. Moreover $H^0(\mathcal{L}(H)) \to \mathcal{L}(H) \otimes \mathcal{O}_p/\mathfrak{m}_p^2$ is surjective for each p (very ampleness), and s_Y^m is a unit at $p \notin Y$, so the sections of $\mathcal{M}$ separate tangent vectors at points of $X - Y$. As in Step 3 of the proof of (5.7) (II, 7.3 and 7.8.2): the morphism $f_1 : X - Y = f_1^{-1}(X_1 - P_1) \to X_1 - P_1$ is proper, injective on points and on tangent vectors, hence a closed immersion; it is surjective, and $X_1 - P_1$ is a variety, so it is an isomorphism.

Step 4. Let $\nu : X_0 \to X_1$ be the normalization (II, Ex. 3.8); it is finite, so X_0 is projective. Since X is nonsingular, hence normal, f_1 factors through a morphism $f : X \to X_0$, birational onto X_0. Over $X_1 - P_1 \cong X - Y$, which is nonsingular and hence normal, ν is an isomorphism; so $f : X - Y \to \nu^{-1}(X_1 - P_1)$ is an isomorphism. The morphism f is surjective (it is proper and dominant), so $\nu^{-1}(P_1) = f(Y)$; as Y is irreducible and $f(Y) \subseteq \nu^{-1}(P_1)$, a finite set, $f(Y)$ is a single point P and $f^{-1}(P) = Y$. Thus $f : X - Y \to X_0 - P$ is an isomorphism, i.e. Y is contractible to the point $P \in X_0$. □

(If X_0 were nonsingular at P, it would be nonsingular everywhere ($X - Y \cong X_0 - P$), and then f, a birational morphism of nonsingular surfaces contracting only Y, would be a monoidal transformation by (5.4) and $Y^2 = -1$. So for $Y^2 < -1$ the point P is necessarily singular, as announced.)

Exercise 5.3

> **Problem.** If $\pi : \widetilde{X} \to X$ is a monoidal transformation with center P, show that $H^1(\widetilde{X}, \Omega_{\widetilde{X}}) \cong H^1(X, \Omega_X) \oplus k$. This gives another proof of (5.8). [*Hints*: Use the projection formula (III, Ex. 8.3) and (III, Ex. 8.1) to show that $H^i(X, \Omega_X) \cong H^i(\widetilde{X}, \pi^*\Omega_X)$ for each i. Next use the exact sequence
>
> $$0 \to \pi^*\Omega_X \to \Omega_{\widetilde{X}} \to \Omega_{\widetilde{X}/X} \to 0$$
>
> and a local calculation with coordinates to show that there is a natural isomorphism $\Omega_{\widetilde{X}/X} \cong \Omega_E$, where E is the exceptional curve. Now use the cohomology sequence of the above sequence (you will need every term) and Serre duality to get the result.]

Solution. Write $\Omega_X = \Omega_{X/k}$, and $i : E \hookrightarrow \widetilde{X}$ for the inclusion.

Step 1: $H^i(\widetilde{X}, \pi^*\Omega_X) \cong H^i(X, \Omega_X)$. By Prop. 3.4, $\pi_*\mathcal{O}_{\widetilde{X}} = \mathcal{O}_X$ and $R^j\pi_*\mathcal{O}_{\widetilde{X}} = 0$ for $j > 0$. By the projection formula (III, Ex. 8.3), since Ω_X is locally free, $R^j\pi_*(\pi^*\Omega_X) \cong R^j\pi_*\mathcal{O}_{\widetilde{X}} \otimes \Omega_X$, which is Ω_X for $j = 0$ and 0 for $j > 0$. By III, Ex. 8.1 (the degenerate Leray spectral sequence), $H^i(\widetilde{X}, \pi^*\Omega_X) \cong H^i(X, \pi_*\pi^*\Omega_X) = H^i(X, \Omega_X)$.

Step 2: the exact sequence and the local calculation. Choose an affine neighbourhood $V = \operatorname{Spec} A$ of P and $x, y \in A$ with $(x, y) = \mathfrak{m}_P$ such that Ω_V is free on dx, dy (II, 8.15). As in the proof of (3.6), $\pi^{-1}(V)$ is the closed subscheme of $V \times \mathbf{P}^1$ given by $ty = ux$, covered by two

charts. In the chart $t = 1$ it is $\operatorname{Spec} A[u]/(y - ux)$ with exceptional curve $E = \{x = 0\}$, $\mathcal{O}_E = A[u]/(x, y-ux) = k[u]$. Since $\Omega_{V\times\mathbf{A}^1}$ is free on dx, dy, du, the sheaf $\Omega_{\widetilde{X}}$ on this chart is the quotient of the free module on dx, dy, du by $d(y-ux) = dy-u\,dx-x\,du$, so it is free on dx, du, with $dy = u\,dx+x\,du$. The map $\pi^*\Omega_X \to \Omega_{\widetilde{X}}$ sends the basis dx, dy to dx and $u\,dx+x\,du$, i.e. it is given by the matrix with columns $(1, 0)$, (u, x) of determinant $x \neq 0$ in the domain $A[u]/(y - ux)$; so the map is injective. Its cokernel is $(\mathcal{O}\,dx\oplus\mathcal{O}\,du)/(\mathcal{O}\,dx+\mathcal{O}(u\,dx+x\,du)) = \mathcal{O}/(x)\cdot du = \mathcal{O}_E\,du$. The restriction map $\Omega_{\widetilde{X}} \to i_*\Omega_E$ (conormal sequence, II, 8.12) sends $dx \mapsto 0$ (as $x|_E = 0$) and $du \mapsto du$, a basis of $\Omega_E = k[u]\,du$; it kills $\pi^*\Omega_X$ (because $dx \mapsto 0$ and $dy = u\,dx + x\,du \mapsto 0$) and its kernel is $\mathcal{O}\,dx + x\mathcal{O}\,du = \operatorname{im}\pi^*\Omega_X$. The other chart is symmetric, and off E the map $\pi^*\Omega_X \to \Omega_{\widetilde{X}}$ is an isomorphism, as π is. We obtain an exact sequence of sheaves on $\widetilde{X}$

$$0 \to \pi^*\Omega_X \to \Omega_{\widetilde{X}} \to i_*\Omega_E \to 0,$$

so $\Omega_{\widetilde{X}/X} \cong i_*\Omega_E$ (naturally, via the restriction map).

Step 3: cohomology. Since $E \cong \mathbf{P}^1$ (3.1), $\Omega_E \cong \mathcal{O}_{\mathbf{P}^1}(-2)$, so $H^0(E, \Omega_E) = 0$, $H^1(E, \Omega_E) = k$, $H^2 = 0$. The long exact sequence reads

$$0 \to H^0(\pi^*\Omega_X) \to H^0(\Omega_{\widetilde{X}}) \to 0 \to H^1(\pi^*\Omega_X) \to H^1(\Omega_{\widetilde{X}}) \to k \xrightarrow{\delta} H^2(\pi^*\Omega_X) \to H^2(\Omega_{\widetilde{X}}) \to 0.$$

By Step 1, $H^0(\pi^*\Omega_X) = H^0(\Omega_X)$ and $H^2(\pi^*\Omega_X) \cong H^2(\Omega_X)$. By Serre duality (III, 7.7) on a surface with $\omega = \bigwedge^2\Omega$ and $\Omega^\vee \otimes \omega \cong \Omega$ (for a rank 2 bundle F, $F^\vee \cong F \otimes (\det F)^{-1}$) we have $H^2(Z, \Omega_Z) \cong \operatorname{Hom}(\Omega_Z, \omega_Z)^\vee = H^0(Z, \Omega_Z)^\vee$ for $Z = X, \widetilde{X}$. Therefore

$$h^2(\Omega_{\widetilde{X}}) = h^0(\Omega_{\widetilde{X}}) = h^0(\pi^*\Omega_X) = h^0(\Omega_X) = h^2(\Omega_X) = h^2(\pi^*\Omega_X).$$

The surjection $H^2(\pi^*\Omega_X) \to H^2(\Omega_{\widetilde{X}})$ is thus a surjection between finite-dimensional spaces of the same dimension, hence an isomorphism, so $\delta = 0$. Consequently $0 \to H^1(\pi^*\Omega_X) \to H^1(\Omega_{\widetilde{X}}) \to k \to 0$ is exact, and with Step 1

$$H^1(\widetilde{X}, \Omega_{\widetilde{X}}) \cong H^1(X, \Omega_X) \oplus k.$$

(In particular $\dim H^1(\Omega)$ increases by exactly 1 at each monoidal transformation, which is the content of the second proof of Theorem 5.8.) □

Exercise 5.4

Problem. Let $f : X \to X'$ be a birational morphism of nonsingular surfaces.

(a) If $Y \subseteq X$ is an irreducible curve such that $f(Y)$ is a point, then $Y \cong \mathbf{P}^1$ and $Y^2 < 0$.

(b) (Mumford [6].) Let $P' \in X'$ be a fundamental point of f^{-1}, and let $Y_1, \ldots, Y_r$ be the irreducible components of $f^{-1}(P')$. Show that the matrix $\|Y_i.Y_j\|$ is negative definite.

Solution. By Cor. 5.4, f is a composite of $n = n(f)$ monoidal transformations

$$X = X_n \xrightarrow{\pi_n} X_{n-1} \to \cdots \xrightarrow{\pi_1} X_0 = X',$$

where $\pi_j : X_j \to X_{j-1}$ is the monoidal transformation with center $p_j \in X_{j-1}$ and exceptional curve $E_j \subseteq X_j$. Let $g_j : X \to X_j$ be the composite $\pi_{j+1} \circ \cdots \circ \pi_n$ ($g_n = \mathrm{id}$, $g_0 = f$), so $g_{j-1} = \pi_j \circ g_j$. Let $E_j^* = g_j^* E_j \in \operatorname{Pic} X$ be the total transform of E_j in X.

(i) $E_i^*.E_j^* = -\delta_{ij}$. First, $E_j^*.E_j^* = E_j^2 = -1$, because g_j^* preserves intersection numbers (it is a composite of the maps π^* of Prop. 3.2(a)) and $E_j^2 = -1$ (3.1). Next, for $i < j$ write $h = \pi_{i+1} \circ \cdots \circ \pi_j : X_j \to X_i$; then $E_i^* = g_j^*(h^* E_i)$ with $h^* E_i = \pi_j^*(\cdots)$, so by (3.2)(a),(b)

$$E_i^*.E_j^* = (h^* E_i).E_j = (\pi_j^*(\cdots)).E_j = 0.$$

(ii) Strict transforms. For an irreducible curve $C \subseteq X_j$ let $\widetilde{C}_{(l)}$ denote its strict transform on X_l, $l \geq j$. By (3.6), $\widetilde{C}_{(l)} = \pi_l^* \widetilde{C}_{(l-1)} - m_l E_l$ with $m_l = \operatorname{mult}_{p_l} \widetilde{C}_{(l-1)} \geq 0$ integers. Pulling back by g_l and using $g_{l-1}^* = g_l^* \pi_l^*$ we get, by induction, for the strict transform $\widetilde{E}_j = \widetilde{(E_j)}_{(n)}$ of E_j on X

$$\widetilde{E}_j = E_j^* - \sum_{l>j} m_{jl} E_l^* \qquad (m_{jl} \in \mathbf{Z}_{\geq 0}).$$

(iii) The contracted curves are the $\widetilde{E}_j$. Let $Y \subseteq X$ be an irreducible curve with $f(Y)$ a point, and let $g_l(Y)$ be its image in X_l ($g_n(Y) = Y$ a curve, $g_0(Y)$ a point). If $g_{l-1}(Y)$ is a curve, so is $g_l(Y)$ (as $\pi_l(g_l(Y)) = g_{l-1}(Y)$). Let $j \geq 1$ be minimal with $g_j(Y)$ a curve. Then $g_{j-1}(Y)$ is a point q and $g_j(Y) \subseteq \pi_j^{-1}(q)$ is a curve, so $q = p_j$ and $g_j(Y) = E_j$ (it is an irreducible curve in $E_j \cong \mathbf{P}^1$). For $l > j$, $g_l(Y) \neq E_l$, since E_l maps to the point p_l while $g_{l-1}(Y)$ is a curve; so $g_l(Y)$ is an irreducible curve mapping onto $g_{l-1}(Y)$ and different from E_l, i.e. it is the strict transform of $g_{l-1}(Y)$. Hence $Y = \widetilde{E}_j$.

(a). By (iii), $Y = \widetilde{E}_j$ for some j. It is $\cong \mathbf{P}^1$: $E_j \cong \mathbf{P}^1$, and if $C \subseteq X_{l-1}$ is an irreducible nonsingular curve, its strict transform $\widetilde{C} \subseteq X_l$ is mapped by π_l birationally and with finite fibres (the fibre over p_l is $\widetilde{C} \cap E_l$, finite since $\widetilde{C} \neq E_l$) onto C; a finite birational morphism onto the normal curve C is an isomorphism (locally $A \subseteq B \subseteq K(A)$ with B integral over the integrally closed ring A, so $B = A$; cf. II, Ex. 3.8). So by induction $\widetilde{E}_j \cong E_j \cong \mathbf{P}^1$. By (i),(ii),

$$Y^2 = \widetilde{E}_j^2 = E_j^{*2} + \sum_{l>j} m_{jl}^2 E_l^{*2} = -1 - \sum_{l>j} m_{jl}^2 \leq -1 < 0.$$

(b). Since P' is a fundamental point of f^{-1}, $f^{-1}(P')$ is connected of dimension ≥ 1 (Thm. 5.2 applied to $T = f^{-1}$), so each irreducible component Y_a is an irreducible curve (a point component would be an isolated point of a connected set containing a curve). Each Y_a is contracted by f, so by (iii) $Y_a = \widetilde{E}_{j(a)}$, with $j(a)$ distinct for distinct a (the curves $\widetilde{E}_j$ are distinct, because $g_l(\widetilde{E}_j)$ is a curve for $l \geq j$ while $g_{j-1}(\widetilde{E}_j)$ is a point). Let $V = \bigoplus_{j=1}^n \mathbf{R}\, E_j^* \subseteq \operatorname{Num} X \otimes \mathbf{R}$ (the E_j^* are linearly independent by (i)). By (i) the intersection form on V is $-\sum c_j^2$ in terms of the basis E_j^*, so it is negative definite on V. By (ii), $\widetilde{E}_j = E_j^* + (\text{combination of } E_l^*,\ l > j)$; the matrix expressing $\widetilde{E}_1, \ldots, \widetilde{E}_n$ in terms of $E_1^*, \ldots, E_n^*$ is unitriangular, so the $\widetilde{E}_j$ are linearly independent in V. Hence for real numbers $t_1, \ldots, t_r$ not all 0, the class $z = \sum_a t_a Y_a \in V$ is nonzero, and

$$\sum_{a,b} t_a t_b \, (Y_a.Y_b) = z^2 < 0.$$

Thus the matrix $\|Y_a.Y_b\|$ is negative definite. □

Exercise 5.5

Problem. Let C be a curve, and let $\pi : X \to C$ and $\pi' : X' \to C$ be two geometrically ruled surfaces over C. Show that there is a finite sequence of elementary transformations (5.7.1) which transform X into X'. [*Hints*: First show that if $D \subseteq X$ is a section of π containing a point P, and if $\widetilde{D}$ is the strict transform of D by elm_P, then $\widetilde{D}^2 = D^2 - 1$ (Fig. 23). Next show that X can be transformed into a geometrically ruled surface X'' with invariant $e \gg 0$. Then use (2.12), and study how the ruled surface $\mathbf{P}(\mathcal{E})$ with $\mathcal{E}$ decomposable behaves under elm_P.]

Solution. "Transform X into X'" means: up to isomorphism over C. Recall the construction of elm_P (5.7.1): $\rho : \widetilde{X} \to X$ is the monoidal transformation with center P, E its exceptional curve, $p = \pi(P)$, $L = \pi^{-1}(p)$ the fibre through P, $F = \widetilde{L}$ its strict transform, which is $\cong \mathbf{P}^1$ with $F^2 = -1$; $g : \widetilde{X} \to X'$ contracts F to a point Q (Thm. 5.7), so that $\widetilde{X}$ is the monoidal transformation of X' with center Q and exceptional curve F; $\pi' : X' \to C$ is the morphism with $\pi' \circ g = \pi \circ \rho$, and $X' = \mathrm{elm}_P X$ is again geometrically ruled. The curve $M = g(E)$ lies in $\pi'^{-1}(p) \cong \mathbf{P}^1$, so it is the fibre of π' over p. For a section D of π we write $N_D = \mathcal{L}(D) \otimes \mathcal{O}_D$, regarded as an invertible sheaf on C via $\pi|_D : D \xrightarrow{\sim} C$; $\deg N_D = D^2$.

Step 0: elm *is invertible.* E is an exceptional curve of the first kind on $\widetilde{X}$ meeting F in one point ($E.F = E.(\rho^*L - E) = 1$), $\widetilde{X}$ is the monoidal transformation of X' at Q, $M \ni Q$ is the fibre through Q and its strict transform on $\widetilde{X}$ is E. So $\mathrm{elm}_Q(X')$ is obtained by blowing up Q (getting $\widetilde{X}$), and contracting the strict transform E of the fibre M; this gives X, because the contraction of E is unique up to isomorphism (if f_1, f_2 contract E to nonsingular surfaces X_1, X_2, the birational map $f_2 f_1^{-1}$ is defined at the image point x_1 of E since otherwise its total transform $f_2(E)$ of x_1 would be a point, contradicting Thm. 5.2, and similarly for its inverse). Hence $X \cong \mathrm{elm}_Q(X')$, and a sequence of elementary transformations can be reversed.

Step 1 (effect on a section). Let D be a section of π, $\widetilde{D}$ its strict transform on $\widetilde{X}$ and $D' = g(\widetilde{D})$.

(i) If $P \in D$, then D' is a section of π', $D'^2 = D^2 - 1$, and $N_{D'} \cong N_D(-p)$.

(ii) If $P \notin D$, then D' is a section of π' through Q, $D'^2 = D^2 + 1$, and $N_{D'} \cong N_D(p)$.

Proof. $D \cong C$ is nonsingular and $D.L = 1$ (Prop. 2.3). Let $\varepsilon = 1$ if $P \in D$ and $\varepsilon = 0$ otherwise; by (3.6), $\widetilde{D} = \rho^*D - \varepsilon E$ and $F = \rho^*L - E$. By (3.2),

$$\widetilde{D}.F = (\rho^*D - \varepsilon E).(\rho^*L - E) = D.L + \varepsilon E^2 = 1 - \varepsilon.$$

Also $\rho|_{\widetilde{D}} : \widetilde{D} \to D$ is a finite birational morphism onto a nonsingular curve, hence an isomorphism.

(i) $\widetilde{D}.F = 0$ and $\widetilde{D} \neq F$, so $\widetilde{D} \cap F = \emptyset$ (Prop. 1.4). Since g is an isomorphism outside F, it is an isomorphism of a neighbourhood of $\widetilde{D}$ onto a neighbourhood of D'. Hence $D' \cong \widetilde{D} \cong D$, $\pi'|_{D'} \circ g|_{\widetilde{D}} = \pi|_D \circ \rho|_{\widetilde{D}}$ is an isomorphism onto C, so D' is a section, and $N_{D'} \cong N_{\widetilde{D}}$. Now $\mathcal{L}(\widetilde{D}) = \rho^*\mathcal{L}(D) \otimes \mathcal{L}(-E)$; restricted to $\widetilde{D} \cong D \cong C$, the first factor gives N_D, and $E.\widetilde{D} = 1$ shows $E \cap \widetilde{D}$ is one reduced point lying over P, so the second gives $\mathcal{O}_C(-p)$. Thus $N_{D'} \cong N_D(-p)$, and $D'^2 = \deg N_D - 1 = D^2 - 1$.

(ii) Now ρ is an isomorphism near D, $\widetilde{D} = \rho^*D \cong D$, and $\widetilde{D}.F = 1$: $\widetilde{D}$ meets F transversally in one point R lying over p (on the fibre L). Write $g^*D' = \widetilde{D} + cF$; then $0 = g^*D'.F = \widetilde{D}.F + cF^2 = 1 - c$, so $c = 1$, i.e. $\mathrm{mult}_Q D' = 1$ and D' is nonsingular at Q (3.5.2), and elsewhere

it is isomorphic to $\widetilde{D}$; so $g|_{\widetilde{D}} : \widetilde{D} \to D'$ is an isomorphism (finite birational onto a nonsingular curve). As before D' is a section. Pulling $N_{D'} = \mathcal{L}(D') \otimes \mathcal{O}_{D'}$ back to $\widetilde{D}$:

$$N_{D'} \cong \big(\mathcal{L}(\widetilde{D} + F) \otimes \mathcal{O}_{\widetilde{D}}\big) = N_D \otimes \mathcal{O}_C(p),$$

since $F \cap \widetilde{D} = \{R\}$ is a reduced point over p. Hence $D'^2 = D^2 + 1$. $\square$

Step 2 (two disjoint sections). Suppose X has two disjoint sections D_1, D_2, and let $A = N_{D_1}$. Then:

(a) $N_{D_2} \cong A^{-1}$, and $X \cong \mathbf{P}(\mathcal{O}_C \oplus A^{-1})$ over C. In particular the isomorphism class of X over C is determined by A.

(b) If $P \in D_1$ then $X' = \operatorname{elm}_P X$ has disjoint sections D'_1, D'_2 and $N_{D'_1} \cong A(-p)$. If $P \in D_2$ then X' has disjoint sections D'_1, D'_2 and $N_{D'_1} \cong A(p)$.

Proof of (a). By Prop. 2.2, $X \cong \mathbf{P}(\mathcal{E})$; by Prop. 2.6 the sections D_i correspond to surjections $\mathcal{E} \to \mathcal{M}_i$ with kernels $\mathcal{N}_i$ (invertible), where $\mathcal{L}(D_i) \cong \mathcal{O}_X(1) \otimes \pi^*\mathcal{N}_i^{-1}$ and $\mathcal{M}_i = \sigma_i^*\mathcal{O}_X(1)$. Hence $N_{D_i} \cong \mathcal{M}_i \otimes \mathcal{N}_i^{-1}$. The sections are disjoint, i.e. at each point $c \in C$ the two quotients $\mathcal{E}(c) \to \mathcal{M}_i(c)$ of the 2-dimensional space $\mathcal{E}(c)$ are distinct, so their kernels are distinct lines and $\mathcal{E}(c) \to \mathcal{M}_1(c) \oplus \mathcal{M}_2(c)$ is an isomorphism. Thus $\mathcal{E} \to \mathcal{M}_1 \oplus \mathcal{M}_2$ is an isomorphism of rank 2 bundles, $\mathcal{N}_1 = \mathcal{M}_2$, $\mathcal{N}_2 = \mathcal{M}_1$, $A = \mathcal{M}_1 \otimes \mathcal{M}_2^{-1}$ and $N_{D_2} = A^{-1}$. Finally $\mathbf{P}(\mathcal{E}) \cong \mathbf{P}(\mathcal{E} \otimes \mathcal{M}_1^{-1})$ (II, Ex. 7.9, quoted in Prop. 2.2) $= \mathbf{P}(\mathcal{O} \oplus \mathcal{M}_2\mathcal{M}_1^{-1}) = \mathbf{P}(\mathcal{O} \oplus A^{-1})$.

Proof of (b). Let $P \in D_1$. By Step 1 applied to D_1 (through P) and to D_2 (not through P), D'_1, D'_2 are sections of π', $N_{D'_1} \cong A(-p)$. They are disjoint: $\widetilde{D}_1 \cap \widetilde{D}_2 = \emptyset$ (their images D_1, D_2 in X are disjoint) and $\widetilde{D}_1 \cap F = \emptyset$; if $x \in D'_1 \cap D'_2$ then $x = g(u) = g(w)$ with $u \in \widetilde{D}_1$, $w \in \widetilde{D}_2$; if $x \neq Q$ then $u = w$ (g is an isomorphism off F), contradiction, and if $x = Q$ then $u \in F \cap \widetilde{D}_1 = \emptyset$, contradiction. If $P \in D_2$, exchange the roles: D_1, D_2 disjoint, $\widetilde{D}_2 \cap F = \emptyset$, and Step 1(ii) for D_1 gives $N_{D'_1} \cong A(p)$. $\square$

Step 3: every ruled surface can be transformed into one with two disjoint sections. Let X be any geometrically ruled surface over C, of genus g. It has a section D (by definition), with $d = D^2 = \deg N_D$. Perform successively r elementary transformations at points of the successive images of the section D (first at $P_1 \in D$, then at P_2 on the new section $D^{(1)}$, and so on). By Step 1(i) we get a ruled surface $X^{(r)} = \mathbf{P}(\mathcal{E})$ with a section $D^{(r)}$ of self-intersection $d - r$. By Prop. 2.6, $D^{(r)}$ corresponds to an exact sequence $0 \to \mathcal{N} \to \mathcal{E} \to \mathcal{M} \to 0$ with $\deg(\mathcal{M} \otimes \mathcal{N}^{-1}) = \deg N_{D^{(r)}} = d - r$. The extension class lies in $\operatorname{Ext}^1(\mathcal{M}, \mathcal{N}) \cong H^1(C, \mathcal{N} \otimes \mathcal{M}^{-1})$ (III, Ex. 6.1 for the correspondence between extensions and Ext^1, and Prop. 6.7 with Prop. 6.3(c) for the computation of $\operatorname{Ext}^1(\mathcal{M}, \mathcal{N}) = \operatorname{Ext}^1(\mathcal{O}_C, \mathcal{M}^{-1} \otimes \mathcal{N})$), and $\deg(\mathcal{N} \otimes \mathcal{M}^{-1}) = r - d$. For $r > d + 2g - 2$ this group vanishes: by Serre duality $H^1(C, \mathcal{B}) \cong H^0(C, \omega_C \otimes \mathcal{B}^{-1})^\vee$ and $\deg(\omega_C \otimes \mathcal{B}^{-1}) = 2g - 2 - (r - d) < 0$. So the sequence splits, $\mathcal{E} \cong \mathcal{N} \oplus \mathcal{M}$, and the two projections give two sections of $\mathbf{P}(\mathcal{E})$ (Prop. 2.6), which are disjoint by the argument of Step 2(a). (Thus, in the language of the hint, $e \gg 0$ and $\mathcal{E}$ is decomposable, as in Thm. 2.12(a).)

Step 4: conclusion. By Step 3 there are sequences of elementary transformations from X to a surface X_1 and from X' to a surface X'_1, each having two disjoint sections; let A, A' be the normal sheaves of the first sections $D_1 \subset X_1$, $D'_1 \subset X'_1$. Write $A' \otimes A^{-1} \cong \mathcal{O}_C(B)$ with $B = \sum n_i p_i$ a divisor (II, 6.13). For each i perform $|n_i|$ elementary transformations at a point lying over p_i: at

a point of the second section if $n_i > 0$, at a point of the first section if $n_i < 0$. By Step 2(b), after each of them the surface again has two disjoint sections, and the normal sheaf of the first section is multiplied by $\mathcal{O}_C(p_i)$, resp. $\mathcal{O}_C(-p_i)$. At the end we have a surface X_2 with two disjoint sections whose first section has normal sheaf $A(B) \cong A'$. By Step 2(a), $X_2 \cong \mathbf{P}(\mathcal{O} \oplus A'^{-1}) \cong X_1'$ over C. Hence $X \to \cdots \to X_1 \to \cdots \to X_2 \cong X_1'$, and, by Step 0, X_1' can be transformed back into X' by elementary transformations (the inverses, again elementary). So X can be transformed into X'. □

Exercise 5.6

> **Problem.** Let X be a surface with function field K. Show that every valuation ring R of K/k is one of the three kinds described in (II, Ex. 4.12). [*Hint*: In case (3), let $f \in R$. Use (Ex. 5.1) to show that for all $i \gg 0$, $f \in \mathcal{O}_{X_i}$, so in fact $f \in R_0$.]

Solution. *The three kinds* (II, Ex. 4.12(b)). A valuation ring of K/k is a valuation ring R with $k \subseteq R \subsetneq K$ (the ring K itself is excluded). The three kinds are: (1) $R = \mathcal{O}_{\eta,X}$, the local ring of the generic point η of an irreducible curve Y on X; (2) $R = \mathcal{O}_{\eta',X'}$ for the generic point η' of an irreducible curve Y' on a surface X' with a birational morphism $X' \to X$ mapping Y' to a closed point x_0 of X; (3) R is a valuation ring dominating the local ring $R_0 = \bigcup_{i \geq 0} \mathcal{O}_{x_i,X_i}$, where $X_0 = X$, $x_0 \in X$ is a closed point, $X_{i+1} \to X_i$ is the blowing up of x_i with exceptional curve E_{i+1} and $x_{i+1} \in E_{i+1}$ is a closed point, for all i (so $\mathcal{O}_{x_i,X_i}$ is dominated by $\mathcal{O}_{x_{i+1},X_{i+1}}$ and R_0 is a local ring). We show that each R is of one of these kinds, and in case (3) that in fact $R = R_0$.

Centers. Every surface obtained from X by blowing up points is projective over k, hence proper, so by the valuative criterion of properness (II, 4.7) $\operatorname{Spec} K \to Z$ extends uniquely to $\operatorname{Spec} R \to Z$; the image z of the closed point is the *center* of R on Z: the unique point z with $\mathcal{O}_{z,Z} \subseteq R$ and $\mathfrak{m}_R \cap \mathcal{O}_{z,Z} = \mathfrak{m}_z$ (R dominates $\mathcal{O}_{z,Z}$). By uniqueness, if $\sigma : Z' \to Z$ is a morphism of such surfaces (compatible with the identification of function fields), the center on Z' maps to the center on Z.

Classification. Let x be the center of R on X. If x is the generic point of X, then $K = \mathcal{O}_x \subseteq R$, excluded. If x is the generic point of an irreducible curve Y, then $\mathcal{O}_{x,X}$ is a discrete valuation ring (as X is nonsingular, II, 6.2) with fraction field K, and $\mathcal{O}_{x,X} \subseteq R \subsetneq K$; an overring of a DVR $S = (S, \pi)$ other than K is S (if $r \in R - S$ then $r = u\pi^{-n}$, $n > 0$, u a unit, so $\pi^{-1} = ru^{-1}\pi^{n-1} \in R$, whence $R \supseteq S[\pi^{-1}] = K$). So $R = \mathcal{O}_{x,X}$: kind (1). Otherwise $x = x_0$ is a closed point. Let $X_1 \to X$ be the blowing up of x_0 and let $x_1 \in E_1 = \pi^{-1}(x_0)$ be the center of R on X_1 (it lies over x_0). If x_1 is the generic point of E_1, then, as above, $R = \mathcal{O}_{x_1,X_1}$, which is kind (2) with $X' = X_1$, $Y' = E_1$. If x_1 is a closed point, blow it up and continue. If at some stage i the center x_i is the generic point of the exceptional curve E_i of $X_i \to X_{i-1}$ (the only possibility for a nonclosed point over a closed point), then $R = \mathcal{O}_{x_i,X_i}$, a ring of kind (2) (with $X' = X_i$, $Y' = E_i$ mapped to x_0). Otherwise all centers x_i are closed points with $x_{i+1} \in E_{i+1}$, for all i; since R dominates every $\mathcal{O}_{x_i,X_i}$, it dominates R_0 (R_0 is local with maximal ideal $\bigcup \mathfrak{m}_{x_i} = \mathfrak{m}_R \cap R_0$): kind (3).

In kind (3), $R = R_0$. We must show $R \subseteq R_0$. Write $O_i = \mathcal{O}_{x_i,X_i}$ and w for the valuation of R.

Lemma B. Let $a, b \in O_i$ be nonzero with $b/a \in R$. Then $b/a \in O_{i'}$ for some $i' \geq i$.

Proof of Lemma B. Put $h = b/a$. The ring O_i is a 2-dimensional regular local ring, hence a UFD. Dividing a, b by their gcd we may assume a, b coprime. If a is a unit, then $h \in O_i$. Otherwise $a \in \mathfrak{m}_{x_i} \subseteq \mathfrak{m}_R$, so $w(a) > 0$; if b were a unit then $w(b) = 0 < w(a)$ and $h \notin R$, so b is a nonunit too. Let $\ell = \ell(a,b) = \dim_k O_i/(a,b)$, finite since a, b are coprime, and induct on ℓ (note $\ell \geq 1$ here). Let $\alpha = \mathrm{ord}_{\mathfrak{m}}(a) \geq 1$, $\beta = \mathrm{ord}_{\mathfrak{m}}(b) \geq 1$, and let e generate $\mathfrak{m}_{x_i} O_{i+1}$, so that e is a local equation of E_{i+1} at x_{i+1}. Choose effective divisors C, D on X_i with local equations a, b at x_i and with no common component (take the zero divisors of a, b and delete the common components, which do not pass through x_i). By (3.6), $\pi^* C = \widetilde{C} + \alpha E_{i+1}$, $\pi^* D = \widetilde{D} + \beta E_{i+1}$, so $a = e^\alpha a_2$, $b = e^\beta b_2$ in O_{i+1} with a_2, b_2 local equations of $\widetilde{C}, \widetilde{D}$ at x_{i+1}; they are coprime and not divisible by e, because $\widetilde{C}, \widetilde{D}$ have no common component and neither contains E_{i+1}. By (3.2), (3.7) and (1.4), comparing the local contributions at x_i before and $\bigcup\{\text{points over } x_i\}$ after, $\sum_{x \in E_{i+1}} (\widetilde{C}.\widetilde{D})_x = \ell - \alpha\beta$ and $\sum_{x \in E_{i+1}} (E_{i+1}.\widetilde{D})_x = \beta$, $\sum_x (E_{i+1}.\widetilde{C})_x = \alpha$. Let $c = \min(\alpha, \beta)$ and $a' = a/e^c$, $b' = b/e^c \in O_{i+1}$; then $h = b'/a'$, a', b' are coprime, and, if $\alpha \geq \beta$, by additivity of the length ($\ell(O/(fg,h)) = \ell(O/(f,h)) + \ell(O/(g,h))$ for pairwise coprime f, g, h in a 2-dimensional regular local ring),

$$\ell(a', b') = (\alpha - \beta)(e.b_2)_{x_{i+1}} + (a_2.b_2)_{x_{i+1}} \leq (\alpha - \beta)\beta + \ell - \alpha\beta = \ell - \beta^2 < \ell,$$

and symmetrically $\ell(a', b') \leq \ell - \alpha^2 < \ell$ if $\alpha < \beta$. By induction applied to (a', b') in O_{i+1} we get $h \in O_{i'}$ for some $i' \geq i + 1$. (If $\ell(a', b') = 0$ then one of a', b' is a unit, which by the argument above must be a', and $h \in O_{i+1}$.) $\square$

Proof that $R \subseteq R_0$. Let $f \in R$, $f \neq 0$. By Ex. 5.1 there is a birational morphism $g : X' \to X$, a composite of monoidal transformations $X' = Y_m \to Y_{m-1} \to \cdots \to Y_0 = X$ (with centers $q_1, \ldots, q_m$, $Y_j \to Y_{j-1}$ the blowing up of $q_j \in Y_{j-1}$), such that f defines a morphism $\varphi : X' \to \mathbf{P}^1$. Let y_j be the center of R on Y_j and $A_j = \mathcal{O}_{y_j, Y_j}$.

(1) $f \in A_m$. If $\varphi(y_m) = \infty$ then $1/f = \varphi^*(\text{uniformizer at } \infty) \in \mathfrak{m}_{y_m} \subseteq \mathfrak{m}_R$, so $f \notin R$ (otherwise $1 = f \cdot f^{-1} \in \mathfrak{m}_R$), a contradiction. So $\varphi(y_m) \in \mathbf{A}^1$ and f is regular at y_m.

(2) For each j there is i_j with $A_j \subseteq O_i$ for all $i \geq i_j$. For $j = 0$ take $i_0 = 0$ ($A_0 = O_0 \subseteq O_i$). Assume it for $j - 1$. If $q_j \neq y_{j-1}$, then $Y_j \to Y_{j-1}$ is an isomorphism near y_{j-1} and $A_j = A_{j-1}$. If $q_j = y_{j-1}$, then $A = A_{j-1}$ is a 2-dimensional regular local ring with maximal ideal (u, v), and $u, v \in A \subseteq O_i$ for $i \geq i_{j-1}$. Renaming we may assume $w(v) \geq w(u)$, i.e. $v/u \in R$. By Lemma B (with $a = u$, $b = v$) there is $i_j \geq i_{j-1}$ with $v/u \in O_i$ for $i \geq i_j$. Over $\mathrm{Spec}\, A$ the blowing up of Y_{j-1} at q_j is covered by the charts $\mathrm{Spec}\, A[v/u]$ and $\mathrm{Spec}\, A[u/v]$. Since $A[v/u] \subseteq R$, the prime $\mathfrak{q} = \mathfrak{m}_R \cap A[v/u]$ is the center of R on Y_j, i.e. $A_j = A[v/u]_{\mathfrak{q}}$. For $i \geq i_j$, $A[v/u] \subseteq O_i$; and if $s \in A[v/u] - \mathfrak{q}$ then $s \in R - \mathfrak{m}_R$, so $s \notin \mathfrak{m}_{x_i} \subseteq \mathfrak{m}_R$ and s is a unit of O_i. Hence $A_j \subseteq O_i$ for $i \geq i_j$.

By (1),(2), $f \in A_m \subseteq O_i \subseteq R_0$ for $i \geq i_m$. Hence $R \subseteq R_0 \subseteq R$, so $R = R_0$. This is the statement of the hint (for all $i \gg 0$, $f \in \mathcal{O}_{x_i, X_i}$). In particular a valuation ring of kind (3) is determined by its sequence of centers, and R_0 is itself a valuation ring. $\square$

Exercise 5.7

Problem. Let Y be an irreducible curve on a surface X, and suppose there is a morphism $f : X \to X_0$ to a projective variety X_0 of dimension 2, such that $f(Y)$ is a point P and $f^{-1}(P) = Y$. Then show that $Y^2 < 0$. [*Hint*: Let $|H|$ be a very ample (Cartier) divisor class

on X_0, let $H_0 \in |H|$ be a divisor containing P, and let $H_1 \in |H|$ be a divisor not containing P. Then consider f^*H_0, f^*H_1, and $\widetilde{H}_0 = f^*(H_0 - P)^-$.]

Solution. *Reading of the statement.* We must assume that f is surjective (equivalently dominant, $f(X)$ being closed): otherwise $f(X)$ is a curve or a point and the statement is false (take for f a fibration $X \to \Gamma \subseteq X_0$ and for Y a fibre: then $Y^2 = 0$). With $\dim X_0 = 2$ and f surjective, f is generically finite.

Fix an embedding $X_0 \subseteq \mathbf{P}^N$ with $\mathcal{O}_{X_0}(1) = \mathcal{O}_{X_0}(H)$. For a linear form λ not vanishing identically on X_0, $f^*\lambda$ is a nonzero section of $f^*\mathcal{O}(1)$ (as f is dominant), with effective divisor $f^*(\{\lambda = 0\})$, and all these divisors are linearly equivalent, with $\mathcal{L}(f^*H) = f^*\mathcal{O}(1)$. Put $D = f^*H_1$ for such a hyperplane section.

(1) $D.Y = 0$. Take $H_1 \in |H|$ with $P \notin H_1$ (a hyperplane section of X_0 not through P). Since $f(Y) = P$, the curve f^*H_1 is disjoint from Y, so $f^*H_1.Y = 0$ by Prop. 1.4 (they have no common component and no common point). Any other divisor $D = f^*H'$ with $H' \in |H|$ is linearly equivalent to f^*H_1, so $D.Y = 0$ as well, by Theorem 1.1(4).

(2) $D^2 > 0$. The set $B \subseteq X_0$ of points over which f has a fibre of dimension ≥ 1 is finite. Indeed, the locus $W \subseteq X$ where the local dimension of the fibre is ≥ 1 is closed (II, Ex. 3.22) and proper, as f is generically finite, so $\dim W \leq 1$; every point of W lies on a curve contained in a fibre, which is then a component of W, so W is a finite union of irreducible curves each contracted by f, and $B = f(W)$ is finite. Choose a hyperplane section H_1 of X_0 avoiding B, and then a hyperplane section H_2 avoiding B and containing none of the finitely many irreducible components of H_1 (hyperplanes through a given point, or containing a given curve, form proper closed subsets of the dual projective space); then $H_1 \cap H_2$ is finite. Then f^*H_1, f^*H_2 have no common irreducible component (a common component would be mapped into the finite set $H_1 \cap H_2$, i.e. contracted to a point of B, which H_1 avoids). Since $\dim X_0 = 2$, $H_1 \cap H_2 \neq \emptyset$ (I, 7.2) and, f being surjective, there is a point $x \in f^{-1}(H_1 \cap H_2) \subseteq \operatorname{Supp} f^*H_1 \cap \operatorname{Supp} f^*H_2$. So by Prop. 1.4, $f^*H_1.f^*H_2 \geq (f^*H_1.f^*H_2)_x > 0$, and by linear equivalence $D^2 = f^*H_1.f^*H_2 > 0$.

(3) Conclusion by the Hodge index theorem. Let H' be an ample divisor on X. Some multiple mH' is very ample, and the intersection number of mH' with an irreducible curve is its degree in the corresponding embedding (cf. the proof of Thm. 1.10); so $Y.H' > 0$, and likewise $D.H' > 0$ because D is a nonzero effective divisor. Put $\Delta = (D.H')\,Y - (Y.H')\,D$, so $\Delta.H' = 0$, and by (1)

$$\Delta^2 = (D.H')^2Y^2 - 2(D.H')(Y.H')(Y.D) + (Y.H')^2D^2 = (D.H')^2Y^2 + (Y.H')^2D^2.$$

Suppose $Y^2 \geq 0$. Then $\Delta^2 > 0$ by (2). If $\Delta \not\equiv 0$ this contradicts Theorem 1.9 ($\Delta.H' = 0 \Rightarrow \Delta^2 < 0$); if $\Delta \equiv 0$ then $\Delta^2 = 0$, again a contradiction. Hence $Y^2 < 0$. □

Exercise 5.8

Problem. *A surface singularity.* Let k be an algebraically closed field, and let X be the surface in $\mathbf{A}^3_k$ defined by the equation $x^2 + y^3 + z^5 = 0$. It has an isolated singularity at the origin $P = (0, 0, 0)$.

(a) Show that the affine ring $A = k[x, y, z]/(x^2 + y^3 + z^5)$ of X is a unique factorization domain, as follows. Let $t = z^{-1}$, $u = t^3x$, and $v = t^2y$. Show that z is irreducible in A; $t \in k[u, v]$, and $A[z^{-1}] = k[u, v, t^{-1}]$. Conclude that A is a UFD.

(b) Show that the singularity at P can be resolved by eight successive blowings-up. If $\widetilde{X}$ is the resulting nonsingular surface, then the inverse image of P is a union of eight projective lines, which intersect each other according to the Dynkin diagram E_8: a chain of seven circles with an eighth circle attached to the third circle from one end. Here each circle denotes a line, and two circles are joined by a line segment whenever the lines intersect.

Solution. Let $f = x^2 + y^3 + z^5$.

The singular locus. The Jacobian criterion: the partial derivatives are $2x, 3y^2, 5z^4$. A point of X where all vanish: in characteristic $\neq 2, 3, 5$ this forces $x = y = z = 0$. In characteristic 2: $y^2 = 0$, $z^4 = 0$, hence $y = z = 0$ and then $x^2 = 0$. In characteristic 3: $2x = 0$, $2z^4 = 0$, so $x = z = 0$ and $y^3 = 0$. In characteristic 5: $2x = 0$, $3y^2 = 0$, so $x = y = 0$ and $z^5 = 0$. In all cases P is the only singular point, so the singularity is isolated.

f is irreducible, so A is a noetherian domain. The polynomial f is monic of degree 2 in x over $k[y, z]$. It has no root in $k(y, z)$: a root x_0 would satisfy $x_0^2 = -(y^3 + z^5)$, and since $k[y, z]$ is integrally closed $x_0 \in k[y, z]$; setting $z = 0$ gives $x_0(y, 0)^2 = -y^3$, impossible by parity of degrees. So f is irreducible in $k(y, z)[x]$, hence in $k[y, z][x]$ by Gauss's lemma (f is monic).

(a). z is prime, hence irreducible. $A/zA \cong k[x, y]/(x^2 + y^3)$, and $x^2 + y^3$ is irreducible: it is monic of degree 2 in x over $k[y]$ and has no root in $k(y)$ (the y-adic valuation of x_0^2 would be even, that of $-y^3$ is odd), so it is irreducible in $k(y)[x]$, hence in $k[y][x]$ by Gauss's lemma. Thus A/zA is a domain. So zA is a prime ideal and z is a prime element, in particular irreducible.

$A[z^{-1}] = k[u, v, t^{-1}]$ and $t \in k[u, v]$. Divide $x^2 + y^3 + z^5 = 0$ by $z^6 = t^{-6}$: $t^6x^2 + t^6y^3 + t = 0$, i.e.

$$u^2 + v^3 + t = 0, \qquad \text{so} \qquad t = -(u^2 + v^3) \in k[u, v].$$

Since $u = t^3x$, $v = t^2y$, $t = z^{-1}$ lie in $A[z^{-1}]$, and $z = t^{-1}$, $x = ut^{-3}$, $y = vt^{-2}$, we have $A[z^{-1}] = k[u, v][t^{-1}] = k[u, v]_{u^2+v^3}$, the localization of the polynomial ring $k[u, v]$ at the element $u^2 + v^3$. To be careful that u, v are algebraically independent (so that this is really a localization of a polynomial ring): define $\varphi : A \to B := k[U, V]_{U^2+V^3}$ by $z \mapsto -(U^2 + V^3)^{-1}$, $x \mapsto U\,t^{-3}$, $y \mapsto V\,t^{-2}$ with $t = -(U^2 + V^3)$. Then $\varphi(f) = t^{-6}(U^2 + V^3 + t) = 0$, so φ is well defined, and $\varphi(z)$ is a unit, so it extends to $A[z^{-1}] \to B$. Conversely $\psi : B \to A[z^{-1}]$, $U \mapsto xz^{-3}$, $V \mapsto yz^{-2}$ is defined because $\psi(U^2 + V^3) = (x^2 + y^3)z^{-6} = -z^{-1}$ is a unit. One checks on generators that $\varphi\psi$ and $\psi\varphi$ are the identity (e.g. $\psi\varphi(z) = \psi(-(U^2 + V^3)^{-1}) = z$, $\psi\varphi(x) = \psi(U)\psi(t)^{-3} = xz^{-3}z^3 = x$). So $A[z^{-1}] \cong k[U, V]_{U^2+V^3}$.

A is a UFD. $k[u, v]$ is a UFD and so are its localizations; hence $A[z^{-1}]$ is a UFD. We now apply Nagata's criterion: *a noetherian domain A with a prime element z such that $A[z^{-1}]$ is a UFD is a UFD.* Proof: since A is noetherian, every nonzero nonunit is a product of irreducible elements, so it suffices to show that every irreducible $p \in A$ is prime. If p is associate to z, this holds. Otherwise $z \nmid p$ (if $p = zq$ then q is a unit by irreducibility). Then p is irreducible in $A[z^{-1}]$: it is not a unit there (if $pr = z^n$ with $r \in A$, then $z \mid pr$ gives $z \mid r$ since $z \nmid p$, and repeating, $p \cdot r' = 1$, so p would be a unit of A); and if $p = ab$ with $a = a_0z^{-i}$, $b = b_0z^{-j}$, $a_0, b_0 \in A$, we may take $i + j$ minimal; if $i + j > 0$ then $z \mid a_0b_0$ so $z \mid a_0$ or $z \mid b_0$, and we could lower $i + j$; so $i + j = 0$, $p = a_0b_0$ in A, one of a_0, b_0 is a unit of A, hence one of a, b is a unit of $A[z^{-1}]$. As $A[z^{-1}]$ is a UFD, p is prime in $A[z^{-1}]$, so $\mathfrak{q} = pA[z^{-1}] \cap A$ is a prime ideal of A. Clearly $pA \subseteq \mathfrak{q}$. Conversely, if $a \in \mathfrak{q}$, then $z^n a = pc$ for some $n \geq 0$, $c \in A$; if $n > 0$, $z \mid pc$ and $z \nmid p$ give $z \mid c$, so

$z^{n-1}a = pc'$; by induction $a \in pA$. So $pA = \mathfrak{q}$ is prime and p is a prime element. Therefore A is a UFD.

(b). All blowings-up below are blowings-up of a closed point of the surface at hand, realized (II, 7.15) as the strict transform in the blowing up of the ambient affine space; in the charts of the blowing up of $\mathbf{A}^3$ at the origin with coordinates (a, b, c) we have the charts $(a,\ b = ab',\ c = ac')$, $(b,\ a = ba',\ c = bc')$, $(c,\ a = ca',\ b = cb')$, and the strict transform of $\{F = 0\}$ is given by dividing F by the highest power of the exceptional coordinate. The computations are valid in every characteristic. Exceptional curves are denoted $C_1, \ldots, C_8$ in order of creation; their strict transforms keep the name.

Stage 1 (blowing up P). The three charts of $F = x^2 + y^3 + z^5$ are:

- x-chart $(y = xy', z = xz')$: $x^2(1 + xy'^3 + x^3z'^5)$, strict transform $1 + xy'^3 + x^3z'^5 = 0$, which has no point with $x = 0$;
- y-chart $(x = yx', z = yz')$: $y^2(x'^2 + y + y^3z'^5)$, strict transform $x'^2 + y + y^3z'^5 = 0$, nonsingular along $y = 0$ ($\partial/\partial y = 1$ there), exceptional set $\{y = x' = 0\}$;
- z-chart $(x = zx', y = zy')$: $z^2(x'^2 + zy'^3 + z^3)$, strict transform $S_1 :\ x'^2 + zy'^3 + z^3 = 0$, exceptional set $\{z = x' = 0\}$; the gradient on the exceptional set is $(2x', y'^3 + 3z^2, 3zy'^2) = (0, y'^3, 0)$, so the only singular point there is $y' = 0$.

The exceptional set is the reduced curve $C_1 \cong \mathbf{P}^1$ ($z' = 1/y'$ is a coordinate); the surface is isomorphic to $X - P$ off C_1, hence nonsingular there. So the blown-up surface has exactly one singular point P_1, the origin of the z-chart, with local equation (renaming $(x', y', z) = (a, b, c)$)

$$a^2 + cb^3 + c^3 = 0, \qquad C_1 = \{a = c = 0\} \text{ (the } b\text{-axis)}.$$

Stage 2 (blowing up P_1). Charts: the a-chart gives $1 + a^2b'^3c' + ac'^3 = 0$, with no point on $a = 0$; the c-chart gives $a'^2 + c^2b'^3 + c = 0$, nonsingular ($\partial/\partial c = 1$ at $c = 0$); the b-chart gives

$$a'^2 + b^2c' + bc'^3 = 0,$$

whose gradient on the exceptional set $\{b = a' = 0\}$ is $(2a',\ 2bc' + c'^3,\ b^2 + 3bc'^2) = (0, c'^3, 0)$, singular only at $c' = 0$. The exceptional curve $C_2 \cong \mathbf{P}^1$ is $\{b = a' = 0\}$ in the b-chart (it is $\{c = a' = 0\}$ in the c-chart); the strict transform of C_1 is the b-axis $\{a' = c' = 0\}$ in the b-chart, so C_1 and C_2 meet exactly at the unique singular point P_2 (origin of the b-chart). Local equation (renaming $(a', b, c') = (a, b, c)$):

$$a^2 + bc^3 + b^2c = 0, \qquad C_1 = \{a = c = 0\},\ C_2 = \{a = b = 0\}.$$

Stage 3 (blowing up P_2). The a-chart has no point on the exceptional set. The b-chart $(a = ba',\ c = bc')$ gives $a'^2 + b^2c'^3 + bc' = 0$, with gradient $(2a',\ 2bc'^3 + c',\ 3b^2c'^2 + b) = (0, c', 0)$ on $\{b = a' = 0\}$: singular exactly at the point $R_1 = (0, 0, 0)$. The c-chart $(a = ca',\ b = cb')$ gives $a'^2 + c^2b' + cb'^2 = 0$, with gradient $(2a',\ 2cb' + b'^2,\ c^2 + 2cb') = (0, b'^2, 0)$ on $\{c = a' = 0\}$: singular exactly at $R_2 = (0, 0, 0)$. The exceptional curve $C_3 \cong \mathbf{P}^1$ ($c'b' = 1$) contains both, at the two distinct points $c' = 0$ and $b' = 0$ (i.e. $c' = \infty$). The strict transform of C_1 (direction $(0 : 1 : 0)$) is the b-axis $\{a' = c' = 0\}$ of the b-chart, through R_1; that of C_2 (direction $(0 : 0 : 1)$) is the

c-axis $\{a' = b' = 0\}$ of the c-chart, through R_2. Away from R_1, R_2 the surface is nonsingular. Local equations:

$$R_1: \quad a^2 + bc + b^2c^3 = 0, \quad C_3 = \{a = b = 0\} \ (c\text{-axis}), \ C_1 = \{a = c = 0\} \ (b\text{-axis});$$
$$R_2: \quad a^2 + bc(b + c) = 0, \quad C_3 = \{a = c = 0\} \ (b\text{-axis}), \ C_2 = \{a = b = 0\} \ (c\text{-axis}),$$

(for R_2 in the coordinates $(a, b, c) = (a', b', c)$). C_1 and C_2 no longer meet (they pass through different points of C_3).

Stage 4 (blowing up R_1 and R_2). At R_1: the charts are $1+b'c'+a^3b'^2c'^3$ (a-chart), $a'^2+c'+b^3c'^3$ (b-chart), $a'^2+b'+b'^2c^3$ (c-chart); each is nonsingular along the exceptional set (partial derivative $\neq 0$: respectively $c' + \dots$ with $c' \neq 0$ on $1 + b'c' = 0$, $1 + 3b^3c'^2$, $1 + 2b'c^3$). So the surface is nonsingular above R_1 and the exceptional curve C_4 is the conic $\{a^2 + bc = 0\} \subset \mathbf{P}^2$ (tangent cone), $\cong \mathbf{P}^1$. The strict transforms of the b-axis (C_1) and c-axis (C_3) meet C_4 in the two distinct points $(0 : 1 : 0)$ and $(0 : 0 : 1)$ of the conic. Hence C_1 and C_3 are now disjoint (a common point would lie over R_1, i.e. on C_4, where they meet C_4 in distinct points), and each meets C_4.

At R_2: the a-chart has no point on the exceptional set; the b-chart ($a = ba'$, $c = bc'$) gives $a'^2 + bc'(1 + c') = 0$ with gradient $(2a', c'(1 + c'), b(1 + 2c'))$; the c-chart ($a = ca'$, $b = cb'$) gives $a'^2 + cb'(1 + b') = 0$ with gradient $(2a', b'(1 + b'), c(1 + 2b'))$ in the coordinates (a', c, b'). The exceptional curve $C_5 \cong \mathbf{P}^1$ ($b'c' = 1$) is $\{b = a' = 0\}$, resp. $\{c = a' = 0\}$. The surface is singular exactly at the three points of C_5 with $c'(1+c') = 0$ in the b-chart or $b'(1+b') = 0$ in the c-chart, i.e. $c' \in \{0, -1, \infty\}$:

- Σ_1: $c' = 0$, where the strict transform of C_3 (the b-axis, direction $(0 : 1 : 0)$) passes;
- Σ_2: $b' = 0$ ($c' = \infty$), where the strict transform of C_2 (the c-axis, direction $(0 : 0 : 1)$) passes;
- Σ_3: $c' = -1$, through which no other exceptional curve passes.

Local equations: at Σ_1, $a'^2 + bc'(1 + c') = 0$ with $C_5 = \{a' = b = 0\}$, $C_3 = \{a' = c' = 0\}$; at Σ_2, $a'^2 + cb'(1 + b') = 0$ with $C_5 = \{a' = c = 0\}$, $C_2 = \{a' = b' = 0\}$; at Σ_3, with $\gamma = 1 + c'$, $a'^2 - b\gamma(1-\gamma) = 0$ and $C_5 = \{a' = b = 0\}$. So C_5 meets C_3, C_2 at Σ_1, Σ_2 but only there; C_3 and C_2 stay disjoint (distinct points Σ_1, Σ_2), and C_3 meets C_4 and C_5.

Stage 5 (blowing up $\Sigma_1, \Sigma_2, \Sigma_3$). Each of these three points has a local equation of the form $a^2 + \epsilon\, pq = 0$ in $\mathbf{A}^3$ with coordinates (a, p, q) and ϵ a polynomial with $\epsilon(0) \neq 0$ (for Σ_1: $(p, q, \epsilon) = (b, c', 1+c')$; for Σ_2: $(c, b', 1+b')$; for Σ_3: $(b, \gamma, -(1-\gamma))$), with the curves through the point being among the p-axis $\{a = q = 0\}$ and the q-axis $\{a = p = 0\}$. Blow up the origin: in the p-chart ($a = pa'$, $q = pq'$) the strict transform is $a'^2 + \epsilon q' = 0$, nonsingular along $p = 0$ ($\partial/\partial q' = \epsilon(0) \neq 0$ there); the q-chart is symmetric; in the a-chart ($p = ap'$, $q = aq'$) it is $1 + \epsilon p'q' = 0$, nonsingular along $a = 0$ (there $q' \neq 0$ and $\partial/\partial p' = \epsilon(0)q' \neq 0$). So the new surface is nonsingular above the point, the exceptional curve is the nonsingular conic $a^2 + \epsilon(0)pq = 0$ in $\mathbf{P}^2$, $\cong \mathbf{P}^1$, and the strict transforms of the p-axis and of the q-axis meet it in the two distinct points $(0 : 1 : 0)$, $(0 : 0 : 1)$, so they are separated. Call the new curves C_6 (over Σ_1), C_7 (over Σ_2), C_8 (over Σ_3). Then C_6 meets C_5 and C_3; C_7 meets C_5 and C_2; C_8 meets only C_5; and C_5 no longer meets C_3 or C_2.

Conclusion. We have blown up eight closed points (P, P_1, P_2, R_1, R_2, Σ_1, Σ_2, Σ_3) in succession; the resulting surface $\widetilde{X}$ is nonsingular: it is isomorphic to $X - P$ away from the exceptional

curves, and along the exceptional curves we found at each stage that the only singular points are the ones blown up next. The inverse image of P is $C_1 \cup \cdots \cup C_8$, each $C_i \cong \mathbf{P}^1$ (the strict transform of a nonsingular curve is isomorphic to it). Keeping track of the intersections at every stage (a pair of curves can only meet over a point at which both pass, or at the new exceptional curve; we recorded which curves pass through each blown-up point and that they are separated afterwards), the lines C_i, C_j meet if and only if $\{i,j\}$ is one of

$$\{1,4\},\ \{4,3\},\ \{3,6\},\ \{6,5\},\ \{5,7\},\ \{7,2\},\ \{5,8\}.$$

This is the graph: the chain $C_1 - C_4 - C_3 - C_6 - C_5 - C_7 - C_2$ of seven lines, together with C_8 attached to C_5, which is the third line from the end C_2. That is the Dynkin diagram E_8 (arms of length $1, 2, 4$ from the trivalent vertex C_5). □

6. Classification of surfaces

Conventions. As in Chapter V, k is algebraically closed and a surface is a nonsingular projective surface; K denotes a canonical divisor, $p_g = \dim H^0(X, \mathcal{L}(K)) = \dim H^2(X, \mathcal{O}_X)$, $q = \dim H^1(X, \mathcal{O}_X)$ and $p_a = \chi(\mathcal{O}_X) - 1 = p_g - q$. The text of Section 6 contains two exercises (6.1 and 6.2). Both use results quoted without proof in the section (Thms. 6.2, 6.3, 6.5); the solutions use only the statements.

Exercise 6.1

> **Problem.** Let X be a surface in $\mathbf{P}^n$, $n \geq 3$, defined as the complete intersection of hypersurfaces of degrees $d_1, \ldots, d_{n-2}$, with each $d_i \geq 2$. Show that for all but finitely many choices of $(n, d_1, \ldots, d_{n-2})$, the surface X is of general type. List the exceptional cases, and where they fit into the classification picture.

Solution. As always in Chapter V, X is assumed nonsingular (for every choice of the degrees there exist nonsingular complete intersections, by Bertini's theorem, II, 8.18 and Ex. 8.4(d)).

Step 1: the canonical sheaf. Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbf{P}^n$. Since X is a complete intersection, $\mathcal{I}/\mathcal{I}^2 \cong \bigoplus_{i=1}^{n-2} \mathcal{O}_X(-d_i)$ (the equations form a regular sequence), and as X is nonsingular the conormal sequence $0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{P}^n}|_X \to \Omega_X \to 0$ is an exact sequence of locally free sheaves (II, 8.17). Taking determinants and using $\bigwedge^n \Omega_{\mathbf{P}^n} \cong \mathcal{O}(-n-1)$ (II, 8.13) we get $\mathcal{O}_X(-n-1) \cong \mathcal{O}_X(-\sum d_i) \otimes \omega_X$, i.e.

$$\omega_X \cong \mathcal{O}_X(m), \qquad m = d_1 + \cdots + d_{n-2} - n - 1$$

(adjunction formula for complete intersections, cf. II, Ex. 8.4(e)). Also $q = \dim H^1(X, \mathcal{O}_X) = 0$ (III, Ex. 5.5(c): $H^i(X, \mathcal{O}_X(j)) = 0$ for $0 < i < \dim X$ and all j).

Step 2: $m > 0$ implies general type. If $m \geq 1$, $\mathcal{L}(K) = \mathcal{O}_X(m)$ is ample (the restriction of the ample sheaf $\mathcal{O}_{\mathbf{P}^n}(m)$), so $\mathcal{L}(nK) = \mathcal{O}_X(nm)$ is very ample for $n \gg 0$ (II, 7.6). Then $|nK|$ is the complete linear system of a very ample divisor and determines an embedding $X \hookrightarrow \mathbf{P}^N$, in particular a birational morphism of X onto its image. By Theorem 6.5, $\kappa(X) = 2$, i.e. X is of general type.

Step 3: which (n, d_i) have $m \leq 0$. We need $\sum d_i \leq n+1$ with $n-2$ summands $d_i \geq 2$, so $2(n-2) \leq \sum d_i \leq n+1$, hence $n \leq 5$. Going through the cases (unordered degrees):

- $n = 3$ (one hypersurface of degree $d \geq 2$): $d \leq 4$, i.e. $d = 2, 3, 4$, with $m = -2, -1, 0$;
- $n = 4$ (two hypersurfaces): $d_1 + d_2 \leq 5$: $(2, 2)$ with $m = -1$ and $(2, 3)$ with $m = 0$;
- $n = 5$ (three hypersurfaces): $d_1 + d_2 + d_3 \leq 6$: only $(2, 2, 2)$, with $m = 0$;
- $n \geq 6$: $2(n-2) > n+1$, so no exceptional case.

So there are exactly 6 exceptional cases, and for all other $(n, d_1, \ldots, d_{n-2})$ we have $m \geq 1$ and, by Step 2, X is of general type. (For instance every nonsingular surface of degree ≥ 5 in $\mathbf{P}^3$.)

Step 4: the exceptional cases. First, $H^0(X, \mathcal{O}_X(j)) = 0$ for $j < 0$: if $s \neq 0$ were a section, $\operatorname{div}(s)$ would be an effective divisor linearly equivalent to jH' (H' a hyperplane section), so $\operatorname{div}(s).H' = jH'^2 < 0$ since $H'^2 = \deg X > 0$; but an effective divisor has nonnegative intersection number with an ample divisor.

(i) $m < 0$: the cases $(n; d) = (3; 2), (3; 3), (4; 2, 2)$ are rational. Here $p_g = h^0(\mathcal{O}_X(m)) = 0$, $P_2 = h^0(\mathcal{O}_X(2m)) = 0$, and $q = 0$ so $p_a = p_g - q = 0$. By Castelnuovo's criterion, Theorem 6.2 (X rational $\iff$ $p_a = P_2 = 0$), X is rational, hence has $\kappa = -1$ (Theorem 6.1: $\kappa = -1$ iff X is rational or ruled). Concretely, the quadric in $\mathbf{P}^3$ is $\mathbf{P}^1 \times \mathbf{P}^1$ (1.4.3), the nonsingular cubic surface is $\mathbf{P}^2$ blown up in 6 points (Cor. 4.7 and Remark 4.7.1), and the complete intersection of two quadrics in $\mathbf{P}^4$ is $\mathbf{P}^2$ blown up in 5 points (a del Pezzo surface of degree 4: Remark 4.7.1 and Ex. 4.13(b)).

(ii) $m = 0$: the cases $(3; 4)$, $(4; 2, 3)$, $(5; 2, 2, 2)$ are K3 surfaces. Here $\omega_X \cong \mathcal{O}_X$, so $K = 0$, $P_n = h^0(\mathcal{O}_X) = 1$ for all $n \geq 1$, so the ring $\bigoplus H^0(nK) = k[T]$ has transcendence degree 1 and $\kappa = 0$; and $q = 0$ by Step 1, $p_g = 1$. By definition (Theorem 6.3(1)) X is a K3 surface: the quartic surface in $\mathbf{P}^3$, the complete intersection of a quadric and a cubic in $\mathbf{P}^4$ (degree 6), and the complete intersection of three quadrics in $\mathbf{P}^5$ (degree 8).

Summary: with the six exceptions listed, all complete intersection surfaces are of general type ($\kappa = 2$); the exceptions are 3 rational surfaces ($\kappa = -1$; degrees $(3; 2), (3; 3), (4; 2, 2)$) and 3 K3 surfaces ($\kappa = 0$; degrees $(3; 4), (4; 2, 3), (5; 2, 2, 2)$). □

Exercise 6.2

> **Problem.** Prove the following theorem of Chern and Griffiths. Let X be a nonsingular surface of degree d in $\mathbf{P}^{n+1}_{\mathbf{C}}$, which is not contained in any hyperplane. If $d < 2n$, then $p_g(X) = 0$. If $d = 2n$, then either $p_g(X) = 0$, or $p_g(X) = 1$ and X is a K3 surface. [*Hint*: Cut X with a hyperplane and use Clifford's theorem (IV, 5.4). For the last statement, use the Riemann–Roch theorem on X and the Kodaira vanishing theorem (III, 7.15).]

Solution. Here $n \geq 2$ (as X is a surface in $\mathbf{P}^{n+1}$ not in a hyperplane), H denotes both a hyperplane of $\mathbf{P}^{n+1}$ and the class of a hyperplane section on X, and $d = H^2$ is the degree of X. For a hyperplane H let $s_H \in H^0(X, \mathcal{O}_X(1))$ be the section defined by an equation of H; $C_H = \operatorname{div}(s_H) = X \cap H$.

Step 0: preliminaries on hyperplane sections. By Bertini's theorem (II, 8.18) and the connectedness of ample divisors (III, 7.9, cf. also the proof of Lemma 1.2) there is a dense open subset U of the dual projective space such that for $H \in U$, C_H is an irreducible nonsingular curve (and the divisor $\operatorname{div}(s_H)$ is reduced).

(a) *If $H \neq H'$ then* $\operatorname{div}(s_H) \neq \operatorname{div}(s_{H'})$. Otherwise $s_{H'}/s_H$ is a rational function on X without zeros or poles, hence a constant λ (as X is projective and integral), so the nonzero linear form $\ell_{H'} - \lambda\ell_H$ vanishes on X, i.e. X lies in a hyperplane, contrary to hypothesis. In particular $H \mapsto C_H$ is injective on U.

(b) *For $H \in U$ the curve $C = C_H \subseteq H \cong \mathbf{P}^n$ is nondegenerate, of degree d.* If C lay in a hyperplane of H, that hyperplane is $H \cap H'$ with $H' \neq H$ a hyperplane of $\mathbf{P}^{n+1}$. Then $s_{H'}$ vanishes on C, so $\operatorname{div}(s_{H'}) = C + Z$ with $Z \geq 0$; but $\operatorname{div}(s_{H'}) \sim H \sim C$, so $Z \sim 0$ and Z is effective, hence $Z = 0$ (a nonzero effective divisor on a projective variety is not linearly equivalent to 0), contradicting (a). The degree is $\deg_C \mathcal{O}_X(1)|_C = H.C = H^2 = d$.

Step 1: the key dichotomy. Let $H \in U$, $C = C_H$ of genus g, and let D denote the divisor of hyperplane sections of $C \subseteq \mathbf{P}^n$, so $\mathcal{O}_C(D) = \mathcal{O}_X(1)|_C = \mathcal{L}(C) \otimes \mathcal{O}_C$, $\deg D = d$. Since C is nondegenerate in $\mathbf{P}^n$, the sections $x_0, \ldots, x_n$ restricted to C are linearly independent elements of $H^0(C, \mathcal{O}_C(D))$, so $\dim |D| \geq n$. By the adjunction formula (II, 8.20, Prop. 1.5), $\omega_C \cong (\omega_X \otimes \mathcal{L}(C)) \otimes \mathcal{O}_C$, so

$$\omega_C \otimes \mathcal{O}_C(-D) \cong \omega_X \otimes \mathcal{O}_C, \qquad \text{i.e.} \qquad K_C - D \sim K_X|_C. \tag{1}$$

Let $\omega \in H^0(X, \omega_X)$ be nonzero and $E = \operatorname{div}(\omega) \sim K_X$, an effective divisor. Since E has finitely many components, and by (a) the curves C_H ($H \in U$) are pairwise distinct and irreducible, there are only finitely many $H \in U$ with $C_H \subseteq \operatorname{Supp} E$. For all other $H \in U$ the restriction $\omega|_{C_H} \in H^0(C_H, \omega_X \otimes \mathcal{O}_{C_H})$ is nonzero, so by (1) $l(K_C - D) > 0$: the effective divisor D is *special*. Clifford's theorem (IV, 5.4) gives

$$n \leq \dim |D| \leq \tfrac{1}{2} \deg D = \tfrac{d}{2}. \tag{2}$$

Step 2: $d < 2n$ implies $p_g = 0$. If $d < 2n$, then (2) is impossible, so there is no nonzero ω at all: $p_g = h^0(\omega_X) = 0$.

Step 3: the case $d = 2n$. Suppose $d = 2n$ and $p_g > 0$; take $\omega \neq 0$, $E = \operatorname{div} \omega$, and $H \in U$ with $C = C_H \not\subseteq \operatorname{Supp} E$ (all but finitely many $H \in U$). Then (2) holds with equality throughout: $\dim |D| = n = \frac{1}{2} \deg D$, so $h^0(\mathcal{O}_C(D)) = n + 1$ and $C \hookrightarrow \mathbf{P}^n$ is given by the complete linear system $|D|$. By the equality case of Clifford's theorem, either $D \sim 0$ (excluded, $\deg D = d > 0$), or $D \sim K_C$, or C is hyperelliptic and D is a multiple of the g^1_2, i.e. $D \sim n g^1_2 = \varphi^* \mathcal{O}_{\mathbf{P}^1}(n)$ with $\varphi : C \to \mathbf{P}^1$ the double cover. The last case cannot occur: $H^0(\mathbf{P}^1, \mathcal{O}(n))$ has dimension $n + 1 = h^0(\mathcal{O}_C(D))$, so every section of $\mathcal{O}_C(D)$ is a pull-back by φ, whence the embedding $C \hookrightarrow \mathbf{P}^n$ would factor through φ, which is $2 : 1$; contradiction. Therefore

$$D \sim K_C, \qquad \text{and by (1)} \qquad \omega_X \otimes \mathcal{O}_C \cong \mathcal{O}_C.$$

Hence $E.C = \deg_C(\mathcal{L}(E) \otimes \mathcal{O}_C) = \deg(\omega_X \otimes \mathcal{O}_C) = 0$. But C is a hyperplane section and E is effective, so $E.C$ is the sum of the degrees of the components of E in $\mathbf{P}^{n+1}$ (with multiplicity), which is 0 only if $E = 0$. So $\omega_X \cong \mathcal{L}(E) = \mathcal{O}_X$: $K = 0$, and $p_g = h^0(\mathcal{O}_X) = 1$.

Step 4: $q = 0$, so X is a K3 surface. Keep $H \in U$ as in Step 3, $C = C_H$; recall $\mathcal{L}(C) \cong \mathcal{O}_X(1)$ and $h^0(\mathcal{O}_C(D)) = n + 1$. The sequence $0 \to \mathcal{O}_X \to \mathcal{O}_X(1) \to \mathcal{O}_C(D) \to 0$ ($\mathcal{O}_X(1) \otimes \mathcal{O}_C = \mathcal{O}_C(D)$) gives

$$H^0(X, \mathcal{O}_X(1)) \xrightarrow{r} H^0(C, \mathcal{O}_C(D)) \to H^1(X, \mathcal{O}_X) \to H^1(X, \mathcal{O}_X(1)).$$

Here $\ker r = H^0(X, \mathcal{O}_X) = k$, and $h^0(\mathcal{O}_X(1)) \geq n + 2$ since the $n + 2$ coordinates are linearly independent on X (nondegeneracy). So $\dim \operatorname{im} r \geq n + 1 = h^0(\mathcal{O}_C(D))$, and r is surjective. On the other hand, as $\omega_X \cong \mathcal{O}_X$, Kodaira vanishing in the form $H^1(X, \omega_X \otimes \mathcal{L}) = 0$ for $\mathcal{L}$ ample (this is III, 7.15, $H^1(X, \mathcal{L}^{-1}) = 0$, combined with Serre duality, III, 7.7) gives $H^1(X, \mathcal{O}_X(1)) = H^1(X, \omega_X \otimes \mathcal{O}_X(1)) = 0$. Exactness forces $H^1(X, \mathcal{O}_X) = 0$, so $q = 0$ ($k = \mathbf{C}$ is used here, and for Bertini and Kodaira).

Thus $K = 0$ and $q = 0$: X is a K3 surface (Theorem 6.3(1)), with $p_g = 1$ (equivalently $p_a = 1$). We have shown that for $d = 2n$ either $p_g = 0$ or $p_g = 1$ and X is a K3 surface. □

(*Remark.* The alternative $p_g = 1$ for $d = 2n$ really occurs: the nonsingular quartic in $\mathbf{P}^3$ ($n = 2$), the complete intersection of a quadric and a cubic in $\mathbf{P}^4$ ($n = 3$, $d = 6$) and of three quadrics in $\mathbf{P}^5$ ($n = 4$, $d = 8$) are K3 surfaces (Ex. 6.1). The Riemann–Roch theorem on X was not needed in the argument above: the vanishing of q was obtained from the equality case of Clifford's theorem together with Kodaira vanishing.)